

Resolución Cuántica de Ecuaciones Diferenciales Parciales

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Bogotá, Colombia
2026

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Tesis presentada como requisito parcial para optar por el título de:
Magíster en Matemática Aplicada

Director(a):
PhD. [Nombre del Director]
[Indicar si es Profesor Titular/Asociado] - Departamento de Matemáticas
Facultad de Ciencias
Universidad Nacional de Colombia

Línea de investigación:
Modelamiento Matemático y Computación Científica

Universidad Nacional de Colombia
Facultad de Ciencias
Departamento de Matemáticas
2026

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Bogotá, 2026

Julián Sierra Salamanca

Resumen

Resolución Cuántica de Ecuaciones Diferenciales Parciales

Palabras clave:

Abstract

Quantum Resolution of Partial Differential Equations

Keywords:

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Introducción

Partial differential equations (PDEs) are among the most powerful tools in science, economics, engineering, and applied mathematics. They appear in models such as kinetic dynamic, Fluid and gas dynamics, Granular materials flows, semiconductor, socio-economics, among others [1]. Through this, we can describe and explain the multiple phenomena of concern in science. By the nature of PDEs, these equations are used to model the behavior of fluids, bodies, molecules, proteins, and social individuals governed by physical laws and social behavior.

Differential Galois theory for linear ordinary differential equations (ODEs) shows that only those equations whose differential Galois group is solvable admit solutions expressible in terms of elementary functions (exponentials, integrals, and algebraic functions) [2]. The vast majority of higher-order linear ODEs possess non-solvable Galois groups, which implies that their solutions lack a closed-form expression in terms of elementary functions and must instead be solved via numerical methods such as Euler, Multisteps, Runga-Kutta and extrapolation methods [3, 4].

Hence, numerical methods are the only way to obtain approximate solutions in most cases. The classical approaches are finite difference methods (FDM) [5], finite element methods (FEM) [6], and finite volume methods (FVM) [7]; each of them discretizes the domain into small pieces forming a mesh and replaces derivatives with algebraic operators so that the computation can be carried out. On the other hand, spectral methods consist of projecting the solution onto a finite-dimensional subspace spanned by global basis functions, such as trigonometric or orthogonal polynomials including Legendre and Chebyshev polynomials [8].

Even so, numerical methods suffer from the curse of dimensionality, complications from complex geometries, nonlinearity and stiffness, lack of flexibility, and limitations in data integration [9]. Computational cost grows exponentially with the dimensionality of the domain. Mesh-based methods require special grid treatment in cases of complex, irregular, and time-dependent domains. Highly complex PDEs with multi-scale features often require a fine grid, which in turn leads to high computational costs.

Due to the rapid growth of available data and computing resources, machine learning techniques have become established tools across various sciences and disciplines. These techniques have been used in computer vision to analyze and segment cells, leading to advances in biomedical engineering [10]; AlphaFold resolved the protein folding problem using attention mechanisms to understand the relationships between amino acids in protein chains [11]; and reinforcement learning has been used to control cylinders in fluid flow to minimize aerodynamic resistance [12].

Current engineering challenges involving large amounts of data, complex geometry, and a lack of knowledge about the internal equations governing them, together with the boom in machine learning and deep learning [13--15], have opened the field to the development of tools that can understand behavior alongside data and model it. Three areas have emerged from this: data-driven learning [16--18], operator learning [19, 20], and physics-informed machine learning (PIML) [21, 22].

Many scientific problems require vast amounts of data that are often unavailable. Consequently, a purely data-driven approach is not always the most effective way to address these challenges [23]. Neural operators and physics-informed machine learning do not rely solely on training data; instead, they leverage physical information, such as partial differential equations (PDEs) and the necessary boundary conditions. However, these paradigms are not mutually exclusive. In fact, hybrid approaches have demonstrated their full potential by combining assumptions about the underlying phenomena, available data, and neural network architectures [24--26].

Furthermore, these new approaches have certain limitations associated with optimization [27--31] and architecture [32--34]. There is gradient imbalance caused by the stiffness of the cost function, spectral bias that hinders the representation of sharp peaks, discontinuities, and propagating wave fronts, and causality issues for propagative PDEs. On the other hand, some authors have developed theories concerning convergence [35--37], numerical stability [38, 39], and approximation error [40, 41]. For spectrum analysis and the training process, the Neural Tangent Kernel (NTK) is a powerful tool to explain the ill-conditioning of classical PINNs [28, 42] regarding spectral bias, stiffness, and rigid gradients.

Justificación y objetivos

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1 About Partial Differential Equations

It is now well established that partial differential equations (PDEs) are the primary tool for modeling phenomena in physics and engineering, and that there is no general method for solving them, since each case has its own particularities. Modern theory divides them into two broad categories: evolution equations and stationary equations. Traditional hyperbolic and parabolic equations are classified as evolution equations, whereas elliptic equations are classified as stationary equations. Consequently, numerical analysis treats the solution process separately for each case.

This chapter reviews the theoretical aspects that are fundamental when working with PDEs. In particular, as previously mentioned, each case presents its own characteristics; one of these, which we analyze throughout the chapter, is regularity. The degree of smoothness that the solution acquires, retains, or loses in relation to the initial data and source functoin. This question is not only theoretical. A physics-informed neural network (Chapter 2) is a smooth function that is trained by evaluating derivatives of the solution, so the regularity of the theorical solution decides how difficult the problem is for the network. For that reason, every equation of Section 1.1 ends with a short paragraph about its regularization, Section 1.2 introduces the spaces where non-smooth solutions live, Section 1.3 classifies the three families that appear in our benchmark (elliptic, parabolic and hyperbolic), and Sections 1.4 and 1.5 explain why we chose the Klein--Gordon equation as the case study of this thesis.

The conceptual framework of this chapter is based fundamentally on the theory of partial differential equations developed by Evans [43]. Unless otherwise stated, the definitions, notations, and theorems presented in this chapter follow the exposition of that work.

1.1 Linear Partial Differential Equations

The four equations of this section are the prototypes of the theory: the transport equation (first order, hyperbolic), Laplace's equation (elliptic), the heat equation (parabolic) and the wave equation (hyperbolic). For each of them we look at the same three things: the homogeneous problem, the non-homogeneous problem through Duhamel's principle, and the regularity of the solution. The complete derivations of the formulas of this section, with all the intermediate steps, are in Appendix A. Here we present the statement of each problem, its solution, and, above all, what the solution indicates about regularity.

1.1.1 Transport Equation

The transport equation is a natural starting point for the study of first-order partial differential equations: it is simple enough to solve explicitly, yet it already exhibits the main features of the theory [43]. It models

the propagation of a physical quantity in space. Here, we only consider a constant velocity $\mathbf{b} \in \mathbb{R}^n$ for its introduction.

The pure homogeneous or non-homogeneous transport problem is represented by a first-order differential operator L, which is hyperbolic and describes the propagation or advection of a profile along characteristic curves. Its solution, whether classical or weak, is given by a rigid translation of the initial data, $u(\mathbf{x}, t) = g(\mathbf{x} - t\mathbf{b})$.

Homogeneous case

Let $u \in C^1(\mathbb{R}^n \times [0, \infty))$ and consider the initial condition for the homogeneous problem:

$$\begin{cases} \partial_t u(\mathbf{x}, t) + \mathbf{b} \cdot \nabla u(\mathbf{x}, t) = 0 & \text{in } \mathbf{x} \in \mathbb{R}^n,\ t \in (0, \infty), \\ u(\mathbf{x}, 0) = g(\mathbf{x}) & \text{on } \mathbf{x} \in \mathbb{R}^n,\ \{t = 0\}. \end{cases} \tag{1-1}$$

Writing the equation in matrix form with the constant space--time vector field $\mathbf{V} = (\mathbf{b}, 1)^T \in \mathbb{R}^{n+1}$ (Appendix A.1), it reads $D_{\mathbf{V}} u = \nabla_{\mathbf{x},t} u \cdot \mathbf{V} = 0$, which establishes that the directional derivative of u in the direction of the vector $\mathbf{V}$ is identically zero throughout the space--time domain. This means that the velocity vector is perpendicular to the gradient and tangent to u. In other words, if one moves in the direction of $\mathbf{V}$, the value of u remains constant along the characteristic trajectory. The solution to $D_{\mathbf{V}} u = 0$ is $u(\mathbf{x}, t) = g(\mathbf{x} - t\mathbf{b})$.

Non-homogeneous case

To analyze the non-homogeneous case:

$$\begin{cases} \partial_t u(\mathbf{x}, t) + \mathbf{b} \cdot \nabla u(\mathbf{x}, t) = f(\mathbf{x}, t) & \text{in } \mathbf{x} \in \mathbb{R}^n,\ t \in (0, \infty), \\ u(\mathbf{x}, 0) = g(\mathbf{x}) & \text{on } \mathbf{x} \in \mathbb{R}^n,\ \{t = 0\}, \end{cases} \tag{1-2}$$

we can apply the same approach as in the homogeneous case: following the characteristic curve $z(s) = u(\mathbf{x} + s\mathbf{b}, t + s)$, $s \in [-t, 0]$, whose derivative is $dz/\,ds = f(\mathbf{x} + s\mathbf{b}, t + s)$, and integrating from $-t$ to 0 (Appendix A.1), we obtain the expression

$$u(\mathbf{x}, t) = g(\mathbf{x} - t\mathbf{b}) + \int_0^t f\big(\mathbf{x} + (\tau - t)\mathbf{b}, \tau\big)\, d\tau. \tag{1-3}$$

Through Duhamel's principle, this result means that the solution is the sum of the transported initial value $g(\mathbf{x} - t\mathbf{b})$ and the values accumulated from the source f, generated in the past times τ along the same characteristic curve.

Regularization of transport operators

The transport operator $L = \partial_t + \mathbf{b} \cdot \nabla$ has no regularizing effect in the interior of the domain. This means that it conserves any roughness, sharp edge, or discontinuity of the initial condition g along the characteristic curves. To handle this problem, one can regularize the data through mollification, or the operator through the vanishing viscosity method, which allows working smoothly in the Sobolev spaces $W^{k,p}$ or the Lebesgue spaces L^p (Section 1.2). We will explore mollification here; for the vanishing viscosity method, which adds a small diffusion $\varepsilon \Delta u$ and therefore turns the transport equation into a parabolic one, the reader is referred to

Evans [43]. Given $g \in L^p(\mathbb{R}^n)$ and a standard mollifier $\eta_\epsilon \in C_c^\infty(\mathbb{R}^n)$, we define the mollified initial condition $g^\epsilon \in C^\infty(\mathbb{R}^n) \cap L^p(\mathbb{R}^n)$ through the convolution

$$g^\epsilon(\mathbf{x}) = (\eta_\epsilon * g)(\mathbf{x}) = \int_{\mathbb{R}^n} \eta_\epsilon(\mathbf{x}-\mathbf{y})\, g(\mathbf{y})\, d\mathbf{y}, \tag{1-4}$$

and the corresponding regularized solution

$$u^\epsilon(\mathbf{x},t) = g^\epsilon(\mathbf{x}-t\mathbf{b}) = (\eta_\epsilon * g)(\mathbf{x}-t\mathbf{b}). \tag{1-5}$$

As $\epsilon \to 0$, the mollified g^ϵ converges to g in $L^p(\mathbb{R}^n)$ for $1 \leq p < \infty$. That is, mollification solves the regularization problem through the smoothing function η, and u is also in L^p. The smoothness depends only on the regularity of the initial data g; the operator itself does not add any.

1.1.2 Laplace's Equation

To introduce Laplace's equation, we define the Laplacian operator $\Delta u := \nabla^2 u$. Laplace's equation is used to model equilibrium in the stationary state. Let the flux be $\mathbf{F}(\mathbf{x},t) = -b\nabla u(\mathbf{x},t)$ with $b > 0$, and let $u(\mathbf{x},t)$ represent a physical scalar variable such as temperature, an electrostatic potential, or a mass density. In stationary equilibrium, there is no temporal change of u; thus, the term $\partial_t u(\mathbf{x},t)$ is zero. For convenience, we write $u(\mathbf{x},t) = u(\mathbf{x})$ for any value of $t > 0$. The conservation of mass requires that the flux divergence be zero, $\operatorname{div}(\mathbf{F}) = 0$; consequently, the divergence of $\mathbf{F}$ is expressed as $-b\Delta u = 0$. By the divergence theorem, the integral of the Laplacian can be expressed in terms of the flux through the boundary ∂U of any subregion U:

$$\int_U \Delta u \, d\mathbf{x} = \int_{\partial U} \nabla u \cdot \mathbf{n} \, dS, \tag{1-6}$$

which gives meaning to the Laplacian operator Δ as a measure of local disequilibrium, based on the properties of the Laplace operator. On the other hand, this means that the function u in the interior of the domain neither adds nor removes flux; that is, as we will see later, u is the average of its surrounding points.

Homogeneous case

In searching for a solution to $\Delta u(\mathbf{x}) = 0$ for $\mathbf{x} \in \mathbb{R}^n$, Evans [43] describes the *fundamental solution of Laplace's equation*, defined by

$$\Phi(\mathbf{x}) = \begin{cases} -\dfrac{1}{2\pi} \log \|\mathbf{x}\|_2 & (n=2), \\ \dfrac{1}{n(n-2)\alpha(n)} \dfrac{1}{\|\mathbf{x}\|_2^{n-2}} & (n \geq 3), \end{cases} \tag{1-7}$$

where $\alpha(n)$ is the volume of the unit ball in $\mathbb{R}^n$ (for $n = 1$ the analogous function is $-\frac{1}{2}|x|$). Some important aspects to analyze are:

1. The function Φ has a singular point at $\mathbf{0}$ for $n \geq 2$.
2. $\Delta\Phi = 0$ for $\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$, but not at the point $\mathbf{0}$. Conceptually, Φ represents the fundamental building block; thus, at the singular point the fundamental solution must be understood as a distribution that contains the information of the point, and with the sign convention of Evans [43] it satisfies $-\Delta\Phi = \delta_{\mathbf{0}}$.
3. Equation (1-7) does not contain boundary information, which is essential for engineering and science applications.

Consequently, it is necessary to introduce more information from the problem itself through a Dirichlet boundary condition, which specifies the values of the function on the boundary through an arbitrary function $g(\mathbf{x})$. Thus, consider the following boundary value problem for $\Omega \subset \mathbb{R}^n$ bounded and $u \in C^2(\bar{\Omega})$:

$$\begin{cases} \Delta u(\mathbf{x}) = 0 & \text{in } \mathbf{x} \in \Omega, \\ u(\mathbf{x}) = g(\mathbf{x}) & \text{on } \mathbf{x} \in \partial\Omega. \end{cases} \tag{1-8}$$

The main idea is to find the solution u using Φ. In stationary problems, the value of u is the result of relaxing the system; in this case no source of perturbation exists, and the only information that acts as a perturbation source is the boundary data g. Thus, the solution may be expressed as an integral of g over the boundary. Following Evans [43], Green's second identity applied to u and Φ gives a representation of u that still contains the unknown normal derivative $\partial u/\partial \mathbf{n}$ on the boundary. To eliminate it, it is necessary to build an auxiliary (corrector) function $\phi^{\mathbf{x}}$ that is harmonic in Ω and matches the fundamental solution on the boundary (the complete derivation is in Appendix A.2). The unknown normal derivative cancels and we finally find the solution u through Green's function $G(\mathbf{x}, \mathbf{y}) = \Phi(\mathbf{y} - \mathbf{x}) - \phi^{\mathbf{x}}(\mathbf{y})$:

$$u(\mathbf{x}) = -\int_{\partial\Omega} g(\mathbf{y}) \frac{\partial G}{\partial \mathbf{n}}(\mathbf{x}, \mathbf{y})\, dS(\mathbf{y}), \tag{1-9}$$

where $\partial G/\partial \mathbf{n} = \nabla_{\mathbf{y}} G \cdot \mathbf{n}$. Finally, we have an analytical solution through an auxiliary function that imposes the cancellation on the boundary:

$$\begin{cases} -\Delta_{\mathbf{y}} G(\mathbf{x}, \mathbf{y}) = \delta(\mathbf{x} - \mathbf{y}) & \text{in } \mathbf{y} \in \Omega, \\ G(\mathbf{x}, \mathbf{y}) = 0 & \text{on } \mathbf{y} \in \partial\Omega. \end{cases} \tag{1-10}$$

Non-homogeneous case

The non-homogeneous case is the *Poisson equation*. Consider the bounded domain Ω; the problem is given by

$$\begin{cases} -\Delta u(\mathbf{x}) = f(\mathbf{x}) & \text{in } \mathbf{x} \in \Omega, \\ u(\mathbf{x}) = h(\mathbf{x}) & \text{on } \mathbf{x} \in \partial\Omega. \end{cases} \tag{1-11}$$

The source term f acts as a local curvature generator and induces the perturbation of a local point $\mathbf{x}$. With respect to the homogeneous problem, the source is added as a new term to the previous equation by the same Green method:

$$u(\mathbf{x}) = \int_{\Omega} G(\mathbf{x}, \mathbf{y}) f(\mathbf{y})\, d\mathbf{y} - \int_{\partial\Omega} h(\mathbf{y}) \frac{\partial G}{\partial \mathbf{n}}(\mathbf{x}, \mathbf{y})\, dS(\mathbf{y}). \tag{1-12}$$

Regularity and some properties

The Green's function and the Laplacian operator smooth the function through the mean-value property. Applying Equation (1-9) to a ball $B(\mathbf{x}, r) \subset \Omega$ centered at a point $\mathbf{x}$ inside the domain, where the normal derivative of G is constant on the sphere $\partial B(\mathbf{x}, r)$ and its denominator is the surface area $|\partial B(\mathbf{x}, r)| = n\alpha(n) r^{n-1}$ (Appendix A.2), we obtain

$$u(\mathbf{x}) = \frac{1}{|\partial B(\mathbf{x}, r)|} \int_{\partial B(\mathbf{x}, r)} u(\mathbf{y})\, dS(\mathbf{y}) = \fint_{\partial B(\mathbf{x}, r)} u\, dS \qquad \text{(mean-value property)}. \tag{1-13}$$

This argument is a heuristic version of the proof; the rigorous proof differentiates the spherical average with respect to r [43, Section 2.2.2]. The mean-value property is responsible for the smoothness of u in

Ω, because the value of u at any point $\mathbf{x}$ is averaged over its neighborhood. The smoothness theorem for Laplace's equation states that if $u \in C(\bar{\Omega})$ satisfies the mean-value property for each ball $B(\mathbf{x}, r) \subset \Omega$, then $u \in C^\infty(\Omega)$. Furthermore, its derivatives are controlled by the derivative estimates in Evans [43]: with $k = |\alpha|$, if u is harmonic in Ω then

$$|D^\alpha u(\mathbf{x})| \leq \frac{C_k}{r^{n+k}} \|u\|_{L^1(B(\mathbf{x},r))}, \tag{1-14}$$

which means that the α-order derivatives of u are bounded by the L^1 norm on the ball $B(\mathbf{x}, r)$ and the constant C_k. An aspect of interest to analyze is the factor r^{-n-k}: if α is of high order, the bound grows as a power of $1/r$, so the scale r acts as a penalization term. In other words, harmonic functions are infinitely smooth, but measuring finer and finer variations on a fixed scale requires larger and larger constants.

1.1.3 Heat Equation

Consider an open domain $\Omega \subset \mathbb{R}^n$ that contains a continuous conductor. Let $\mathbf{F}$ be the flux vector of the scalar function $u(\mathbf{x}, t)$, which changes for $t > 0$. The conservation of energy yields $u_t + \operatorname{div} \mathbf{F} = 0$. If $\mathbf{F}$ is defined as $-b\nabla u$ with $b > 0$, the equation is transformed to $u_t - b\Delta u = 0$, obtaining the homogeneous heat equation (in what follows $b = 1$).

The temporal derivative is of first order while the spatial derivative is of second order. Looking for solutions of the form $u(\mathbf{x}, t) = t^{-\alpha} v(\mathbf{x}/t^\beta)$, invariant under the scaling $(\mathbf{x}, t) \mapsto (\lambda^\beta \mathbf{x}, \lambda t)$, that also conserve the thermal mass (Appendix A.3), Evans [43] shows that $\beta = 1/2$ and $\alpha = n/2$. Thus, we obtain *the fundamental solution of the heat equation*, defined as

$$\Phi(\mathbf{x}, t) = \begin{cases} \dfrac{1}{(4\pi t)^{n/2}} e^{-\frac{\|\mathbf{x}\|_2^2}{4t}} & (\mathbf{x} \in \mathbb{R}^n,\ t > 0), \\ 0 & (\mathbf{x} \in \mathbb{R}^n,\ t < 0). \end{cases} \tag{1-15}$$

In this manner, this function works similarly to the Laplace fundamental solution; that is, it also has a singular point at $(\mathbf{0}, 0)$, and $\Phi(\cdot, t) \to \delta_{\mathbf{0}}$ in the sense of distributions as $t \to 0^+$.

Homogeneous case

As we saw in Laplace's equation, additional information (here, an initial condition) is necessary to find a unique solution which is not trivial. Let $u \in C^{2,1}(\mathbb{R}^n \times (0, \infty))$ and consider the Cauchy problem

$$\begin{cases} \partial_t u(\mathbf{x}, t) - \Delta u(\mathbf{x}, t) = 0 & \text{in } \mathbf{x} \in \mathbb{R}^n,\ t \in (0, \infty), \\ u(\mathbf{x}, 0) = h(\mathbf{x}) & \text{on } \mathbf{x} \in \mathbb{R}^n,\ \{t = 0\}. \end{cases} \tag{1-16}$$

As a first approach, the solution is given by the convolution of the fundamental solution Φ and the initial condition h:

$$u(\mathbf{x}, t) = (\Phi(\cdot, t) * h)(\mathbf{x}) = \frac{1}{(4\pi t)^{n/2}} \int_{\mathbb{R}^n} h(\mathbf{y})\, e^{-\frac{\|\mathbf{x}-\mathbf{y}\|_2^2}{4t}}\, d\mathbf{y} \qquad (\mathbf{x} \in \mathbb{R}^n,\ t > 0). \tag{1-17}$$

However, we need a closed analytical form for a bounded domain $\Omega \subset \mathbb{R}^n$:

$$\begin{cases} \partial_t u(\mathbf{x}, t) - \Delta u(\mathbf{x}, t) = 0 & \text{in } \mathbf{x} \in \Omega,\ t \in (0, \infty), \\ u(\mathbf{x}, 0) = h(\mathbf{x}) & \text{on } \mathbf{x} \in \Omega,\ \{t = 0\}, \\ u(\mathbf{x}, t) = g(\mathbf{x}, t) & \text{on } \mathbf{x} \in \partial\Omega,\ t \in (0, \infty), \end{cases} \tag{1-18}$$

which the previous formula does not include. Again, we must appeal to a Green's function. Applying the second Green's identity to the heat operator $L = \partial_s - \Delta$ and to its adjoint (backward) operator $L^* = -\partial_s - \Delta$, and building, as in Laplace's equation, an auxiliary function $\phi^{\mathbf{x}}(\mathbf{y}, \tau)$ that solves the heat equation with boundary values $\Phi(\mathbf{x}-\mathbf{y}, \tau)$ (Appendix A.3), we define the Green's function $G(\mathbf{x}, \mathbf{y}, \tau) = \Phi(\mathbf{x}-\mathbf{y}, \tau) - \phi^{\mathbf{x}}(\mathbf{y}, \tau)$, which vanishes for $\mathbf{y} \in \partial\Omega$, and the unknown term disappears. We obtain the closed form

$$u(\mathbf{x}, t) = \int_\Omega h(\mathbf{y}) G(\mathbf{x}, \mathbf{y}, t)\, d\mathbf{y} - \int_0^t \int_{\partial\Omega} g(\mathbf{y}, s) \frac{\partial G}{\partial \mathbf{n}}(\mathbf{x}, \mathbf{y}, t-s)\, dS(\mathbf{y})\, ds, \tag{1-19}$$

where $\partial G/\partial \mathbf{n} = \nabla_{\mathbf{y}} G \cdot \mathbf{n}$. The structure is exactly the same as Equation (1-9): an integral of the data against a kernel that already knows the boundary.

Non-homogeneous equation – Duhamel's principle

Consider the non-homogeneous problem

$$\begin{cases} \partial_t u(\mathbf{x}, t) - \Delta u(\mathbf{x}, t) = f(\mathbf{x}, t) & \text{in } \mathbf{x} \in \mathbb{R}^n,\ t \in (0, T], \\ u(\mathbf{x}, 0) = h(\mathbf{x}) & \text{on } \mathbf{x} \in \mathbb{R}^n,\ \{t = 0\}. \end{cases} \tag{1-20}$$

Duhamel's principle interprets the forcing term $f(\mathbf{x}, t)$ as an accumulation of initial conditions for the homogeneous problem. The solution is constructed from

$$u(\mathbf{x}, t) = \int_{\mathbb{R}^n} \Phi(\mathbf{x} - \mathbf{y}, t) h(\mathbf{y})\, d\mathbf{y} + \int_0^t \int_{\mathbb{R}^n} \Phi(\mathbf{x} - \mathbf{y}, t-s) f(\mathbf{y}, s)\, d\mathbf{y}\, ds, \tag{1-21}$$

where, for example, $f \in C_1^2(\mathbb{R}^n \times [0, \infty))$ with compact support guarantees that this formula is a classical solution [43]. In a bounded domain we encounter the same issue as in the homogeneous case, and the Green's function is used again to finally obtain

$$u(\mathbf{x}, t) = \int_\Omega h(\mathbf{y}) G(\mathbf{x}, \mathbf{y}, t)\, d\mathbf{y} - \int_0^t \int_{\partial\Omega} g(\mathbf{y}, s) \frac{\partial G}{\partial \mathbf{n}}(\mathbf{x}, \mathbf{y}, t-s)\, dS(\mathbf{y})\, ds + \int_0^t \int_\Omega f(\mathbf{y}, s) G(\mathbf{x}, \mathbf{y}, t-s)\, d\mathbf{y}\, ds. \tag{1-22}$$

Regularization and some properties

The fundamental solution of the heat equation is a Gaussian function (the heat kernel). Some of its properties are smoothness, boundedness, and decay to 0 as $\|\mathbf{x}\| \to \infty$. For those reasons, the heat kernel propagates with apparently infinite velocity: a change in the value of u at a point $\mathbf{x}$ has an immediate effect at any point $\mathbf{y}$ with $\operatorname{dist}(\mathbf{x}, \mathbf{y}) > 0$, as if it were moving with infinite velocity.

This kernel acts as a filter for corners, edges, and discontinuities. To see it, take a compact set K inside an open set V of space--time, $K \subset\subset V$, and a cutoff function $\zeta \in C_c^\infty(\mathbb{R}^n \times (0, T))$ that equals one in K and zero outside V. The function $v = u\zeta$ solves the heat equation with a source that only contains u, ∇u and the derivatives of ζ. Writing the Duhamel solution (1-21) for v and moving the derivative from u to the known functions (Appendix A.3), we arrive at

$$v(\mathbf{x}, t) = \int_0^t \int_{\mathbb{R}^n} u(\mathbf{y}, s)\, \mathcal{K}(\mathbf{x}, \mathbf{y}, t, s)\, d\mathbf{y}\, ds, \qquad \mathcal{K} = \Phi\left(\partial_s \zeta + \Delta\zeta\right) + 2\nabla\Phi \cdot \nabla\zeta, \tag{1-23}$$

where $\Phi = \Phi(\mathbf{x} - \mathbf{y}, t - s)$ and all derivatives act on $\mathbf{y}$. Here is the key point: the kernel $\mathcal{K}$ only contains derivatives of ζ, which vanish near K. Therefore, for $(\mathbf{x}, t)$ inside K the kernel never touches the singularity

of Φ and $\mathcal{K}$ is C^∞ in $(\mathbf{x}, t)$. Since $v = u$ in K, Equation (1-23) says that u is infinitely smooth there, whatever the roughness of u outside. The estimation theorem in the heat equation section of Evans [43] establishes, moreover, that its derivatives are bounded on parabolic cylinders $C(\mathbf{x}, t; r)$:

$$\max_{C(\mathbf{x},t;r/2)} \left|D_{\mathbf{x}}^k D_t^l u\right| \leq \frac{C_{kl}}{r^{k+2l+n+2}} \|u\|_{L^1(C(\mathbf{x},t;r))}. \tag{1-24}$$

Equation (1-24) offers a clear perspective on the regularization of the heat equation. The structure here takes the form of

$$\begin{aligned} &\text{Maximum change in some region} \\ \leq\ &\text{Total heat of a bigger region} \times \text{Geometric constant;} \end{aligned}$$

the space--time derivatives of u on a parabolic cylinder are bounded by the local L^1 norm of u and a constant C_{kl}, meaning that the variation of u at an arbitrary point of $C(\mathbf{x}, t; r/2)$ is governed by the total heat of the region where it is contained. The bound depends on the geometric factor $r^{-(k+2l+n+2)}$; taking higher derivatives makes this factor increasingly larger, reflecting that measuring finer variations requires a larger bound when the scale r is fixed. This is how the equation handles any irregular behavior of the data g or h: after any positive time, it has been filtered out.

1.1.4 Wave Equation

The heat equation introduces a first-order time derivative alongside the Laplacian operator, giving rise to a local property: the update of the local disequilibrium measure, that is to say, a continuous adaptation to the surrounding points (diffusion). Increasing the order of the time derivative by one causes a wave propagation phenomenon:

$$\partial_t^2 u(\mathbf{x}, t) - \Delta u(\mathbf{x}, t) = 0 \qquad \text{in } \mathbf{x} \in \mathbb{R}^n,\ t \in [0, T]. \tag{1-25}$$

Homogeneous case

Let $u \in C^2(\mathbb{R}^n \times [0, \infty))$ be a function that solves the wave equation for $\mathbf{x} \in \mathbb{R}^n$ and $t > 0$. Given the initial data g and h, the homogeneous problem is defined as

$$\begin{cases} \partial_t^2 u(\mathbf{x}, t) - \Delta u(\mathbf{x}, t) = 0 & \text{in } \mathbf{x} \in \mathbb{R}^n,\ t \in (0, T], \\ u(\mathbf{x}, 0) = g(\mathbf{x}) & \text{on } \mathbf{x} \in \mathbb{R}^n,\ \{t = 0\}, \\ \partial_t u(\mathbf{x}, 0) = h(\mathbf{x}) & \text{on } \mathbf{x} \in \mathbb{R}^n,\ \{t = 0\}. \end{cases} \tag{1-26}$$

With the aim of solving this, Evans [43] shows that the closed forms depend on the dimension.

Case $n = 1$: d'Alembert's formula. D'Alembert's formula takes advantage of the factorization of the wave operator into two transport operators, $\Box = \partial_t^2 - \partial_x^2 = (\partial_t + \partial_x)(\partial_t - \partial_x)$. Solving one homogeneous and one non-homogeneous transport problem with Equation (1-3) (Appendix A.4), we obtain

$$u(x, t) = \frac{1}{2}\big[g(x + t) + g(x - t)\big] + \frac{1}{2}\int_{x-t}^{x+t} h(y)\, dy. \tag{1-27}$$

This is d'Alembert's formula.

Case $n \geq 2$: spherical averages. For finding a solution in dimensions $n \geq 2$, it is necessary to return to the concept of average introduced in Laplace's equation, to analyze what happens with the surrounding points of a particular point. Evans [43] works with the mean value of u over the sphere $\partial B(\mathbf{x}, r)$, which

satisfies the Euler--Poisson--Darboux equation (Appendix A.4), and distinguishes the solution between odd and even dimensions.

Solution for odd n. Suppose $u \in C^2(\mathbb{R}^n \times [0,\infty))$ is a solution to the Cauchy problem for the wave equation for $\mathbf{x} \in \mathbb{R}^n$ and $t > 0$, where $n \geq 3$ is odd. Also suppose that $g \in C^{m+1}(\mathbb{R}^n)$ and $h \in C^m(\mathbb{R}^n)$, where $m = (n+1)/2$. The closed-form analytical expression that solves this case is given by [43]

$$u(\mathbf{x},t) = \frac{1}{\gamma_n}\left[\partial_t\Big(\frac{1}{t}\partial_t\Big)^{\frac{n-3}{2}}\Big(t^{n-2}\fint_{\partial B(\mathbf{x},t)} g\,dS\Big) + \Big(\frac{1}{t}\partial_t\Big)^{\frac{n-3}{2}}\Big(t^{n-2}\fint_{\partial B(\mathbf{x},t)} h\,dS\Big)\right], \quad (1\text{-}28)$$

where $\gamma_n = 1\cdot 3\cdot 5\cdots(n-2)$. It only requires knowing the functions g and h over the sphere $\partial B(\mathbf{x},t) = \{\mathbf{y} \in \mathbb{R}^n \mid \|\mathbf{y}-\mathbf{x}\|_2 = t\}$, which is the base of the backward light cone with vertex $(\mathbf{x},t)$. This represents a light front because the Minkowski space--time interval is $s^2 = (ct)^2 - \|\mathbf{y}-\mathbf{x}\|_2^2$. For a light-like path the interval of wave propagation is $s^2 = 0$, assuming the wave speed is $c = 1$. This means that the distance between two points of the spatial domain is equal to the time. Geometrically, this boundary is a sphere in the spatial domain, centered at $\mathbf{x}$ with radius t. The fact that the solution depends only on this surface, and not on the interior of the ball, is the mathematical statement of Huygens' principle: information propagates exactly at the speed c without leaving a residual trail. Whatever happens inside the ball does not affect $u(\mathbf{x},t)$.

Solution for even n. Suppose $u \in C^2(\mathbb{R}^n \times [0,\infty))$ is a solution to the Cauchy problem for the wave equation for $\mathbf{x} \in \mathbb{R}^n$ and $t > 0$, where n is even. Also suppose that $g \in C^{m+1}(\mathbb{R}^n)$ and $h \in C^m(\mathbb{R}^n)$, where $m = (n+2)/2$. The closed-form analytical expression that solves this case is given by the method of descent [43]:

$$u(\mathbf{x},t) = \frac{1}{\gamma_n}\left[\partial_t\Big(\frac{1}{t}\partial_t\Big)^{\frac{n-2}{2}}\Big(t^n\fint_{B(\mathbf{x},t)}\frac{g(\mathbf{y})}{(t^2-\|\mathbf{y}-\mathbf{x}\|_2^2)^{1/2}}\,d\mathbf{y}\Big) + \Big(\frac{1}{t}\partial_t\Big)^{\frac{n-2}{2}}\Big(t^n\fint_{B(\mathbf{x},t)}\frac{h(\mathbf{y})}{(t^2-\|\mathbf{y}-\mathbf{x}\|_2^2)^{1/2}}\,d\mathbf{y}\Big)\right], \quad (1\text{-}29)$$

where $\gamma_n = 2\cdot 4\cdots(n-2)\cdot n$. It requires knowing the functions g and h over the entire volume of the solid ball $B(\mathbf{x},t)$; here, what happens inside the solid ball does affect $u(\mathbf{x},t)$. This is reflected in the non-zero weight given by $(t^2-\|\mathbf{y}-\mathbf{x}\|_2^2)^{-1/2}$ inside the ball. Thus, $s^2 = (ct)^2 - \|\mathbf{y}-\mathbf{x}\|_2^2 > 0$, meaning that the elapsed time is greater than the spatial distance. This implies that the wave leaves a wake and interacts with points inside the solid ball.

Non-homogeneous case

By the linearity of the wave operator $\Box = \partial_t^2 - \Delta$, the solution to the non-homogeneous Cauchy problem can be decomposed by superposition as $u = u_{\text{hom}} + u_{\text{part}}$, where u_{hom} solves the homogeneous problem determined by the initial data g and h, and u_{part} solves the problem with source f and zero initial conditions. To construct u_{part}, we apply Duhamel's principle. Rather than treating the source f as a global function in time, we interpret it as a continuous superposition of instantaneous impulses. For each instant $s \in [0,t]$, consider the following auxiliary initial value problem:

$$\begin{cases} \partial_t^2 u(\mathbf{x},t;s) - \Delta u(\mathbf{x},t;s) = 0 & \text{in } \mathbf{x} \in \mathbb{R}^n,\ t \in (s,\infty), \\ u(\mathbf{x},s;s) = 0 & \text{on } \mathbf{x} \in \mathbb{R}^n,\ \{t = s\}, \\ \partial_t u(\mathbf{x},s;s) = f(\mathbf{x},s) & \text{on } \mathbf{x} \in \mathbb{R}^n,\ \{t = s\}. \end{cases} \quad (1\text{-}30)$$

By Duhamel's principle, the solution to the non-homogeneous problem is obtained by integrating all these auxiliary solutions over all past times s. The micro-source $f(\mathbf{x}, s)$ can be seen as a local property: a micro-perturbation occurring at time s gives a pulse to the system that is converted into velocity, while the position at that instant is not changed (the initial position of each auxiliary problem is zero). The solution is expressed as

$$u_{\text{part}}(\mathbf{x}, t) = \int_0^t u(\mathbf{x}, t; s)\, ds. \tag{1-31}$$

About the problem of solutions with closed forms

Commenting on closed forms, Evans [43, p. 82] argues: ''the explicit formulas demonstrate the necessity of making more and more smoothness assumptions upon the data g and h to ensure the existence of a C^2 solution of the wave equation for larger and larger n''. The wave operator, unlike the Laplace and heat operators, does not produce a smoothing estimate like the ones in Equations (1-14) and (1-24), the regularity of $u(\mathbf{x}, t)$ is exactly the regularity of the initial data, and singularities travel along the light cone. Thus, the modern theory of PDEs is needed to handle problems without so many assumptions about the regularity of the initial data g and h, and of course about u.

1.2 Modern Theory of Partial Differential Equations

The classical theory of PDEs faces limitations when handling real problems. As discussed in the previous sections, the existence of classical C^2 solutions imposes strict regularity requirements on the initial data and boundary data. For the wave equation, highly smooth data $g \in C^{m+1}$ and $h \in C^m$ are required. For Laplace's equation, the domain boundaries must be smooth enough for the Green's function to exist, and corners produce singularities. Even for the heat equation, discontinuous initial data or sources prevent the existence of a classical solution at $t = 0$. These strict requirements fail to represent most cases that science or engineering must address, where data often present discontinuities, geometric corners, or lack high regularity. It is necessary to formulate a more flexible framework. To respond to that, modern PDE theory requires tools that can handle them. In this section, we address this through Hölder spaces, Lebesgue spaces, weak derivatives, Sobolev spaces and weak solutions.

1.2.1 Hölder Spaces

Some functions are perfect candidates for the classical framework, but irregular functions are better suited to represent the real world. How can one measure the derivative of a function that is not differentiable? To address this question, the Hölder norm avoids differentiation and measures, instead, how fast the function can oscillate [43]:

Definición 1.2.1: Hölder norm.

Let $\Omega \subset \mathbb{R}^n$ be an open set and $u : \Omega \to \mathbb{R}$ a bounded and continuous function. The supremum norm is defined as

$$\|u\|_{C^0(\bar{\Omega})} := \sup_{\mathbf{x} \in \Omega} |u(\mathbf{x})| .$$

The Hölder seminorm of u with exponent $0 < \gamma \leq 1$ is given by

$$[u]_{C^{0,\gamma}(\bar{\Omega})} := \sup_{\substack{\mathbf{x},\mathbf{y}\in\Omega \\ \mathbf{x}\neq\mathbf{y}}} \frac{|u(\mathbf{x}) - u(\mathbf{y})|}{|\mathbf{x} - \mathbf{y}|^{\gamma}},$$

and the Hölder norm is then defined as $\|u\|_{C^{0,\gamma}(\bar{\Omega})} := \|u\|_{C^0(\bar{\Omega})} + [u]_{C^{0,\gamma}(\bar{\Omega})}$.

Definición 1.2.2: Hölder space.

The Hölder space $C^{k,\gamma}(\bar{\Omega})$ consists of all functions $u \in C^k(\bar{\Omega})$ for which the norm

$$\|u\|_{C^{k,\gamma}(\bar{\Omega})} := \sum_{|\alpha|\leq k} \|D^{\alpha}u\|_{C^0(\bar{\Omega})} + \sum_{|\alpha|=k} [D^{\alpha}u]_{C^{0,\gamma}(\bar{\Omega})} \tag{1-32}$$

is finite.

The case $\gamma = 1$ is Lipschitz continuity, so Hölder continuity is a generalization of it in which the ''slope'' is allowed to blow up in a controlled way. Evans [43] establishes that every space $C^{k,\gamma}(\bar{\Omega})$ is a Banach space, that is, it is complete, which is what one needs for convergence and fixed-point arguments. However, this is not enough. Having a function that is smooth everywhere except at a few points relaxes the rigid regularity assumptions on the initial and boundary data, but it does not resolve the PDE problem, because the equation still asks for classical derivatives. A Hölder norm only gives knowledge of the local oscillations of the function and of its supremum value. We will meet the Hölder seminorm again in Chapter 2, where it is used to regularize the loss function of a PINN [37].

1.2.2 Lebesgue Spaces L^p

To rigorously define the modern formulation, we introduce the Lebesgue spaces, following Castillo [44].

Definición 1.2.3: L^p space.

Let $\Omega \subset \mathbb{R}^n$ be an open domain and $p \in [1, \infty)$. The space $L^p(\Omega)$ consists of all (classes of) measurable functions whose p-th power is Lebesgue integrable:

$$L^p(\Omega) := \Big\{ f : \Omega \to \mathbb{R} \;\Big|\; \int_{\Omega} |f(\mathbf{x})|^p \, d\mathbf{x} < \infty \Big\}. \tag{1-33}$$

In particular, the space $L^2(\Omega)$ is a Hilbert space equipped with the inner product $\langle f, g \rangle_{L^2} = \int_{\Omega} fg \, d\mathbf{x}$, meaning that it contains functions with finite energy that may present discontinuities. This is exactly the space in which the loss functions of Chapter 2 are measured.

1.2.3 Weak Derivatives

Consider the Poisson equation (1-11) with $h = 0$. From Evans [43], the solution uses the Green's function G and integrates it against the source f:

$$u(\mathbf{x}) = \int_{\Omega} G(\mathbf{x}, \mathbf{y}) f(\mathbf{y}) \, d\mathbf{y}.$$

However, this representation requires knowledge of the auxiliary function $\phi^{\mathbf{x}}$ (the corrector), which is not an easy task because geometries are complex and finding such functions is difficult. Even if a solution can be found, we must solve a homogeneous Dirichlet problem infinitely many times, one for each point $\mathbf{x}$, which requires a high computational cost. Moreover, since Φ is a singular function, the source f must be regular enough (for example, Hölder continuous) for u to be a classical C^2 solution.[VERIFICAR: precise hypothesis on f, e.g. Gilbarg–Trudinger]

Instead of looking for u through the kernel G, we test the equation against a smooth function ϕ that is compactly supported, $\phi \in C_c^\infty(\Omega)$ [45]:

$$\int_\Omega -\Delta u(\mathbf{y})\,\phi(\mathbf{y})\,d\mathbf{y} = \int_\Omega f(\mathbf{y})\phi(\mathbf{y})\,d\mathbf{y}. \tag{1-34}$$

For the identity $\phi\Delta u = \operatorname{div}(\phi\nabla u) - \nabla u\cdot\nabla\phi$:

$$\int_\Omega \nabla u\cdot\nabla\phi\,d\mathbf{y} - \int_\Omega \operatorname{div}(\phi\nabla u)\,d\mathbf{y} = \int_\Omega f\phi\,d\mathbf{y},$$
$$\int_\Omega \nabla u\cdot\nabla\phi\,d\mathbf{y} - \int_{\partial\Omega} \phi\frac{\partial u}{\partial \mathbf{n}}\,dS(\mathbf{y}) = \int_\Omega f\phi\,d\mathbf{y}.$$

Since $\phi = 0$ on the boundary,

$$\int_\Omega \nabla u\cdot\nabla\phi\,d\mathbf{y} = \int_\Omega f\phi\,d\mathbf{y}. \tag{1-35}$$

If we now use $u\Delta\phi = \operatorname{div}(u\nabla\phi) - \nabla u\cdot\nabla\phi$ and apply it,

$$\int_\Omega u\Delta\phi\,d\mathbf{y} = \int_{\partial\Omega} u\frac{\partial\phi}{\partial\mathbf{n}}\,dS(\mathbf{y}) - \int_\Omega \nabla u\cdot\nabla\phi\,d\mathbf{y} = -\int_\Omega \nabla u\cdot\nabla\phi\,d\mathbf{y},$$

and finally it comes to

$$\int_\Omega u\Delta\phi\,d\mathbf{y} = -\int_\Omega f\phi\,d\mathbf{y}. \tag{1-36}$$

In Equation (1-36) there is no derivative of u at all: all of them were transferred to the test function. And, applying the Cauchy--Schwarz inequality [44],

$$\Big|\int_\Omega f\phi\,d\mathbf{y}\Big| \le \Big(\int_\Omega f^2\,d\mathbf{y}\Big)^{1/2}\Big(\int_\Omega \phi^2\,d\mathbf{y}\Big)^{1/2} = \|f\|_{L^2(\Omega)}\,\|\phi\|_{L^2(\Omega)}\,, \tag{1-37}$$

so the right-hand side makes sense as soon as $f \in L^2(\Omega)$. Here we observe that this weak formulation relaxes the regularity requirements: instead of requiring a classical C^2 solution and a Hölder continuous source f, as demanded by the classical Green's function representation, the weak theory only requires $f \in L^2(\Omega)$ (or even $f \in H^{-1}(\Omega)$) and seeks the solution u in a Sobolev space [44]. Being able to represent the solution through a source f residing in a less regular space, while still guaranteeing the existence of a solution, is a major advance of the modern theory. What we did with Δ is a particular case of the following definition.

Definición 1.2.4: Weak derivative (derivative in the sense of distributions).

Given $u \in L^1_{\mathrm{loc}}(\Omega)$ and a multi-index α, we say that $v_\alpha \in L^1_{\mathrm{loc}}(\Omega)$ is the α-th weak derivative of u, and we write $D^\alpha u := v_\alpha$, if for every test function $\phi \in C_c^\infty(\Omega)$ it holds that [45]

$$\int_\Omega u\,D^\alpha\phi\,d\mathbf{y} = (-1)^{|\alpha|}\int_\Omega v_\alpha\,\phi\,d\mathbf{y}. \tag{1-38}$$

In our Poisson case we already have this result, but seen in another way. If we organize it a little, with $|\alpha| = 2$ the sign is $(-1)^2 = 1$ and

$$\int_\Omega u\Delta\phi\,d\mathbf{y} = (-1)^2\int_\Omega \Delta u\,\phi\,d\mathbf{y} = -\int_\Omega f\phi\,d\mathbf{y}, \tag{1-39}$$

which is Equation (1-36). The weak Laplacian of u is $-f$, even if u has no classical second derivatives.

1.2.4 Sobolev Spaces

While the Lebesgue spaces $L^p(\Omega)$ provide a framework for functions with finite energy, they do not account for the behavior of their derivatives. To rigorously measure the regularity of a function alongside its weak derivatives, we introduce the Sobolev spaces.

Definición 1.2.5: Sobolev spaces.

Let $\Omega \subset \mathbb{R}^n$ be an open domain, $k \in \mathbb{N}$ and $1 \leq p < \infty$. The Sobolev space $W^{k,p}(\Omega)$ consists of all functions $u \in L^p(\Omega)$ whose weak derivatives $D^\alpha u$ up to order k also belong to $L^p(\Omega)$. That is:

$$W^{k,p}(\Omega) := \big\{ u \in L^p(\Omega) \;\big|\; D^\alpha u \in L^p(\Omega) \;\; \forall\, |\alpha| \leq k \big\}. \tag{1-40}$$

In the specific case $p = 2$, we denote this space as $H^k(\Omega) := W^{k,2}(\Omega)$. The space $H^1(\Omega)$ is a Hilbert space equipped with the inner product

$$\langle u, v \rangle_{H^1(\Omega)} := \int_\Omega uv \, d\mathbf{y} + \int_\Omega \nabla u \cdot \nabla v \, d\mathbf{y}. \tag{1-41}$$

The space $H_0^1(\Omega)$ is defined as the closure of $C_c^\infty(\Omega)$ in $H^1(\Omega)$. It means that every function of $H_0^1(\Omega)$ can be approximated, in the energy norm, by smooth functions with compact support; this is how the space encodes the zero boundary condition (in the sense of traces) without asking the function to be smooth: it may have corners or kinks, as long as its energy does not blow up. Since the weak formulation Equation (1-35) involves the gradients ∇u and $\nabla \phi$, $H_0^1(\Omega)$ is the natural space for the solution of the Poisson problem with zero boundary data. On the other hand, the space $H^2(\Omega)$ consists of functions whose weak derivatives up to second order belong to $L^2(\Omega)$; this is the natural space for operators involving the Laplacian Δu or second temporal derivatives $\partial_{tt} u$ in strong form, as it guarantees that both the function and its curvature have finite energy. Finally, $H^{-1}(\Omega)$ denotes the topological dual space of $H_0^1(\Omega)$. While $H_0^1(\Omega)$ is the space of states with finite energy, $H^{-1}(\Omega)$ is the space of forces or sources.

This difference between H^1 (weak form) and H^2 (strong form) will be important later: a standard PINN minimizes the *strong* residual in L^2 (Chapter 2), which implicitly asks the solution to live in H^2, while the weak theory only needs H^1.

1.3 Second-Order Partial Differential Equations

The four prototypes of Section 1.1 are particular cases of three general families. In this section we write each family in divergence form, with variable coefficients, and we collect the properties that Evans [43] proves for their weak solutions: existence, regularity, maximum principle and spectral representation. Throughout, $\Omega \subset \mathbb{R}^n$ is a bounded open set with smooth boundary and $\Omega_T = \Omega \times (0, T]$.

1.3.1 Second-Order Elliptic Equations

The mathematical definition of ellipticity

Let us consider the second-order linear operator in divergence form

$$Lu := -\nabla \cdot \big(A(\mathbf{x}) \nabla u \big) + \mathbf{b}(\mathbf{x}) \cdot \nabla u + c(\mathbf{x}) u, \tag{1-42}$$

where the first term, with the symmetric matrix $A(\mathbf{x}) = [a_{ij}(\mathbf{x})]$ that measures the conductivity or tension of the medium at the point $\mathbf{x}$, is responsible for diffusion; the second term, with $\mathbf{b}(\mathbf{x})$, is responsible for the drift of the medium; and finally, the third term, with $c(\mathbf{x})$, acts as a local source or absorption. We say that L is *uniformly elliptic* if there exists a constant $\theta > 0$ such that

$$\sum_{i,j=1}^{n} a_{ij}(\mathbf{x})\xi_i\xi_j \geq \theta \left|\boldsymbol{\xi}\right|^2 \qquad \text{for almost every } \mathbf{x} \in \Omega \text{ and } \forall\, \boldsymbol{\xi} \in \mathbb{R}^n. \tag{1-43}$$

The ellipticity condition ensures that the medium exhibits a baseline stiffness θ in all directions. This means there are no preferred directions where the diffusion or tension drops to zero, which prevents the operator from degenerating. Laplace's and Poisson's equations correspond to $A = I$, $\mathbf{b} = 0$, $c = 0$.

Some properties of elliptic theory

Once the problem $Lu = f$, $u = 0$ on $\partial\Omega$, is tested against $v \in H_0^1(\Omega)$, we obtain the bilinear form

$$B[u, v] = \int_\Omega \Big((A(\mathbf{x})\nabla u) \cdot \nabla v + (\mathbf{b}(\mathbf{x}) \cdot \nabla u)v + c(\mathbf{x})uv \Big)\, d\mathbf{x}, \tag{1-44}$$

and $u \in H_0^1(\Omega)$ is a weak solution if $B[u, v] = (f, v)_{L^2}$ for all $v \in H_0^1(\Omega)$. Some properties are [43]:

1. **Existence and uniqueness (Lax--Milgram).** If the drift is zero ($\mathbf{b} \equiv 0$) and $c(\mathbf{x}) \geq 0$, ellipticity directly implies coercivity:

$$B[u, u] = \int_\Omega \left((A\nabla u) \cdot \nabla u + c\, u^2 \right) d\mathbf{x} \geq \int_\Omega (A\nabla u) \cdot \nabla u \, d\mathbf{x}$$

$$\geq \theta \int_\Omega \left|\nabla u\right|^2 \, d\mathbf{x} = \theta \left\|\nabla u\right\|^2_{L^2(\Omega)},$$

   and, by Poincaré's inequality, $\|\nabla u\|_{L^2}$ controls the whole H_0^1 norm. By the Lax--Milgram theorem, this guarantees that for any source $f \in L^2(\Omega)$ there exists a unique weak solution $u \in H_0^1(\Omega)$.

2. **Regularity.** A weak solution $u \in H_0^1(\Omega)$ starts with low regularity: only one weak derivative in L^2. The elliptic operator gains two derivatives with respect to the source. If the coefficients are smooth and $f \in L^2(\Omega)$, interior regularity guarantees $u \in H^2_{\text{loc}}(\Omega)$; that is, its second derivatives are controlled in the Sobolev norm. More generally, $f \in H^m$ gives $u \in H^{m+2}_{\text{loc}}$, and smooth data give smooth solutions.

3. **The weak maximum principle.** If $c \equiv 0$ and $Lu \leq 0$ in Ω, the solution cannot have a strict interior maximum: $\max_{\bar{\Omega}} u = \max_{\partial\Omega} u$ (for $c \geq 0$ the same holds for u^+). The maximum must be achieved at the boundary.

4. **Functional bases.** In elliptic theory, solutions can be built using functional bases. For the symmetric operator ($\mathbf{b} \equiv 0$), these basis functions ϕ_k arise by solving the eigenvalue problem associated with the operator,

$$L\phi_k = \lambda_k \phi_k, \qquad 0 < \lambda_1 \leq \lambda_2 \leq \cdots \to \infty, \tag{1-45}$$

   and $\{\phi_k\}$ is an orthonormal basis of $L^2(\Omega)$. Solving this problem is typically performed through the method of separation of variables [43] or via projection methods like Galerkin and finite elements in complex geometries [6]. Once the basis is known, the full solution is simply a linear combination of these functional bases, $u(\mathbf{x}) = \sum_k c_k \phi_k(\mathbf{x})$ with $c_k = (f, \phi_k)/\lambda_k$.

1.3.2 Second-Order Parabolic Equations

The mathematical definition of parabolicity

Let us now consider the second-order linear operator in divergence form, with coefficients that may depend on time,

$$Lu := -\nabla \cdot \big(A(\mathbf{x},t)\nabla u\big) + \mathbf{b}(\mathbf{x},t)\cdot\nabla u + c(\mathbf{x},t)u, \tag{1-46}$$

where the matrix $A(\mathbf{x},t) = [a_{ij}(\mathbf{x},t)]$ measures the diffusion at the point $\mathbf{x}$ and time t, the vector $\mathbf{b}(\mathbf{x},t)$ represents a drift or convection, and the function $c(\mathbf{x},t)$ acts as a local reaction or absorption term. We say that the evolution operator $\partial_t + L$ is *uniformly parabolic* if there exists a constant $\theta > 0$ such that

$$\sum_{i,j=1}^{n} a_{ij}(\mathbf{x},t)\xi_i\xi_j \geq \theta\,|\boldsymbol{\xi}|^2 \qquad \text{for almost every } (\mathbf{x},t)\in\Omega_T \text{ and } \forall\,\boldsymbol{\xi}\in\mathbb{R}^n. \tag{1-47}$$

Similarly to the elliptic case, this condition ensures that the medium exhibits a baseline diffusion θ in all spatial directions. The heat equation corresponds to $A = I$, $\mathbf{b} = 0$, $c = 0$.

Some properties of parabolic theory

Consider the problem $\partial_t u + Lu = f$ in Ω_T, $u = 0$ on $\partial\Omega\times[0,T]$, $u = g$ on $\Omega\times\{t=0\}$. We formulate it in the weak sense by looking for $u \in L^2(0,T;H_0^1(\Omega))$ with $\partial_t u \in L^2(0,T;H^{-1}(\Omega))$, tested against functions $v \in H_0^1(\Omega)$. For almost every time $t\in(0,T]$, the associated bilinear form is

$$B[u,v;t] = \int_\Omega \Big((A(\mathbf{x},t)\nabla u)\cdot\nabla v + (\mathbf{b}(\mathbf{x},t)\cdot\nabla u)v + c(\mathbf{x},t)uv\Big)\,d\mathbf{x}. \tag{1-48}$$

And some fundamental properties are [43]:

1. **Existence and uniqueness (Galerkin method).** Unlike the elliptic case, where we use Lax--Milgram directly, evolution equations require more construction. The basic idea is to approximate the solution through the basis functions $\{w_k\}_{k=1}^{\infty}$ of $H_0^1(\Omega)$, $u_m(t) = \sum_{k=1}^{m} d_k(t)w_k$, and substitute them into the weak form of the PDE to convert it into a system of first-order ordinary differential equations (ODEs) for the coefficients $d_k(t)$. Energy estimates, uniform in m, allow passing to the limit $m\to\infty$.
2. **Regularity.** Here the situation is even better than in the elliptic case. For smooth coefficients and any strictly positive time $t > 0$, the solution becomes infinitely differentiable in space, $u(\cdot,t)\in C^\infty(\Omega)$, even if the initial data are irregular. The spatial derivatives are regularized instantly as time moves forward; this is the general version of Equation (1-24).
3. **The weak maximum principle.** If $c\equiv 0$ and $\partial_t u + Lu \leq 0$ in Ω_T, the solution cannot have a strict local maximum in the interior of the space--time domain. The maximum must be on the parabolic boundary $\Gamma_T = (\bar\Omega\times\{t=0\})\cup(\partial\Omega\times[0,T])$. Physically, heat cannot spontaneously concentrate in the middle of the domain; it must originate from a boundary or from the initial data.
4. **Functional bases and spectral decomposition.** In parabolic theory, solutions are also built using functional bases. By separation of variables, we solve the eigenvalue problem Equation (1-45) for the spatial operator (time-independent and symmetric), and once the basis is found, the full solution is a linear combination where each mode decays exponentially in time:

$$u(\mathbf{x},t) = \sum_{k=1}^{\infty} c_k\, e^{-\lambda_k t}\phi_k(\mathbf{x}). \tag{1-49}$$

   This spectral decay is the mathematical reason why the higher frequencies (the roughness) disappear almost instantly: the larger λ_k, the faster the mode is killed.

1.3.3 Second-Order Hyperbolic Equations

The mathematical definition of hyperbolicity

Let us now study the second great class of evolution equations: the second-order hyperbolic model, whose prototypical example is the wave equation. Let L be the spatial second-order differential operator in divergence form,

$$Lu := -\nabla \cdot \big(A(\mathbf{x},t)\nabla u\big) + \mathbf{b}(\mathbf{x},t)\cdot\nabla u + c(\mathbf{x},t)u. \tag{1-50}$$

The evolution operator is defined as $\partial_t^2 + L$. We say that this operator is *uniformly hyperbolic* if the spatial operator L satisfies the uniform ellipticity condition, that is, there exists a constant $\theta > 0$ such that

$$\sum_{i,j=1}^{n} a_{ij}(\mathbf{x},t)\xi_i\xi_j \geq \theta\,|\boldsymbol{\xi}|^2 \qquad \text{for almost every } (\mathbf{x},t)\in\Omega_T \text{ and } \forall\,\boldsymbol{\xi}\in\mathbb{R}^n. \tag{1-51}$$

The wave equation corresponds to $A = I$, $\mathbf{b} = 0$, $c = 0$, and the Klein--Gordon equation of Section 1.5 to $A = I$, $\mathbf{b} = 0$, $c = m^2$. Unlike parabolic equations, which describe irreversible diffusion and dissipate energy, hyperbolic equations govern propagation. The total energy is conserved, and initial singularities are rigidly transported along characteristic curves at a finite speed.

Some properties of hyperbolic theory

Consider the space--time cylinder $\Omega_T = \Omega\times(0,T]$. The problem requires two initial conditions because of the second-order time derivative: an initial position $u(0) = g$ and an initial velocity $\partial_t u(0) = h$. The weak formulation seeks a function $u \in L^2(0,T;H_0^1(\Omega))$ with $\partial_t u \in L^2(0,T;L^2(\Omega))$ and $\partial_t^2 u \in L^2(0,T;H^{-1}(\Omega))$, such that for almost every $t\in(0,T]$

$$\langle \partial_t^2 u(t), v\rangle + B[u(t),v;t] = (f(t),v)_{L^2(\Omega)} \qquad \forall\, v\in H_0^1(\Omega), \tag{1-52}$$

where $B[u,v;t]$ is the spatial bilinear form Equation (1-48) and $\langle\cdot,\cdot\rangle$ is the pairing between H^{-1} and H_0^1. Some fundamental properties are [43]:

1. **Existence and uniqueness (Galerkin method).** Projecting the PDE onto a finite-dimensional subspace spanned by basis functions $\{w_k\}_{k=1}^m$, whose coefficients change with time, transforms the problem into a system of second-order ODEs in time. Thanks to the energy structure, the energy estimates are uniform in m. This bounds the sequence of Galerkin approximations, allowing us to take the limit $m\to\infty$ and recover the unique weak solution.

2. **Regularity.** This is where the hyperbolic model contrasts sharply with the parabolic one. The hyperbolic operator does not add any order of differentiability to the solution. If the initial position is $g\in H_0^1(\Omega)\setminus H^2(\Omega)$, meaning that g vanishes on $\partial\Omega$ and the energy of its first-order weak derivatives is finite but its second-order weak derivatives are not square integrable, the solution $u(t)$ remains in $H_0^1(\Omega)\setminus H^2(\Omega)$ for all $t>0$ (the equation is reversible in time, so it cannot gain regularity). Spatial irregularities do not dissolve; they remain tied to the energy norm and propagate through space indefinitely.

3. **Energy conservation and finite propagation speed.** For the homogeneous case $f\equiv 0$, with $\mathbf{b}\equiv 0$, $c\equiv 0$ and coefficients independent of t, the total energy is strictly conserved:

$$E(t) := \frac{1}{2}\int_\Omega \Big(\,|\partial_t u(\mathbf{x},t)|^2 + (A(\mathbf{x})\nabla u)\cdot\nabla u\Big)\,d\mathbf{x} = E(0) \qquad \forall\, t\in(0,T). \tag{1-53}$$

(With $c \geq 0$ constant, the term $c\,u^2$ is added inside the integral.) A localized version of this identity on cones shows that the speed of propagation is finite: the wave cannot travel faster than the coefficients of the medium allow.

4. **Functional bases and spectral decomposition.** Just like in the previous models, solutions are built using functional bases ϕ_k obtained from the spatial eigenvalue problem $L\phi_k = \lambda_k \phi_k$. The crucial difference is the time evolution of each mode. Instead of the exponential decay of the parabolic case, the hyperbolic modes oscillate harmonically over time:

$$u(\mathbf{x}, t) = \sum_{k=1}^{\infty} \Big(c_k \cos\big(\sqrt{\lambda_k}\, t\big) + d_k \sin\big(\sqrt{\lambda_k}\, t\big) \Big) \phi_k(\mathbf{x}). \tag{1-54}$$

This oscillatory behavior, combined with the preservation of high frequencies, is precisely what makes hyperbolic equations the ultimate stress test for physics-informed neural networks (PINNs), as standard architectures often suffer from spectral bias and struggle to capture traveling wavefronts [28, 46].

1.4 Regularity Across the Three Classes and its Consequences for Learning

While the classical representation formulas (Green, Poisson, Kirchhoff) provide exact solutions for smooth data, neural networks approximate solutions by minimizing a loss function evaluated in L^2 norms. This is why the modern Sobolev spaces introduced in this chapter are the natural framework for analyzing PINNs: they allow us to measure the approximation error of the network in terms of weak derivatives, bridging the gap between the classical theory of Evans [43] and the machine learning architectures discussed in the next chapter. Table **1-1** summarizes the three classes, and Table **1-2** summarizes the weak formulations of the four prototypes.

Tabla 1-1: Comparison of second-order linear PDEs.

Property	Elliptic	Parabolic	Hyperbolic
Operator	$Lu = f$	$\partial_t u + Lu = f$	$\partial_t^2 u + Lu = f$
Physics	Equilibrium	Diffusion	Propagation
Existence	Lax--Milgram	Galerkin (1st-order ODEs)	Galerkin (2nd-order ODEs)
Regularity	Gains 2 derivatives	Instant smoothing	No smoothing
Max. principle	Yes (on $\partial\Omega$)	Yes (on Γ_T)	No (energy is conserved)
Time evolution of modes	--	Decay $e^{-\lambda_k t}$	Oscillation $\cos(\sqrt{\lambda_k} t)$
Propagation speed	--	Infinite	Finite

Reading Table **1-1** from the point of view of a neural network, we can extract three intuitions that guide the rest of this thesis.

1. **Smoothing helps the learner.** In elliptic and parabolic problems the operator itself filters out high frequencies (gain of two derivatives, decay $e^{-\lambda_k t}$). A smooth approximator such as a neural network with tanh activations is well suited to represent such solutions. In hyperbolic problems, on the contrary, every frequency present in the data survives for all times (Equation (1-54)), so the network must represent high frequencies along the whole space--time domain. This is where the spectral bias of neural networks [32, 46] hurts the most.

2. **The maximum principle is a tool for error analysis.** The convergence analysis of Shin et al. [37] for PINNs relies on the regularity estimates and on the maximum principle of elliptic and parabolic

Tabla 1-2: Weak formulations via distributional derivatives, with $\phi \in C_c^\infty(\Omega)$ or $\phi \in C_c^\infty(V)$, $V = \Omega \times (0,\infty)$.

Equation	Classical form	Weak formulation
Transport	$\partial_t u + \mathbf{b}\cdot\nabla u = f$	$\int_V u\,(-\partial_t\phi - \mathbf{b}\cdot\nabla\phi)\,d\mathbf{x}\,dt = \int_V f\phi\,d\mathbf{x}\,dt$
Laplace	$-\Delta u = 0$	$-\int_\Omega u\Delta\phi\,d\mathbf{x} = 0$
Poisson	$-\Delta u = f$	$-\int_\Omega u\Delta\phi\,d\mathbf{x} = \int_\Omega f\phi\,d\mathbf{x}$
Heat	$\partial_t u - \Delta u = f$	$\int_V u\,(-\partial_t\phi - \Delta\phi)\,d\mathbf{x}\,dt = \int_V f\phi\,d\mathbf{x}\,dt$
Wave	$\partial_t^2 u - \Delta u = f$	$\int_V u\,(\partial_t^2\phi - \Delta\phi)\,d\mathbf{x}\,dt = \int_V f\phi\,d\mathbf{x}\,dt$

Note. The formulation applies the distributional definition $\int_\Omega uD^\alpha\phi\,d\mathbf{y} = (-1)^{|\alpha|}\int_\Omega (D^\alpha u)\phi\,d\mathbf{y}$. For second-order derivatives $|\alpha| = 2$, so the sign is positive; for first-order derivatives $|\alpha| = 1$, so the sign is negative. Initial data enter as additional boundary terms when ϕ does not vanish at $t = 0$.

equations. Hyperbolic equations do not have a maximum principle, so a different tool is needed: the energy identity (1-53). We use it in Chapter 2 to bound the error of a PINN for the Klein--Gordon equation.

3. **Strong residuals ask for more regularity than the weak theory.** A PINN minimizes the strong residual in L^2, which is meaningful when the solution has second derivatives in L^2 (H^2). For elliptic and parabolic problems this is guaranteed by the regularity theory, even with rough data; for hyperbolic problems it is only true if the data are already in H^2. This is one of the reasons why weak and variational PINNs [47, 48] have been developed for hyperbolic problems.

Our benchmark (Chapter 3) contains elliptic, parabolic and hyperbolic problems precisely to test these three intuitions; the next section explains why, among the hyperbolic ones, we chose the Klein--Gordon equation as the main case study.

1.5 The Klein–Gordon Equation as a Case Study

The physicists Oskar Klein and Walter Gordon proposed in 1926 the Klein--Gordon equation as the equation that models the dynamics of relativistic electrons; nevertheless, in later years it was understood that the Klein--Gordon equation (a dispersive wave equation) describes perturbations (waves) of the field of spin-0 bosonic particles [49]. In other words, Klein and Gordon discovered an equation that describes the behavior of relativistic bosonic (quantum) particles. Moreover, each component of the Dirac spinor satisfies the Klein--Gordon equation individually [50] (Appendix B.2), so the dynamics of relativistic wave packets can be studied validly with the scalar formalism. Dispersive hyperbolic PDEs such as the Klein--Gordon equation are widely used in cosmology and high-energy physics [51], in the study of solitons and quantum physics [52], in dispersive phenomena and fluid physics [53], as well as in condensed matter [54].

Mathematically, we are talking about a hyperbolic partial differential equation for a scalar field $\phi(\mathbf{x},t)$:

$$\underbrace{\frac{1}{c^2}\frac{\partial^2}{\partial t^2}\phi(\mathbf{x},t)}_{\text{temporal evolution}} \underbrace{-\nabla^2\phi(\mathbf{x},t)}_{\text{spatial coupling}} + \underbrace{\left(\frac{mc}{\hbar}\right)^2\phi(\mathbf{x},t)}_{\text{restoring force}} = 0, \tag{1-55}$$

where we find the second-order time derivative ∂_{tt}, which induces the change and the dynamics of the system; the Laplacian term ∇^2, which couples each point with its neighbors (in a hyperbolic equation it is not a

diffusion but the term that propagates the perturbation); and the mass term $\mu^2 = (mc/\hbar)^2$, which acts as a restoring force. In one spatial dimension, the Klein--Gordon equation can be interpreted as a string that, besides its tension, is attached to an elastic medium that pulls every point back to the equilibrium. In the language of Section 1.3.3, it is the uniformly hyperbolic operator with $A = I$, $\mathbf{b} = 0$ and $c = \mu^2 > 0$. From now on we use natural units, $c = \hbar = 1$, so that $\mu = m$.

Plane waves and dispersion relation

We want the solution to contain the frequencies $e^{\pm i\mathbf{k}\cdot\mathbf{x}}$. This solution can be found through the method of separation of variables proposed by Fourier, $\phi(\mathbf{x}, t) = \Psi(\mathbf{x})T(t)$. We find the temporal and spatial derivatives (in one dimension, for simplicity),

$$\frac{\partial^2\phi}{\partial t^2} = \Psi(x)T''(t), \qquad \frac{\partial^2\phi}{\partial x^2} = \Psi''(x)T(t), \tag{1-56}$$

and replace them in the differential equation:

$$\frac{1}{c^2}\Psi(x)T''(t) - \Psi''(x)T(t) + \mu^2\Psi(x)T(t) = 0. \tag{1-57}$$

Separating variables,

$$\frac{1}{c^2}\frac{T''(t)}{T(t)} + \mu^2 = \frac{\Psi''(x)}{\Psi(x)}.$$

For this identity to hold, the left and the right sides of the equality must be equal to the same constant, $-k^2$. In this way we obtain an ODE for each function separately:

$$T''(t) + \omega_k^2 T(t) = 0, \qquad \Psi''(x) + k^2\Psi(x) = 0, \tag{1-58}$$

where

$$\omega_k^2 = c^2(k^2 + \mu^2) \quad \big(= k^2 + m^2 \text{ in natural units}\big). \tag{1-59}$$

Both are ODEs of the simple harmonic oscillator; therefore their solutions are $\Psi(x) = Ae^{ikx} + Be^{-ikx}$ and $T(t) = Ce^{i\omega_k t} + De^{-i\omega_k t}$. Putting the solutions of each function together, we obtain plane-wave solutions of the Klein--Gordon field,

$$\phi_k(\mathbf{x}, t) = K_1 e^{i(\mathbf{k}\cdot\mathbf{x} - \omega_k t)} + K_2 e^{-i(\mathbf{k}\cdot\mathbf{x} - \omega_k t)} \quad (+ \text{ the counter-propagating pair}), \tag{1-60}$$

with $\mathbf{x} = (x, y, z)$ and $\mathbf{k} = (k_x, k_y, k_z)$. Equation (1-59) is the *dispersion relation*. For the wave equation ($\mu = 0$) it is linear, $\omega_k = c\,|k|$, and every frequency travels at the same speed. For the Klein--Gordon equation it is not linear: the group velocity

$$v_g(k) = \frac{d\omega_k}{dk} = \frac{c^2 k}{\omega_k} = \frac{c\,k}{\sqrt{k^2 + \mu^2}} < c \tag{1-61}$$

depends on k. High wavenumbers travel almost at the speed of light, while low wavenumbers stay behind: a localized wave packet is *dispersed*. This is exactly the behavior that we will see in the reference solution of Chapter 3: the fastest components draw the light cone $|x| = t$, and the slower ones fill its interior.

General solution

By linearity, the general solution is a superposition of plane waves with all wavenumbers. Let $\hat{\phi}_0(k)$ and $\hat{\phi}_1(k)$ be the Fourier transforms of the initial position $\phi(x, 0) = \phi_0(x)$ and of the initial velocity $\partial_t\phi(x, 0) = \phi_1(x)$. Then, for each k, the coefficient $\hat{\phi}(k, t)$ solves the harmonic oscillator Equation (1-58), and

$$\phi(x, t) = \frac{1}{2\pi}\int_{\mathbb{R}} \Big(\hat{\phi}_0(k)\cos(\omega_k t) + \hat{\phi}_1(k)\frac{\sin(\omega_k t)}{\omega_k}\Big)e^{ikx}\,dk. \tag{1-62}$$

The exact solution depends on the initial position and velocity and, in addition, on a temporal Fourier basis $\{\cos(\omega_k t), \sin(\omega_k t)\}$. When the initial velocity is zero, only the cosine term remains. Since ω_k depends on k, this is *not* the initial condition multiplied by a single $\cos(\omega t)$: each Fourier component of the initial pulse oscillates with its own frequency, which is the analytic form of dispersion. This makes us think that a sinusoidal activation (SIREN) or a Fourier embedding would be natural candidates to give representation power to the network (Chapter 2), and it also gives us an exact way of computing a reference solution: the pseudo-spectral propagator of Section 3.2 evaluates Equation (1-62) with the discrete Fourier transform.

Energy

Multiplying Equation (1-55) by $\partial_t \phi$ and integrating by parts (as in Appendix B.1), the Klein--Gordon field conserves the energy

$$E_\phi(t) = \frac{1}{2} \int_\Omega \left(|\partial_t \phi|^2 + |\nabla \phi|^2 + m^2 |\phi|^2 \right) d\mathbf{x}, \tag{1-63}$$

for periodic or homogeneous boundary conditions. Unlike the wave equation, the mass term makes this energy control the L^2 norm of ϕ itself, not only its derivatives.

Why the Klein–Gordon equation

With all the elements of this chapter, we can now state the reasons for our choice.

1. **It is hyperbolic: there is no smoothing.** Everything in the initial pulse is transported for all times along the light cone (Table **1-1**), so the difficulty for the network does not disappear with time as in a parabolic problem.

2. **It is dispersive: it is multi-frequency by construction.** Because of the nonlinear dispersion relation (1-59), the components of a pulse separate, and by choosing an initial condition with several wavenumbers $k_1 < k_2 < \ldots < k_n$ we can control how many frequencies the network has to learn simultaneously. This turns spectral bias into a controllable experimental variable.

3. **It has an exact reference solution.** For periodic boundary conditions the propagator is diagonal in Fourier space and exact in time, so the reference solution has no time-discretization error (Section 3.2). The errors we measure belong to the network, not to the numerical method used for comparison.

4. **It has an energy identity.** Since there is no maximum principle, the energy Equation (1-63) is the natural tool to bound the error of a PINN (Section 2.3.6), and the mass term gives control of the L^2 error itself.

5. **It appears inside standard benchmarks.** In the PINNacle benchmark [55], the reference solution of the task *Wave2d-MS*, $u = \sin(\pi x)\sinh(\pi y)\cos(\pi t) + \sinh(3\pi x)\sin(2\pi y)\cos(\pi t)$, does not satisfy the stated wave equation $u_{tt} - (u_{xx} + 2u_{yy}) = 0$. A direct computation shows that both terms satisfy $u_{tt} - (u_{xx} + 2u_{yy}) = -2\pi^2 u$, that is, the Klein--Gordon equation with mass term $2\pi^2 u$ (Appendix D). The Klein--Gordon structure is therefore not a toy problem: it is hidden in the problems used to rank PINN architectures.

6. **It connects with the quantum part of this thesis.** As the equation of a relativistic spin-0 field, it is a natural bridge with the quantum formulation of Chapter 4.

1.6 Chapter Summary

In this chapter we followed the regularity of the solution through the four prototypes of linear PDEs. The transport and wave equations keep the roughness of the data; Laplace's and the heat equations remove it, through the mean-value property and the Gaussian kernel, respectively. The modern theory (Hölder, Lebesgue and Sobolev spaces, weak derivatives) allows us to work with non-smooth data, and the classification into elliptic, parabolic and hyperbolic families summarizes the behavior of each class (Table **1-1**). Hyperbolic equations do not smooth, do not have a maximum principle and conserve all their frequencies; among them, the Klein--Gordon equation adds dispersion, an exact spectral reference solution and an energy identity, which is why it is the case study of this thesis.

2 Physics-Informed Neural Networks - PINN

Solving partial differential equations has traditionally relied on classical numerical methods, most prominently the finite difference method (FDM), the finite element method (FEM), and the finite volume method (FVM). Because analytical solutions are not available for most real-world problems, these approaches discretize the spatial and temporal domains into meshes or grids, replacing continuous derivatives with algebraic operators and often requiring simplified approximations. Although these methods allow one to estimate the prediction accuracy, or to improve convergence by increasing the mesh density, the number of basis functions, or the simulation time, they are typically computationally expensive and present technical difficulties in their implementation and adaptation to a specific problem, requiring deep knowledge of the particular system to determine whether the technique is suitable.

This chapter presents physics-informed neural networks (PINNs) as an alternative. It is organized following the three sources of error of a PINN (approximation, generalization and optimization): Section 2.1 places PINNs among the numerical methods; Section 2.2 gives the formal definition; Section 2.3 collects the modern theory that bounds each source of error, including an energy estimate for the Klein--Gordon equation; Sections 2.4 and 2.5 analyze the optimization problem and its pathologies; and Sections 2.6, 2.7, 2.8 and 2.9 present the training strategies and architectures that are compared in Chapter 3, ending with our proposal, MABUR-PINN.

2.1 From Classical Numerical Methods to Neural Networks

Table **2-1** summarizes the classical methods. All of them are well understood and, for problems of moderate size and simple geometry, they remain the reference. However, classical mesh-based methods face severe limitations when addressing real-world problems:

1. **Curse of dimensionality.** The high dimensionality of the domain implies an exponential growth of the computational and memory cost.

2. **Complex mesh generation.** For geometrically complex, irregular or time-evolving domains, building and refining the computational mesh is technically difficult and costly.

3. **Data.** Measurements are not incorporated in a natural way into classical numerical methods.

4. **High data cost of pure networks.** On the other side, purely data-driven neural networks require large amounts of labeled data.

This rigidity contrasts with the flexibility of emerging deep learning paradigms, such as physics-informed neural networks (PINNs), which can combine data with physical knowledge expressed as PDEs and are conceptually general and independent of the complexity of the underlying equations.

Tabla 2-1: Classical numerical methods for PDEs.

Method	Representative authors	Key idea
Finite differences (FDM)	Causon and Mingham [56]; Strikwerda [5]	Replaces derivatives by algebraic differences on a space--time grid.
Finite elements (FEM)	Reddy [57]; Zienkiewicz et al. [6]	Subdivides the domain and solves the weak form with piecewise polynomials.
Finite volumes (FVM)	Eymard et al. [58]; Moukalled et al. [7]	Conserves physical fluxes in discrete control volumes.
Spectral methods	Hesthaven et al. [59]; Shen et al. [8]	Projects the solution onto global bases: Chebyshev, Legendre or Fourier.

Mathematical foundations and first ideas. The use of neural networks as numerical methods rests on the universal approximation theorems. Cybenko [60] and Hornik et al. [61] proved that a multilayer perceptron (MLP) with a single hidden layer and sigmoidal activation functions can approximate any continuous function on a compact set, and Leshno et al. [62] extended the result to any non-polynomial, locally bounded activation. Dissanayake and Phan-Thien [63] proposed, very early, a neural network to approximate the solution of a PDE by minimizing its residual, and Raissi et al. [21, 64, 65] structured the modern framework of PINNs for forward and inverse problems. Since then, many variants have been proposed; Table **2-2** groups the main ones that are relevant for this thesis.

Tabla 2-2: Main advances and variants in the field of PINNs.

Category	Variant	Authors	Key idea
Loss weighting	LRA / GradNorm	Wang et al. [27]	Balancing based on the norm of the gradients.
Loss weighting	NTK weights	Wang et al. [28]	Calibrates weights via the neural tangent kernel.
Adaptive sampling	RAR	Lu et al. [66]; Wu et al. [67]	Adds points in zones of high residual.
Architecture	LAAF / GAAF	Jagtap et al. [68, 69]	Local and global adaptive activations.
Architecture	Random Fourier features	Tancik et al. [70]; Wang et al. [46]	Overcomes the spectral bias at high frequencies.
Architecture	RWF, modified MLP	Wang et al. [71, 72]	Improves representation in nonlinear problems.
Architecture	SIREN	Sitzmann et al. [73]	Periodic sine activations.
Variational	VPINNs / hp-VPINNs	Kharazmi et al. [47, 74]	Polynomial test functions inspired by hp-FEM.
Variational	wPINNs	De Ryck et al. [48]	Entropy solutions of hyperbolic conservation laws.
Operator learning	DeepONet / FNO	Lu et al. [75]; Li et al. [20]	Learns operators between function spaces.
Geometric optimization	Preconditioning	De Ryck et al. [76]	Conditioning of the differential operator.
RKHS kernels	ANaGRAM / AMStraMGRAM	Schwencke and Furtlehner [77] and Schwencke et al. [78]	The empirical natural gradient is related to the Green's function in an RKHS.

To address the limitations above, PINNs have emerged as a mesh-free, nonlinear parametric numerical method [21]. The fundamental premise is to use a nonlinear model, such as an MLP, to capture both the linear and the nonlinear dynamics. Instead of discretizing the domain into a mesh, PINNs substitute the exact solution $u(\mathbf{x}, t)$ with a parametric approximator $u_\theta(\mathbf{x}, t)$, where $\theta \in \Theta \subset \mathbb{R}^P$ represents the trainable parameters of the network. Consequently, the problem of finding the solution of the PDE is shifted into an optimization problem over the parameter space.

2.2 Formulation of Physics-Informed Neural Networks

Definición 2.2.1: PDE formulation.

Let $\Omega \subset \mathbb{R}^d$ be a bounded spatial domain with smooth boundary $\partial\Omega$, and let $T > 0$. Let $\mathcal{D}$ be a differential operator that contains the time derivative of order $\alpha \in \{1, 2\}$ and a spatial operator, and let $\mathcal{B}$ be a boundary operator. We seek a solution $u(\mathbf{x}, t)$ satisfying the initial--boundary value problem

$$\begin{cases} \mathcal{D}[u](\mathbf{x}, t) = f(\mathbf{x}, t) & \text{in } \Omega \times (0, T), \\ u(\mathbf{x}, 0) = g_0(\mathbf{x}) & \text{in } \Omega, \\ \partial_t u(\mathbf{x}, 0) = g_1(\mathbf{x}) & \text{in } \Omega \quad (\text{only if } \alpha = 2), \\ \mathcal{B}[u](\mathbf{x}, t) = h(\mathbf{x}, t) & \text{on } \partial\Omega \times (0, T). \end{cases}$$

For the second-order problems of Section 1.3.3, the natural (weak) setting is $u \in L^2(0, T; H_0^1(\Omega))$ with $\partial_t u \in L^2(0, T; L^2(\Omega))$, $f \in L^2(0, T; H^{-1}(\Omega))$, $g_0 \in H_0^1(\Omega)$, $g_1 \in L^2(\Omega)$ and $h \in L^2(0, T; H^{1/2}(\partial\Omega))$. For the Klein--Gordon equation, $\mathcal{D} = \partial_{tt} - \nabla^2 + m^2$ [79].

Definición 2.2.2: Approximation to the PDE formulation.

Let $u_\theta(\mathbf{x}, t)$ be a parameterized function approximator, where $\theta \in \mathbb{R}^P$ denotes the set of learnable parameters. The objective of u_θ is to approximate and solve the PDE formulation of Definition 2.2.

The existence theorems for elliptic, parabolic and hyperbolic PDEs of Section 1.3 establish that solutions can be approximated using basis functions through projections (Galerkin) or collocation methods. Definition 2.2 extends this classical framework by replacing the fixed set of basis functions with a parameterized global function approximator u_θ, whose parameters are learned directly from the governing equations.

Definición 2.2.3: Physics-informed neural network as approximator to the PDE formulation.

Let $\Omega \subset \mathbb{R}^d$ be a bounded spatial domain with boundary $\partial\Omega$, and let $[0, T]$ be the time interval. A physics-informed neural network (PINN) is a global, parameterized function approximator $u_\theta : \Omega \times [0, T] \to \mathbb{R}^m$, represented by a neural network, usually a multilayer perceptron (MLP) with an element-wise activation function $\sigma : \mathbb{R} \to \mathbb{R}$. The network is defined by the parameters $W^{[l]} \in \mathbb{R}^{n_l \times n_{l-1}}$ and $\mathbf{b}^{[l]} \in \mathbb{R}^{n_l}$, $l = 1, \ldots, L$, where L is the total number of layers. The forward propagation is given by [14]

$$\mathbf{a}^{[0]} = (\mathbf{x}, t) \in \mathbb{R}^{d+1}, \qquad \mathbf{z}^{[l]} = W^{[l]}\mathbf{a}^{[l-1]} + \mathbf{b}^{[l]}, \qquad \mathbf{a}^{[l]} = \sigma(\mathbf{z}^{[l]}),$$

with a linear last layer, and the output of the network is $u_\theta(\mathbf{x}, t) = \mathbf{a}^{[L]} \in \mathbb{R}^m$. The set of all learnable parameters is $\theta = \{W^{[l]}, \mathbf{b}^{[l]}\}_{l=1}^L \in \mathbb{R}^P$, where $P = \sum_{l=1}^L (n_l n_{l-1} + n_l)$ with $n_0 = d + 1$. It is found by solving a minimization problem for the PINN loss function, which is built from the residuals of the PDE and of the initial and boundary data:

$$\mathcal{L}(\theta; \mathcal{T}) = \mathcal{L}_{\text{data}}(\theta) + \mathcal{L}_f(\theta) = \mathcal{L}_{bc}(\theta; \mathcal{T}_{bc}) + \mathcal{L}_{ic}(\theta; \mathcal{T}_{ic}) + \mathcal{L}_f(\theta; \mathcal{T}_f),$$

where the individual loss terms $\mathcal{L}_{bc}$, $\mathcal{L}_{ic}$, $\mathcal{L}_f$ correspond to the boundary condition, the initial condition and the PDE, respectively, and $\mathcal{T}_f$, $\mathcal{T}_{bc}$, $\mathcal{T}_{ic}$ are the corresponding sets of collocation points. The term $\mathcal{L}_{ic}$ can be divided into the initial position $\mathcal{L}_{icp}$ and the initial velocity $\mathcal{L}_{icv}$.

The cost functions are defined through their residuals as follows:

$$\mathcal{L}_f(\theta;\mathcal{T}_f) = \frac{1}{|\mathcal{T}_f|}\sum_{(\mathbf{x},t)\in\mathcal{T}_f} |R_{\text{PDE}}(\mathbf{x},t;\theta)|^2, \tag{2-1}$$

$$\mathcal{L}_{icp}(\theta;\mathcal{T}_{ic}) = \frac{1}{|\mathcal{T}_{ic}|}\sum_{\mathbf{x}\in\mathcal{T}_{ic}} |R_{ic0}(\mathbf{x};\theta)|^2, \tag{2-2}$$

$$\mathcal{L}_{icv}(\theta;\mathcal{T}_{ic}) = \frac{1}{|\mathcal{T}_{ic}|}\sum_{\mathbf{x}\in\mathcal{T}_{ic}} |R_{ic1}(\mathbf{x};\theta)|^2, \tag{2-3}$$

$$\mathcal{L}_{bc}(\theta;\mathcal{T}_{bc}) = \frac{1}{|\mathcal{T}_{bc}|}\sum_{(\mathbf{x},t)\in\mathcal{T}_{bc}} |R_{bc}(\mathbf{x},t;\theta)|^2, \tag{2-4}$$

and the residuals are defined as

$$R_{\text{PDE}}(\mathbf{x},t;\theta) = \mathcal{D}[u_\theta](\mathbf{x},t) - f(\mathbf{x},t), \tag{2-5}$$

$$R_{bc}(\mathbf{x},t;\theta) = \mathcal{B}[u_\theta](\mathbf{x},t) - h(\mathbf{x},t), \tag{2-6}$$

$$R_{ic0}(\mathbf{x};\theta) = u_\theta(\mathbf{x},0) - g_0(\mathbf{x}), \tag{2-7}$$

$$R_{ic1}(\mathbf{x};\theta) = \partial_t u_\theta(\mathbf{x},0) - g_1(\mathbf{x}). \tag{2-8}$$

Here $\mathcal{D}$ is the differential operator governing the PDE, $\mathcal{B}$ defines the boundary conditions $\mathcal{B}[u](\mathbf{x},t) = h(\mathbf{x},t)$, and $u(\mathbf{x},0) = g_0(\mathbf{x})$ and $\partial_t u(\mathbf{x},0) = g_1(\mathbf{x})$ are the initial conditions for position and velocity. The derivatives of u_θ with respect to $\mathbf{x}$ and t are computed exactly by automatic differentiation. Each sum is a Monte Carlo (or quadrature) approximation of the corresponding L^2 norm, for example

$$\mathcal{L}_f \approx |\Omega\times(0,T)|^{-1}\,\|R_{\text{PDE}}\|^2_{L^2(\Omega\times(0,T))}.$$

When experimental data are available, a data term of the same form is added to the cost function. The optimal parameters are found by solving the minimization problem

$$\theta^*(\mathcal{T}) = \arg\min_{\theta\in\Theta} \mathcal{L}(\theta;\mathcal{T}). \tag{2-9}$$

Three errors. A good approximation is one in which the residual is as close to zero as possible, but a small loss is not the same as a small error. Following De Ryck and Mishra [38], we distinguish:

- the *total error* $\mathcal{E}(\theta) = \|u - u_\theta\|_X$, typically with $X = L^2(\Omega)$;
- the *generalization error* $\mathcal{E}_G(\theta) = \|R_{\text{PDE}}\|_{L^2} + \dots$, the continuous physics-informed residual (all the terms of the loss measured in their continuous norms);
- the *training error* $\mathcal{E}_T(\theta;\mathcal{T}) = \sqrt{\mathcal{L}(\theta;\mathcal{T})}$, the discretized residual that the optimizer actually sees.

In the language of statistical learning (Figure **2-1**), the optimization error is the difference between the network that we have trained and the best network that the optimizer could find in the chosen class; the approximation error is the difference between the exact function (or the reference numerical method) and the best network of the class; and the estimation (generalization) error is the error associated with points of the domain that were not part of the training. When we speak of these three errors, we assume that there exists an ideal network, theoretically or empirically better trained, with respect to which our approximation is compared.

[VERIFICAR: permission / "adapted from" for the reproduced figure]

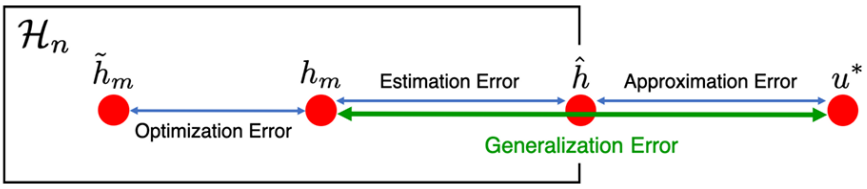


Figura 2-1: Diagram of the different errors that appear when training a PINN: optimization, estimation and approximation errors.
Fuente: Shin et al. [37].

2.3 Modern Theory of PINNs

This section collects the results that bound each of the three errors. We do not prove them; the goal is to understand what each one says about our experiments. The theorem numbers refer to the original works.[VERIFICAR: theorem/lemma numbers against the published versions]

2.3.1 Error Decomposition

Teorema 2.3.1: Error decomposition [38, 80, Lemma 3.1 and Eq. 3.11].

Suppose that the PDE is stable in the sense that $\|u - u_\theta\|_X \le C\,\mathcal{E}_G(\theta)^\alpha$ for some $C, \alpha > 0$, and let $\hat{\theta}$ be the parameter of the ideal model. Then the total error $\mathcal{E}^* = \|u - u_{\theta^*(\mathcal{T})}\|_X$ satisfies

$$\mathcal{E}^* \le C\Big(\underbrace{\mathcal{E}_G(\hat{\theta})}_{\text{approximation}} + \underbrace{2\sup_{\theta\in\Theta} |\mathcal{E}_T(\theta;\mathcal{T}) - \mathcal{E}_G(\theta)|}_{\text{generalization gap}} + \underbrace{\mathcal{E}_T(\theta^*(\mathcal{T});\mathcal{T}) - \mathcal{E}_T(\hat{\theta};\mathcal{T})}_{\text{optimization}} \Big)^\alpha .$$

The stability hypothesis is exactly where the PDE theory of Chapter 1 enters: for elliptic and parabolic problems it follows from coercivity and the maximum principle; for hyperbolic problems it follows from energy estimates (Section 2.3.6).

2.3.2 Approximation Error

Teorema 2.3.2: Approximation with tanh networks [81, Theorem 4.4].

Let $\Omega = [0,1]^d$. For every $N \in \mathbb{N}$ and every $f \in W^{s,\infty}(\Omega)$ there exists a network with tanh activation, two hidden layers and width N, $\hat{u}$, such that for all $0 \le k < s$

$$\|f - \hat{u}\|_{W^{k,\infty}(\Omega)} \le C\big(\ln(cN)\big)^k N^{-(s-k)/d}.$$

Consequence. The network approximates not only f but also its derivatives up to order k, which are the ones demanded by the operator $\mathcal{D}$. The rate, however, deteriorates with the dimension d.

Teorema 2.3.3: Sobolev--Barron approximation [38, Theorem 4.33].

Let σ be smooth with $C_1 := \max_{m\le\ell} \left\|\sigma^{(m)}\right\|_{L^\infty(\mathbb{R})} < \infty$. If f belongs to the spectral Barron space $\mathcal{B}_1^R(\Omega)$ (index 1, radius $R \ge 1$), then for every width $k \in \mathbb{N}$ there exists a shallow network $u_k(\mathbf{x}) = \frac{1}{k}\sum_{i=1}^k a_i \sigma(\mathbf{w}_i^\top \mathbf{x} + b_i)$ with $(a_i, \mathbf{w}_i, b_i) \in \mathbb{R} \times B_R^d \times \mathbb{R}$ such that

$$\|u_k - f\|_{H_\mu^\ell(\Omega)} \le \frac{2C_1\big(R\sqrt{ed}\big)^\ell \, \|f\|_{\mathcal{B}_1^R(\Omega)}}{\sqrt{k}}.$$

The rate $k^{-1/2}$ is independent of d: in the Sobolev--Barron regime the network overcomes the curse of dimensionality. Notice that the activation function only enters through the constant C_1, the maximum of its derivatives up to the order ℓ of the PDE. We will use this observation to analyze our adaptive activations in Section 2.9.

2.3.3 Generalization Gap

Teorema 2.3.4: Generalization gap [82, 83, Theorem 6.2].

Let $\mathcal{T}$ consist of N points drawn i.i.d. according to a measure μ, let $\Theta = [-R, R]^n$ be compact, and let $\mathfrak{L}$ be the Lipschitz constant of the map $\theta \mapsto \mathcal{E}_G(\theta)^2$. Then

$$\mathbb{E}\big[\mathcal{E}_G(\theta^*(\mathcal{T}))^2\big] \le \mathbb{E}\big[\mathcal{E}_T(\theta^*(\mathcal{T}); \mathcal{T})^2\big] + \sqrt{\frac{2c^2(n+1)}{N} \ln\Big(\frac{R\,\mathfrak{L}\sqrt{N}}{c}\Big)}.$$

The average gap between the training and the generalization errors is of order

$$\mathcal{O}\Big(\frac{K_1}{\sqrt{N}} \ln\big(K_2\sqrt{N}\big)\Big).$$

For our experiments this means that the number of collocation points must grow with the number of parameters n; this is one of the reasons for the stochastic resampling of Section 2.6, which shows the network new points at every epoch.

2.3.4 Optimization: Conditioning and Spectral Bias

Teorema 2.3.5: Convergence in the NTK regime [76, Theorem 7.9].

In the neural tangent kernel (NTK) regime (wide, over-parameterized networks), the number of gradient descent steps needed to reduce the error to ϵ is

$$N(\epsilon) = \mathcal{O}\big(\kappa(A)\ln(1/\epsilon)\big), \qquad \kappa(A) = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)},$$

with $A_{ij} = \langle \mathcal{D}\varphi_i, \mathcal{D}\varphi_j\rangle_{L^2(\Omega)} + \lambda\langle \varphi_i, \varphi_j\rangle_{L^2(\partial\Omega)}$ and $\varphi_i(\mathbf{x}) = \nabla_{\theta_i} u_\theta(\mathbf{x}; \theta_0)$.

Teorema 2.3.6: Spectral bias and conditioning [76, Theorem 7.10].

$\kappa(A)$ is governed by the condition number of the operator $\mathcal{A} \circ TT^*$, where $\mathcal{A} = \mathcal{D}^*\mathcal{D}$ and TT^* is the integral operator whose kernel is associated with the architecture.

Pathology of conditioning. Since $\mathcal{D}^*\mathcal{D}$ contains derivatives of twice the order of the PDE, high frequencies make $\kappa(A)$ blow up rapidly: a Fourier mode e^{ikx} is amplified by a factor of order k^4 for a second-order operator. For the Klein--Gordon operator, $\mathcal{D}^*\mathcal{D}$ multiplies the mode $e^{i(kx-\omega t)}$ by $(\omega^2 - k^2 - m^2)^2$, which is why initial conditions with several wavenumbers are a hard test.

Spectral convergence rate. In the theory of Jacot et al. [42] and its extension to PINNs by Wang et al. [28], it is shown that, under an infinitesimal learning rate, the evolution in training time τ of the vector of predictions $\mathbf{f}_\theta(\tau)$ with respect to the targets $\mathbf{Y}$ satisfies the linear ODE

$$\frac{d\mathbf{f}_\theta(\tau)}{d\tau} \approx -\mathbf{K}\big(\mathbf{f}_\theta(\tau) - \mathbf{Y}\big), \tag{2-10}$$

where $\mathbf{K}$ is the NTK. Projecting the error onto the eigenvectors $\mathbf{q}_i$ of $\mathbf{K}$, with eigenvalues λ_i, we obtain the analytical solution

$$\mathbf{q}_i^T\big(\mathbf{f}_\theta(\tau) - \mathbf{Y}\big) = e^{-\lambda_i \tau}\, \mathbf{q}_i^T\big(\mathbf{f}_\theta(0) - \mathbf{Y}\big). \tag{2-11}$$

The components of the error associated with large eigenvalues (low frequencies, for standard MLPs) are learned first, while those associated with small eigenvalues (high frequencies) are learned exponentially slower. This is the spectral bias, and it is the counterpart, inside the optimizer, of the parabolic decay $e^{-\lambda_k t}$ of Equation (1-49): the network ‘‘diffuses’’ its error mode by mode.

2.3.5 Regularized Loss and the Hölder Seminorm

For linear second-order elliptic and parabolic PDEs, Shin et al. [37] proved that the PINN loss with m_r residual points and m_b boundary points satisfies

$$\mathrm{Loss}^{\mathrm{PINN}}(h;\lambda) \le C_m \cdot \mathrm{Loss}_m(h;\lambda,\hat{\lambda}_m^R) + C'\big(m_r^{-\alpha/d} + m_b^{-\alpha/(d-1)}\big), \tag{2-12}$$

where $\mathrm{Loss}_m^{\mathrm{PINN}}(h;\lambda) = \mathcal{L}(\theta)$, h denotes the functions involved (u_θ, $\partial_t u_\theta$, $\mathcal{D}[u_\theta]$), and $\lambda, \hat{\lambda}$ are the coefficients of the loss function.[VERIFICAR: exponents of m_r and m_b against the original] This tells us that if we reduce the loss function and the values $m_r^{-\alpha/d}$ and $m_b^{-\alpha/(d-1)}$, it is possible to find a good estimation. The bound applies to a loss that is regularized through the Hölder seminorm,

$$\mathrm{Loss}_m(h;\lambda,\hat{\lambda}_m^R) = \mathrm{Loss}_m^{\mathrm{PINN}}(h;\lambda) + \hat{\lambda}_f^R[\mathcal{R}_f[h]]_{\alpha;U}^2 + \hat{\lambda}_{bc}^R[\mathcal{R}_{bc}[h]]_{\alpha;U}^2 + \hat{\lambda}_{ic}^R[\mathcal{R}_{ic}[h]]_{\alpha;U}^2, \tag{2-13}$$

where $\mathcal{R}_i$ are the residuals of Equations (2-5)--(2-8) and the Hölder seminorm (Definition 1.2.1) is

$$[\mathcal{R}_i[h]]_{\alpha;U} = \sup_{\substack{\mathbf{x},\mathbf{y}\in U \\ \mathbf{x}\neq\mathbf{y}}} \frac{\|\mathcal{R}_i[h](\mathbf{x}) - \mathcal{R}_i[h](\mathbf{y})\|}{\|\mathbf{x}-\mathbf{y}\|^\alpha}, \qquad 0 < \alpha \le 1. \tag{2-14}$$

It is natural to try to regularize the loss function through L^1 or L^2 penalties of the parameters; however, this does not make the network represent the behavior of the solution more faithfully. The Hölder regularization is different: it penalizes the oscillation of the residual, and it assumes that the unknown residuals are Hölder continuous, which is the kind of uniform equicontinuity needed to pass from the discrete loss to the continuous one. The proof of Shin et al. [37] uses the regularity estimates and the maximum principle of elliptic and parabolic equations. As we saw in Section 1.4, hyperbolic PDEs have particularities, such as not having a maximum principle, which does not allow us to find a convergence bound in the same way. A natural alternative is to work with the energy conservation that this type of equation satisfies.

2.3.6 Energy Estimate for the Klein–Gordon Equation

Since the Klein--Gordon equation is dispersive and conservative, the energy must be conserved (Equation (1-63)). This allows an energy analysis in Sobolev spaces and, therefore, an extension of Equation (2-13) with a term coming from the energy. The estimate for the error is computed through the error $e = u_\theta - u$, which, by linearity, satisfies the same operator with the residual as source:

$$\partial_{tt} e - \nabla^2 e + m^2 e = r \quad \text{in } \Omega \times (0,T), \qquad e(\cdot,0) = I_0, \quad \partial_t e(\cdot,0) = I_1, \quad e|_{\partial\Omega} = B, \tag{2-15}$$

where $r = R_{\text{PDE}}$, $I_0 = R_{ic0}$, $I_1 = R_{ic1}$ and $B = R_{bc}$ are the residuals of Equations (2-5)--(2-8). The energy of the error is

$$E(t) = \frac{1}{2}\int_\Omega \left(|\partial_t e|^2 + |\nabla e|^2 + m^2 |e|^2 \right) d\mathbf{x}. \tag{2-16}$$

Teorema 2.3.7: Energy estimate for the PINN error.

Let e be a sufficiently smooth solution of Equation (2-15). Then, for all $t \in [0,T]$,

$$E(t) \le \Big(E(0) + \frac{1}{2}\int_0^t \Big[\|r\|^2_{L^2(\Omega)} + \|\partial_t B\|^2_{L^2(\partial\Omega)} + \Big\| \frac{\partial e}{\partial \mathbf{n}} \Big\|^2_{L^2(\partial\Omega)} \Big] ds \Big) e^t, \tag{2-17}$$

with initial energy $E(0) = \frac{1}{2}\big(\|I_1\|^2_{L^2} + \|\nabla I_0\|^2_{L^2} + m^2 \|I_0\|^2_{L^2} \big)$.

The proof (Appendix B.1) uses integration by parts, Young's inequality and Gronwall's inequality. The bound says that the energy of the error is controlled by the initial energy of the error, plus the accumulated residual, plus the accumulated error flux through the boundary, times an amplification factor. Each of these terms can be approximated by the loss functions Equations (2-1)--(2-4), weighted by the coefficients of the learning rate annealing (Section 2.5.2), except two of them: the initial energy needs the *gradient* of the initial position residual, $\|\nabla I_0\|^2_{L^2}$, which a standard PINN does not penalize, and the boundary needs $\partial_t B$ and the normal flux of the error. Therefore, to control the energy of the error, the loss must include the term $\|\nabla I_0\|^2_{L^2}$ (and $\|\partial_t B\|^2$ for non-periodic boundaries). For periodic boundary conditions, as in our experiments, the boundary terms cancel and this part of the regularization is not necessary. In one spatial dimension, $E(t) \to 0$ implies convergence in H^1 and, by the Sobolev embedding $H^1 \subset C^{0,1/2}$, uniform convergence.

Remark (which residual gradients are needed)

One could think that the gradients of all residuals, $\{\|\nabla B\|, \|\nabla I_0\|, \|\nabla I_1\|, \|\nabla r\|\}$, must be added to the loss. The estimate (2-17) only requires ∇I_0 (and $\partial_t B$); the terms ∇I_1 and ∇r would be needed for a higher-order energy (controlling $\partial_t \nabla e$ and $\nabla^2 e$), and ∇r is costly because it needs third derivatives of u_θ, as in gradient-enhanced PINNs [84]. This is consistent with Experiment 2 (Section 3.5), where adding derivative regularization did not improve the error.

2.4 The Optimization Problem: Hard vs. Soft Constraints

The formulation of the PINN problem can be rigorously analyzed through the lens of constrained optimization. If we wish to minimize the data loss subject to the strict constraint that the physics loss is zero, we formulate a minimization with an equality constraint (hard constraint) [29]:

$$\min_\theta \ \mathcal{L}_{ic}(\theta) + \mathcal{L}_{bc}(\theta), \tag{2-18}$$

$$\text{s.t.} \ \ \mathcal{L}_f(\theta) = 0. \tag{2-19}$$

According to Krishnapriyan et al. [29], this problem is costly to solve. To see why, we introduce the Lagrangian function with the dual variable $v \in \mathbb{R}$:

$$\mathscr{L}(\theta, v) = \mathcal{L}_{\text{data}}(\theta) + v\,\mathcal{L}_f(\theta) = \big(\mathcal{L}_{ic}(\theta) + \mathcal{L}_{bc}(\theta)\big) + v\,\mathcal{L}_f(\theta). \tag{2-20}$$

The Karush--Kuhn--Tucker (KKT) conditions for this system are [85]

$$\nabla_\theta \mathcal{L}_{ic}(\theta^*) + \nabla_\theta \mathcal{L}_{bc}(\theta^*) + v\nabla_\theta \mathcal{L}_f(\theta^*) = 0, \tag{2-21}$$

$$\mathcal{L}_f(\theta^*) = 0. \tag{2-22}$$

Expanding these gradients (Appendix C.1), the gradients of the initial and boundary terms contain $2u_\theta \nabla_\theta u_\theta$ and $2g_0 \nabla_\theta u_\theta$, which are nonlinear in θ, because u_θ is nonlinear and so is its derivative. For the physics loss, the gradient is even more complex because of the differential operator: for the Klein--Gordon operator $\mathcal{D}_{KG} = \partial_{tt} - \nabla^2 + m^2$ it requires differentiating with respect to the parameters through the temporal and spatial derivatives of the network (∂_{tt}, ∇^2), which increases the computational complexity compared with a standard neural network. Written in matrix form, the KKT conditions are a system of high-dimensional nonlinear equations, where the dimensionality is induced by the neural network.

A degenerate constraint. There is a subtle problem with Equation (2-19). Since $\mathcal{L}_f \geq 0$, every feasible point ($\mathcal{L}_f(\theta^*) = 0$) is a global minimizer of $\mathcal{L}_f$, and therefore $\nabla_\theta \mathcal{L}_f(\theta^*) = 0$. The gradient of the only constraint vanishes at every feasible point, so the usual constraint qualification (linear independence of the constraint gradients) fails, and the KKT conditions are not guaranteed to characterize the optimum.[VERIFICAR: add a reference on constraint qualifications, e.g. Nocedal & Wright] This suggests that there should be one Lagrange multiplier per constraint, that is, one per collocation point (Appendix C.1); whether that system has a solution presupposes that the chosen architecture has enough features to give existence to that optimal point.

Non-convexity. To analyze the convexity of the loss functions, one should compute the Hessian of each loss function $\nabla^2_\theta \mathcal{L}_i(\theta)$ and its eigenvalues (Appendix C.2). The Hessian has a Gauss--Newton part, which is always positive semidefinite, and a second part that contains the residual times $\nabla^2_\theta u_\theta$, whose sign is not known. Already for a single layer the dependence on the parameters passes through an activation that is neither convex nor concave, so the cost function is non-convex, and non-convex optimization methods must be used.

Computational cost of Newton's method. Since $\mathcal{L}_{\text{data}}$ and $\mathcal{L}_f$ are not convex, and we do not know where the feasible region is, Boyd and Vandenberghe [85] suggest the infeasible-start Newton method, which applies the Newton step $\nabla^2 \mathscr{L}\,\delta = -\nabla \mathscr{L}$ to both variables (θ, v) with the linearized constraint (Appendix C.3). Applying this method consumes significant computational resources, because it must compute the Hessian of the approximating function u_θ for the losses of the initial and boundary data, and the Hessian of the physics loss, which in turn contains spatial and temporal derivatives of u_θ. To estimate the cost, recall that the universal approximation theorem says that a large enough network can approximate a function with any desired accuracy, but not how large [86], and that deep networks can represent certain families of functions with exponentially fewer units than shallow ones [87]; so the PINN should be deep as well as wide. For the architecture of Wang et al. [27], with 50 neurons per layer and 6 hidden layers, each layer has 50×50 weights plus 50 biases, that is, 2 550 parameters, and the network has about $6 \times 2\,550 = 15\,300$ parameters (ignoring the input and output layers). The Hessian has $P^2 \approx 2.34 \times 10^8$ entries, which in single precision is close to one gigabyte of memory, at every iteration. Moreover, Wang et al. [27] observed that the eigenvalues of the Hessian of the physics loss are very dispersed (very ill-conditioned), so Newton's method would need many iterations to satisfy the physics. As a result, the hard-constrained approach is intractable at scale.

2.4.1 Soft Constraints and Multi-Objective Optimization

Therefore, the hard constraint is relaxed into a soft constraint: a regularization constant is placed in the objective function, which transforms the constrained problem into an unconstrained multi-objective (MO) problem [29]. To understand what is lost, recall the vector optimization problem of Boyd and Vandenberghe [85]:

$$\text{minimize (w.r.t. } K) \;\; f_0(\mathbf{x}) \qquad \text{s.t. } f_i(\mathbf{x}) \leq 0, \;\; h_j(\mathbf{x}) = 0, \tag{2-23}$$

where $\mathbf{x} \in \mathbb{R}^n$ is the optimization variable, $f_0 : \mathbb{R}^n \to \mathbb{R}^q$ is the vector objective, $K \subseteq \mathbb{R}^q$ is a proper cone ($K = \mathbb{R}^q_+$ in our case), f_i are the inequality constraints and h_j the equality constraints. The set of achievable objective values is

$$\mathcal{O} = \{f_0(\mathbf{x}) \mid \exists\, \mathbf{x} \in \mathcal{D},\; f_i(\mathbf{x}) \leq 0,\; i = 1, \ldots, m;\; h_j(\mathbf{x}) = 0,\; j = 1, \ldots, p\}. \tag{2-24}$$

A feasible point $\mathbf{x}^*$ is optimal if its objective can be compared with that of every other feasible point and is better or equal, that is, $\mathcal{O} \subseteq f_0(\mathbf{x}^*) + K$. The set $f_0(\mathbf{x}^*) + K$ is the cone centered at $f_0(\mathbf{x}^*)$ that defines the order relation: it points to where the worse results are. If K points to the worse results, then $-K$ points to the better ones; that is, if we stand at $f_0(\mathbf{x}^*)$, look in the direction $-K$ and see no other achievable value of $\mathcal{O}$, then we are at a *Pareto optimal* value. The Pareto optimal values are minimal elements of $\mathcal{O}$, and they lie on its boundary,

$$\mathcal{P} \subseteq \mathcal{O} \cap \operatorname{bd} \mathcal{O}. \tag{2-25}$$

In this way, we introduce the minimization problem of Krishnapriyan et al. [29]:

$$\min_{\theta} \;\; \mathcal{L}_{ic}(\theta) + \mathcal{L}_{bc}(\theta) + \lambda_f \mathcal{L}_f(\theta). \tag{2-26}$$

If $\lambda_f = 1$, this is what is known as the *vanilla PINN*. Here, a weight vector $\lambda \succeq_{K^*} 0$ represents a supporting hyperplane of $\mathcal{O}$ at the optimum found, which is the point of contact. This presupposes a problem: if the set $\mathcal{O}$ is not convex, there are Pareto values in the regions that cannot be supported by a hyperplane, so the scalarization will not find all the Pareto points, but only some of them that also minimize the objective function. The formulation of the optimization problem can be generalized further [88]:

$$\min_{\theta} \sum_{i \in \{f, ic, bc\}} \lambda_i \mathcal{L}_i(\theta). \tag{2-27}$$

If $\lambda_f = 1$, this is the loss function proposed by Wang et al. [27], which weights the data losses with respect to the physics loss:

$$\min_{\theta} \;\; \mathcal{L}_f(\theta) + \sum_{i \in \{ic, bc\}} \lambda_i \mathcal{L}_i(\theta). \tag{2-28}$$

In this way, a balance between the gradients of the data and physics losses is sought, preventing one term from governing the other. Rohrhofer et al. [88] named the related phenomenon the *apparent Pareto front*: gradient methods (Newton, ADAM, SGD, etc.) reach a set of locally optimal trade-offs between the loss terms, which is the same gradient problem found by Wang et al. [27].

2.5 Gradient Pathologies and Optimizers

Training a PINN involves navigating a highly non-convex and stiff loss landscape. For Equation (2-28) the only KKT condition is the first-order condition for local optima,

$$\underbrace{\nabla_\theta \big(\lambda_{ic}\mathcal{L}_{ic}(\theta) + \lambda_{bc}\mathcal{L}_{bc}(\theta)\big)}_{\text{gradient of the data}} = \underbrace{-\nabla_\theta \mathcal{L}_f(\theta)}_{\text{gradient of the physics}}, \tag{2-29}$$

so at a stationary point the two gradients must be equal in magnitude and opposite in direction. If $\lambda_{ic} = \lambda_{bc} = 1$, we have the problem of gradient imbalance: the gradients coming from the PDE loss ($\nabla_\theta \mathcal{L}_f$) are usually larger and stiffer than those coming from the initial and boundary conditions ($\nabla_\theta \mathcal{L}_i$). Because ADAM attempts to minimize the global error, it dedicates most of its effort to the term with the largest gradient, that is, the gradient that governs the cost function [27].

2.5.1 Stiffness Analysis

To analyze this stiffness, Wang et al. [27] used the loss function of the vanilla PINN and the steepest descent method,

$$\theta_{n+1} = \theta_n - \eta \nabla_\theta \mathcal{L}(\theta_n) = \theta_n - \eta\big(\nabla_\theta \mathcal{L}_f(\theta_n) + \nabla_\theta \mathcal{L}_{\text{data}}(\theta_n)\big), \tag{2-30}$$

where η is the learning rate (the step). They apply a Taylor expansion of the loss function around the current parameters, up to second order, and diagonalize the Hessian $\nabla^2_\theta \mathcal{L}(\xi) = Q^T \operatorname{diag}(\lambda_1, \ldots, \lambda_P) Q$ by the spectral theorem [89]. With $\mathbf{y} = Q \nabla_\theta \mathcal{L}(\theta_n) / \|\nabla_\theta \mathcal{L}(\theta_n)\|$ (Appendix C.4), the loss difference after one step becomes

$$\mathcal{L}(\theta_{n+1}) - \mathcal{L}(\theta_n) = \eta \|\nabla_\theta \mathcal{L}(\theta_n)\|^2 \Big(-1 + \frac{1}{2}\eta \sum_{i=1}^{P} \lambda_i y_i^2 \Big). \tag{2-31}$$

For the optimization method to work, we want $\mathcal{L}(\theta_{n+1}) < \mathcal{L}(\theta_n)$, so the term in parentheses must be negative, which requires $1 > \frac{1}{2}\eta \sum_i \lambda_i y_i^2$. This is only possible if the eigenvalues are small (the loss function near the minimum is smooth and well behaved); if the landscape is complicated, the eigenvalues are large and, instead of following the descent path, the iterate jumps away to another valley. Decomposing the loss into its objectives, the same procedure gives

$$\mathcal{L}_j(\theta_{n+1}) - \mathcal{L}_j(\theta_n) \approx \eta \|\nabla_\theta \mathcal{L}_j(\theta_n)\|^2 \Big(-1 + \frac{1}{2}\eta \sum_{i=1}^{P} \lambda_i^j (y_i^j)^2 \Big), \qquad j \in \{f, ic, bc\}, \tag{2-32}$$

where λ_i^j are the eigenvalues of $\nabla^2_\theta \mathcal{L}_j$. According to Wang et al. [27], the cause of the gradient imbalance is in the eigenvalues λ_i^f associated with the physics loss, which are very large near a minimum.

Moreover, Wang et al. [27] bound the size of each gradient by the largest eigenvalue of the curvature of its loss (Appendix C.5), which reaffirms the imbalance between gradients.

2.5.2 Learning Rate Annealing

To mitigate this, learning rate annealing (LRA) is used to dynamically balance the gradients [27]. We start from the objective (2-28) and minimize it with a first-order method:

$$\theta_{n+1} = \theta_n - \eta\Big(\nabla_\theta \mathcal{L}_f(\theta_n) + \sum_{i \in \{ic, bc\}} \lambda_i \nabla_\theta \mathcal{L}_i(\theta_n)\Big). \tag{2-33}$$

The weights of the initial and boundary losses are computed as

$$\hat{\lambda}_i = \frac{\max_\theta |\nabla_\theta \mathcal{L}_f(\theta_n)|}{\overline{|\nabla_\theta \mathcal{L}_i(\theta_n)|}}, \tag{2-34}$$

where the numerator is the maximum (over the components of θ) of the gradient of the physics loss and the denominator is the mean of the absolute gradient of the loss i.[VERIFICAR: whether the denominator includes λ_i, as in the original Algorithm 1] By using the maximum, the numerator captures the largest stiffness of the differential equation, acting as an upper limit that represents the worst-case scenario, while the average in the denominator works as a regularizer. A general analysis of the possible situations gives:

- If $\hat{\lambda}_i \approx 1$, the worst-case gradient of the physics and the average gradient of the data term i have the same magnitude: the gradients are balanced and no reweighting is needed.
- If $\hat{\lambda}_i > 1$ (the usual situation), the physics gradient dominates, and the weight of the term i is increased so that the optimizer does not neglect it; geometrically, the scalarization moves to another region of the (apparent) Pareto front.
- If $\hat{\lambda}_i < 1$, the average gradient of the data term i is larger than the worst case of the physics gradient, and its weight is decreased, giving room to the physics.

The weights are then updated with an exponential moving average to remove the noise generated by the stochastic gradients during training:

$$\lambda_i \leftarrow (1-\alpha)\lambda_i + \alpha\hat{\lambda}_i. \tag{2-35}$$

The algorithm proposed by Wang et al. [27] is given in Algorithm 1, with the modification of using ADAM instead of plain gradient descent.

Algorithm 1 Learning rate annealing with ADAM.

Require: Recommended hyperparameters $\eta = 10^{-3}$, $\alpha = 0.9$; number of steps S; number of constraint terms M; initial weights λ_i, $i = 1, \dots, M$.

for $n = 1$ to S **do**
  (a) Gradient statistics
  $G_{\max} \leftarrow \max_\theta\{|\nabla_\theta \mathcal{L}_f(\theta_n)|\}$ ▷ maximum gradient of the physics residual
  $G^i_{\text{mean}} \leftarrow \overline{|\nabla_\theta \mathcal{L}_i(\theta_n)|}$ for each term i
  $\hat{\lambda}_i \leftarrow G_{\max}/G^i_{\text{mean}}$ ▷ Equation (2-34)
  (b) Weight update (moving average)
  $\lambda_i \leftarrow (1-\alpha)\lambda_i + \alpha\hat{\lambda}_i$ ▷ Equation (2-35)
  (c) Network parameters update
  $\nabla_{\text{total}} \leftarrow \nabla_\theta \mathcal{L}_f(\theta_n) + \sum_{i=1}^{M} \lambda_i \nabla_\theta \mathcal{L}_i(\theta_n)$
  $\theta_{n+1} \leftarrow \text{ADAM}(\theta_n, \nabla_{\text{total}}, \eta)$ ▷ Equation (2-37)
end for
return optimized parameters θ_{S+1}

2.5.3 ADAM as an Approximation of the Natural Gradient

To understand the ADAM method, one can start from the natural gradient [90, 91]. This means that the step no longer moves through the common Euclidean space of parameters, but through the space of probability distributions, under the metric induced by the Kullback--Leibler (KL) divergence. For this, the Fisher information matrix F is introduced in the gradient descent method:

$$\theta_{n+1} = \theta_n - \eta\, F^{-1}\nabla_\theta \mathcal{L}(\theta_n), \qquad F = \mathbb{E}\big[\nabla_\theta \log P(\mathbf{x} \mid \theta)\, \nabla_\theta \log P(\mathbf{x} \mid \theta)^T\big]. \tag{2-36}$$

The Fisher matrix is the second moment of $\nabla_\theta \log P(\mathbf{x} \mid \theta)$, which looks like the gradient of a typical loss function for a likelihood [90]; it contains the variances and covariances of the stochastic steps generated by the mini-batches, and it is the local quadratic approximation of the KL divergence, $\text{KL}(\theta \,\|\, \theta+\delta) \approx \frac{1}{2}\delta^T F \delta$ (Appendix C.6). The ADAM method assumes, in practice, that the parameters are independent of each other, so only the diagonal elements of the Fisher matrix survive. In addition, ADAM keeps the exponential decay of the momentum method [92] for the first moment, and it adds the second moment as a proxy of the diagonal of the Fisher matrix [93]:

$$\begin{gathered} \theta_{n+1} = \theta_n - \eta \frac{\hat{m}_n}{\sqrt{\hat{v}_n} + \epsilon}, \\ m_n = \beta_1 m_{n-1} + (1-\beta_1)\nabla_\theta \mathcal{L}(\theta_n), \qquad v_n = \beta_2 v_{n-1} + (1-\beta_2)\big(\nabla_\theta \mathcal{L}(\theta_n)\big)^2, \end{gathered} \tag{2-37}$$

where $\hat{m}_n, \hat{v}_n$ are the bias-corrected moments. Kingma and Ba [93] explain that they use the square root of the inverse of the variance because it preserves invariance to the scale of the gradient and it is a more conservative preconditioner. In the natural gradient method, if the Fisher curvature is small, the method assumes that it is in a flat basin and takes a very large step to cross it; since our functions are non-convex, this can skip local minima. To mitigate this, ADAM does not use the whole variance, but its square root, something like a standard deviation. We must say that, for a PINN loss (which is not a likelihood), this analogy is heuristic; the recent natural-gradient methods for PINNs [9, 77] make it precise by using the metric of the model in function space.

Second-order refinement (L-BFGS). After ADAM with LRA has found a region of the parameter space where a good trade-off between losses is located, a quasi-Newton method can be used as a final minimizer. L-BFGS builds an approximation of the inverse Hessian from the last pairs of differences of parameters and gradients, so it uses curvature information without storing the $P \times P$ Hessian.[VERIFICAR: add reference for L-BFGS (Liu & Nocedal, 1989)] We use it in the preliminary experiment of Section 3.3.

2.6 Training Strategies for Hyperbolic PDEs

Until now we have attacked the gradient stiffness, which induces the optimizer to fall into local minima; however, we have not attacked the specific problems of hyperbolic PDEs, such as the propagation of wave fronts. Beyond optimization, specific techniques are required.

Causal training. Causal training puts a blindfold on the physics loss, forcing it to focus on the present before moving to the future [94]. Since propagation phenomena depend on the past, the neural network must respect this process. The time domain is divided into N_t chunks $\{t_i\}$, and the physics loss is weighted by a factor that depends on the residuals of the past:

$$\mathcal{L}_f = \frac{1}{N_t}\sum_{i=1}^{N_t} w_i \mathcal{L}_f(\theta; t_i), \qquad w_i = \exp\Big(-\epsilon \sum_{k=1}^{i-1} \mathcal{L}_f(\theta; t_k)\Big). \tag{2-38}$$

This imposes a penalization factor ϵ: the residual at time t_i is only taken into account when the residuals of all previous times are small. Wang et al. [94] propose values $\epsilon \in \{10^{-2}, 10^{-1}, 10^{0}, 10^{1}, 10^{2}\}$. Without this mechanism, the network can find minima that satisfy the PDE but do not have physical meaning (for example, the trivial solution far from $t = 0$), that is, minima that deceive the network.

Focused stochastic sampling. Since the domain is large and the computing time grows with the amount of data, a stochastic sampling scheme is used: at each epoch, new sets of N_f, N_{bc}, N_{ic} points are drawn. Because of the gradient stiffness, more collocation points are assigned to the physics than to the initial and boundary conditions. In dispersive equations, the wave fronts carry the most critical dynamics; the physics happens inside these fronts, and the exterior must keep the initial (resting) state. For this reason, we designed a localized scheme that assigns 80 % of the physics collocation points inside the light cone $\{|x| \le t\}$ (where the wave propagates, since the group velocity is bounded by $c = 1$, Equation (1-61)) and 20 % outside, to guarantee that the network does not violate the physics outside the cone.

Input normalization. Normalizing the variables allows comparing them on the same scale, ensuring a uniform contribution of each input during training, which leads to more accurate results [95]. We use a modified

min--max scaling to the range $[-1, 1]$:

$$x_{\text{norm}} = 2\frac{x - x_{\min}}{x_{\max} - x_{\min}} - 1 = \alpha_x(x - x_{\min}) - 1, \qquad t_{\text{norm}} = 2\frac{t - t_{\min}}{t_{\max} - t_{\min}} - 1 = \alpha_t(t - t_{\min}) - 1, \quad (2\text{-}39)$$

with $\alpha_x = 2/(x_{\max} - x_{\min})$ and $\alpha_t = 2/(t_{\max} - t_{\min})$. Since the loss functions express the physics in the original domain, the derivatives must be scaled by the chain rule, $\partial_x = \alpha_x \partial_{x_{\text{norm}}}$, $\partial_x^2 = \alpha_x^2 \partial_{x_{\text{norm}}}^2$, and analogously for t. The Klein--Gordon operator becomes

$$\big(\alpha_t^2 \partial_{tt} - \alpha_x^2 \nabla^2 + m^2\big) u_\theta(x_{\text{norm}}, t_{\text{norm}}) = 0, \quad (2\text{-}40)$$

where the derivatives are taken with respect to the normalized variables.

2.7 Representation: Fourier Features and Activation Functions

Random Fourier features. To overcome the spectral bias of MLPs (their tendency to learn low frequencies first, Equation (2-11)), transformations of the input coordinates based on Fourier features have been proposed. The most common technique is random Fourier features [70], which apply the mapping

$$\gamma(\mathbf{v}) = \begin{pmatrix} \cos(B\mathbf{v}) \\ \sin(B\mathbf{v}) \end{pmatrix}, \quad (2\text{-}41)$$

where each entry of $B \in \mathbb{R}^{m \times d}$ is sampled independently from $\mathcal{N}(0, \sigma^2)$. This mapping modifies the spectrum of the NTK and allows fully connected networks to learn high frequencies more effectively. In the context of PINNs, Wang et al. [46] extended this idea with multi-scale Fourier embeddings to represent wave phenomena, taking advantage of the separability of variables: several mappings with different values σ_i are applied separately to the spatial and temporal coordinates; the outputs of each branch enter a shallow but wide MLP, and the outputs are combined by point-wise multiplication before the output layer. This strategy captures periodic or rapidly varying behaviors at different scales, as shown in wave, reaction--diffusion and convection equations. The price is a new hyperparameter, σ, whose value must be tuned for each problem.

Sinusoidal activations (SIREN). For training networks on periodic PDEs or on solutions that are approximated by Fourier series, Sitzmann et al. [73] proposed the SIREN methodology, with sinusoidal activation functions:

$$\Phi(\mathbf{x}) = W_n(\varphi_{n-1} \circ \varphi_{n-2} \circ \cdots \circ \varphi_0)(\mathbf{x}) + \mathbf{b}_n, \qquad \varphi_i(\mathbf{x}_i) = \sin(\omega_0 W_i \mathbf{x}_i + \mathbf{b}_i). \quad (2\text{-}42)$$

The authors argued that, when using SIREN, the initialization of the parameters must be adjusted, because the sine is almost linear near zero: if the weights are initialized with very small values, the activation loses its representation power. To fix this, they analyzed the distribution of the output signal of each layer. At the input, the distribution is assumed uniform; after the sinusoidal activation the output follows an arcsine distribution, and, after several layers, by the central limit theorem, the pre-activations become normally distributed. To keep this distribution stable across layers they proposed the initialization

$$w_i \sim \mathcal{U}\Big(-\sqrt{6/n},\ \sqrt{6/n}\Big), \quad (2\text{-}43)$$

where n is the number of inputs of the neuron, and a large factor ω_0 that scales the frequencies to attack the problem of small values (low frequencies). The original work uses $\omega_0 = 30$; in our experiments we use $\omega_0 = 15$ for the Klein--Gordon equation and $\omega_0 = 20$ for the benchmark (Chapter 3).

Rational activations. The need to find an activation function that is expressive, differentiable of high order (C^n with $n \geq 2$, since the PDE is of second order), and whose derivatives do not vanish quickly, led us to consider rational activation functions. In particular, the bell-shaped rational function $f(x) = 1/(1+x^2)$ is C^∞, bounded with bounded derivatives, and its derivatives decay only algebraically, like $|x|^{-3}$, instead of exponentially as for tanh. Boullé et al. [96] showed that networks with (learnable) rational activations can approximate functions with exponentially smaller depth than ReLU networks while keeping C^∞ smoothness, which mitigates the vanishing gradient problem; later variants addressed the numerical instability of the original rational activations [97]. Because of these properties (expressivity, high-order differentiability and non-vanishing derivatives), the function $1/(1+x^2)$ is a candidate for contexts where stable training is required, and it appears as one of the branches of our adaptive activation.

2.8 State of the Art in Architectures

Next, we describe five representative architectures of the PINN literature, which are the baselines of Chapter 3.

Vanilla PINN. The original formulation of Raissi et al. [21] approximates the solution with a fully connected MLP with tanh activations. The network is trained by minimizing the soft-constrained loss with weights $\lambda = (1,1,1)$, $\mathcal{L}(\theta) = \mathcal{L}_{ic} + \mathcal{L}_{bc} + \mathcal{L}_f$. This architecture suffers from gradient stiffness, problems of propagation and spectral bias. It works well for smooth and low-frequency problems, but it fails for nonlinear solutions, for functions with high and low frequencies, and for hyperbolic PDEs, because of its inability to capture the propagation and because of the vanishing gradient of some of the terms involved.

Multi-scale Fourier features (MSF). To address the spectral bias, Wang et al. [46] propose two architectures based on Fourier features. The first one (multi-scale Fourier feature architecture) applies several mappings $\gamma^{(i)}$ with different σ_i to the input, passes each one through the same fully connected network, and concatenates the outputs before a final linear layer. The second one (spatio-temporal multi-scale Fourier feature architecture) treats the spatial and temporal coordinates separately, applies different Fourier mappings to each group, processes them in parallel, and combines them by point-wise multiplication. These embeddings do not add trainable parameters (the matrices B are fixed), and they allow learning high-frequency components that the vanilla PINN cannot capture. The trunk is a vanilla MLP.

Modified MLP (M4). In Wang et al. [27] the pathologies of unbalanced gradients are identified and two improvements are proposed: (i) the learning rate annealing algorithm of Section 2.5.2; and (ii) an improved architecture, the modified MLP (M4), which incorporates two encoders that act as gates, with multiplicative interactions and residual connections. The forward propagation of M4 is

$$\begin{aligned} U = \sigma(XW^1 + \mathbf{b}^1), \qquad V = \sigma(XW^2 + \mathbf{b}^2), \qquad H^{(1)} = \sigma(XW^{z,1} + \mathbf{b}^{z,1}), \\ Z^{(k)} = \sigma(H^{(k)}W^{z,k} + \mathbf{b}^{z,k}), \qquad H^{(k+1)} = (1 - Z^{(k)}) \odot U + Z^{(k)} \odot V, \qquad k = 1, \dots, L, \\ f_\theta(X) = H^{(L+1)}W + \mathbf{b}. \end{aligned} \tag{2-44}$$

The element-wise products between the gates and the hidden representations reduce the stiffness of the gradient flow and improve the accuracy in problems such as the Helmholtz and Klein--Gordon equations.

PirateNet. Proposed by Wang et al. [98], PirateNet (Physics-Informed Residual Adaptive Network) is designed to allow the stable training of deep PINNs. The architecture combines: an initial random Fourier

embedding; the gates U and V, obtained from two dense layers applied to the embedding; and adaptive residual blocks, each with three dense layers, two gating operations (with U and V) and a residual connection with a trainable parameter $\alpha^{(l)}$:

$$\mathbf{x}^{(l+1)} = \alpha^{(l)}\mathbf{h}^{(l)} + (1-\alpha^{(l)})\mathbf{x}^{(l)}. \tag{2-45}$$

The key is to initialize $\alpha^{(l)} = 0$, so that at the beginning the network behaves as a linear combination of the embedding (a shallow network); during training the $\alpha^{(l)}$ grow and introduce nonlinearity progressively. In addition, the last layer is initialized by solving a least-squares problem that fits the available data (initial conditions, the solution of the linearized PDE, etc.). This strategy avoids the initialization pathologies of deep derivatives and lets PirateNet scale well with depth, outperforming MLPs and conventional ResNets in benchmarks such as Allen--Cahn, Korteweg--de Vries, Gray--Scott and the lid-driven cavity.

AbuPINN (adaptive blending units). Based on Wang et al. [99], AbuPINN learns the optimal activation function for each problem as a linear combination of candidate functions:

$$f(x) = \sum_{i=1}^{N} G(\alpha_i)\, \sigma_i(\beta_i x), \tag{2-46}$$

where σ_i are elementary functions (sin, tanh, GELU, Swish, Softplus, etc.), α_i are learnable coefficients normalized by $G = \text{softmax}$, and β_i are adaptive scale factors. This activation unit can be applied neuron-wise or layer-wise. AbuPINN guarantees continuous derivatives and avoids numerical instabilities. At the same time, it tries to soften the complexity of the loss landscape by giving expressivity to the network through the characteristics of each function involved in the activation. In their experiments, AbuPINN consistently outperforms fixed activations (tanh, sin, etc.) and other adaptive activations in convection, Burgers, Allen--Cahn, Korteweg--de Vries, Cahn--Hilliard and Navier--Stokes problems, and the resulting neural tangent kernel is shown to be learnable, with a more favorable eigenvalue spectrum for convergence.

2.9 Proposed Architecture: MABUR-PINN

Motivation. Traditional PINNs induce three pathologies: *spectral bias*, *gradient imbalance and stiffness*, and *violation of temporal causality*. Previous proposals for the spectral bias (multi-scale Fourier features, PirateNets) introduce a Fourier embedding layer, but they depend critically on the scale hyperparameter σ. Building upon the expressivity of AbuPINN and the capacity of PirateNets for nonlinear problems, we propose the *Modified Adaptive Blending Units Residual* network for PINNs (MABUR-PINN). The idea is to join two characteristics: that the network learns the scales (biases) it needs for its expressivity, through adaptive activations, and that it trains stably in depth, through residual and convex connections. Instead of fixing a Fourier basis through σ, MABUR-PINN uses a set of specialized activation functions (sin, cos, tanh and a rational function, for the Klein--Gordon equation) within an adaptive blending (Abu) layer. This allows the network to learn autonomously a Fourier-like basis (through sin and cos with SIREN initialization), to capture the initial Gaussian condition (through tanh and the rational bell), and to represent the sharp changes of the wave fronts.

Architecture. The architecture (Figure **2-2**) uses the convex gated connections (U and V) of M4 and the adaptive residual blocks of PirateNet, applying the skip connection every two layers. The input $(\mathbf{x}, t)$ feeds the gating layers and the residual trunk. Each neuron contains an adaptive Abu function, learning its optimal scales dynamically, with the aim that the optimizer does not fall into the local minima caused by gradient stiffness. We developed two versions, summarized in Table **2-3**: Version 1 has no Fourier embedding and uses Abu activations everywhere; Version 2 uses a Fourier embedding ($\sigma = 1$) and a fixed tanh in the first layer of the trunk and of the gates.

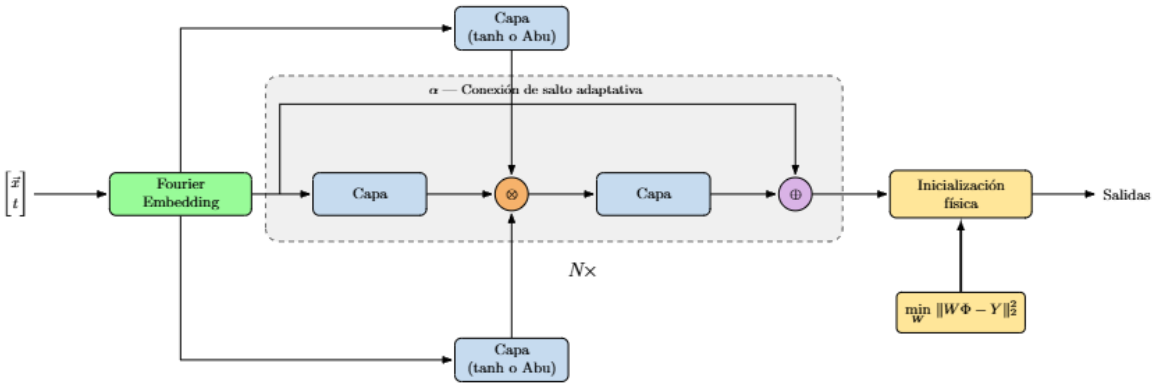


Figura 2-2: MABUR-PINN architecture, based on PirateNet [98]: gates U and V and the residual trunk with adaptive skip connections. Each dense block has three layers with N neurons. In Version 1 every neuron contains an Abu unit with $\{\sin, \cos, \tanh, \text{rational}\}$ and there is no Fourier embedding; in Version 2 the first layer of the trunk and of U, V uses only tanh, and there is a Fourier embedding.

Tabla 2-3: The two versions of MABUR-PINN.

Property	Version 1	Version 2
Implementation	Monolithic class	Modular, four classes (modified MLP of Wang et al. [27])
Fourier embedding	No	Yes, 256 random features, $\sigma = 1.0$
Adaptive activation layer	AbuLayer: softmax blend of $\sin, \cos, \tanh$, rational	AbuLayer_UV: same blend, with U/V gating in a separate layer
Encoders U / V	Adaptive layer (trainable activations)	Dense + tanh (fixed activation)
Initial projection	Adaptive layer with SIREN initialization	Dense + tanh with Glorot initialization
Skip connections	Every 3 layers	Every 2 layers
ω_0 (SIREN)	15.0	15.0

Theoretical intuition. The Sobolev--Barron theorem (Theorem 2.3.2) only sees the activation through the constant $C_1 = \max_{m\le\ell} \left\|\sigma^{(m)}\right\|_{L^\infty}$. This motivates the following assumption on the family of adaptive activations.

Supuesto 2.9.1: Adaptive softmax activations.

Let $\Omega \subset \mathbb{R}^d$ be a bounded open domain, μ a probability measure on Ω, and $\ell \in \mathbb{N}$ the maximum order of the differential operator of the PDE. Consider the family of adaptive softmax activations

$$\mathcal{S}_{\text{softmax}} := \left\{ \sigma_{\text{ad}}(x) = \sum_{i=1}^{N} \frac{e^{\alpha_i}}{\sum_{j=1}^{N} e^{\alpha_j}} \sigma_i(\beta_i x) \;\middle|\; \boldsymbol{\alpha} \in \mathbb{R}^N,\ \beta_i \in [\beta_{\min}, \beta_{\max}] \subset (0,\infty) \right\}.$$

We assume that $\mathcal{S}_{\text{softmax}}$ satisfies:

1. **Uniform C_1 regularity:** $\sup_{\sigma_{\text{ad}} \in \mathcal{S}_{\text{softmax}}} \max_{m\le\ell} \left\|\sigma_{\text{ad}}^{(m)}\right\|_{L^\infty(\mathbb{R})} \le C_{1,\text{ad}} < \infty.$
2. **Sobolev--Barron approximation:** the conclusion of Theorem 2.3.2 holds for shallow networks whose activation is any fixed $\sigma_{\text{ad}} \in \mathcal{S}_{\text{softmax}}$, with C_1 replaced by $C_{1,\text{ad}}$.[VERIFICAR: this condition is stated without content in the presentation; its validity must be checked or proved]

The first condition can actually be verified. Since the softmax weights w_i are positive and sum to one, by the chain rule

$$\left|\sigma_{\text{ad}}^{(m)}(x)\right| = \left|\sum_{i=1}^{N} w_i \beta_i^m \sigma_i^{(m)}(\beta_i x)\right| \le \max_{1\le i\le N} \beta_{\max}^m \left\|\sigma_i^{(m)}\right\|_{L^\infty(\mathbb{R})}, \tag{2-47}$$

so $C_{1,\text{ad}} \le \max_{m\le\ell} \max_i \beta_{\max}^m \left\|\sigma_i^{(m)}\right\|_{L^\infty}$, a bound that is *uniform in* $\boldsymbol{\alpha}$: whatever blend the optimizer learns, it can never be worse than the worst branch at the largest scale. Two intuitions follow. First, condition 1 holds for the branches $\{\sin, \cos, \tanh, 1/(1+x^2)\}$ used for the Klein--Gordon equation (and for the Gaussian branch e^{-x^2}), because they are bounded with bounded derivatives, but it fails for unbounded branches such as Swish, Softplus or $e^{\beta x} - 1$, which appear in some of the benchmark configurations (Table **3-5**). Second, the constant grows like $\beta_{\max}^{\ell}$: for the Klein--Gordon equation ($\ell = 2$) and the SIREN scale $\omega_0 = 15$, the factor is of order 225. Large scales give access to high frequencies, but they make the constant (and, by Theorem 2.3.4, the conditioning) worse. This is the same trade-off that σ controls in a Fourier embedding, now learned inside a bounded range. We stress that these are intuitions: the theorem is for shallow networks with a single activation, while MABUR-PINN is a deep gated network whose neurons have different activations.

Activation functions. Table **2-4** lists the candidate branches considered, with the analytical property that justifies each of them in a PDE.

Tabla 2-4: Candidate branches of the adaptive activation.

Function	Formulation $f(x)$	Analytical property / justification in the PDE
Sine	$\sin(\beta_i x)$	Captures oscillatory dynamics and spatial/temporal harmonic modes.
tanh	$\frac{e^{\beta_i x} - e^{-\beta_i x}}{e^{\beta_i x} + e^{-\beta_i x}}$	Numerical stability; models smooth transition profiles.
Swish / SiLU	$\frac{x}{1 + e^{-\beta_i x}}$	Good behavior of gradients in deep networks, without abrupt cancellations.
Rational	$p(x)/q(x)$, e.g. $1/(1+(\beta_i x)^2)$	Captures poles, asymptotes and boundary layers; high expressivity with few parameters.
Exponential	$e^{\beta_i x} - 1$	Designed to capture boundary layers, strict dissipation and exponential decays.

3 Experimentation and Results

This chapter evaluates the architectures of Chapter 2 along two complementary lines. The first one is a controlled case study, the linear homogeneous Klein--Gordon equation (Section 1.5), where the reference solution is exact in time and the number of frequencies of the initial condition can be controlled; it contains a preliminary optimization experiment (Section 3.3), the comparison of architectures (Section 3.4) and a study of regularization and of the Fourier scale (Section 3.5). The second line is breadth: MABUR-PINN is evaluated on twelve tasks of the PINNacle benchmark, which contain elliptic, parabolic and hyperbolic equations (Section 3.7). Section 3.8 relates the results with the theory of Chapters 1 and 2.

3.1 General Methodology

Metrics. For all experiments we report the relative L^2 error on a reference grid of N_{test} points $(\mathbf{x}_j, t_j)$,

$$\text{L2RE} = \frac{\left(\sum_j |u_\theta(\mathbf{x}_j, t_j) - u(\mathbf{x}_j, t_j)|^2\right)^{1/2}}{\left(\sum_j |u(\mathbf{x}_j, t_j)|^2\right)^{1/2}}, \tag{3-1}$$

which is the discrete version of the total error $\mathcal{E}(\theta)$ of Section 2.2; the mean absolute error; the number of trainable parameters; and the final value of the (weighted) loss, which is the training error $\mathcal{E}_T^2$. Reporting both the loss and the L2RE is important, because Theorem 2.3.1 says that they are different things.

Implementation. All networks were implemented and trained by us under the same conditions. We used mixed precision: 32-bit floating point only for the parameters of the network, and 64-bit floating point for the collocation points, the optimization coefficients, the back-propagated gradients, etc. The code and the results are available in the repositories `https://github.com/Emmaquantum/HPINN` and `https://github.com/Emmaquantum/Optimization_PINN`. [VERIFICAR: framework (PyTorch/JAX), hardware (GPU), training time and number of random seeds per configuration]

3.2 Reference Solution: Pseudo-Spectral Propagator

Problem. We consider the one-dimensional Klein--Gordon equation (1-55) in natural units ($m = c = \hbar = 1$):

$$\begin{cases} \partial_{tt}\phi(x,t) - \partial_{xx}\phi(x,t) + m^2\phi(x,t) = 0, & x \in \Omega = [-L/2, L/2],\ t \in [0,T], \\ \phi(-L/2,t) = \phi(L/2,t), & t \in [0,T], \\ \phi(x,0) = A\exp\Big(-\dfrac{x^2}{2\sigma^2}\Big)\displaystyle\sum_{k\in\mathcal{K}}\big(e^{ikx} + e^{-ikx}\big), & x \in \Omega, \\ \partial_t\phi(x,0) = 0, & x \in \Omega. \end{cases} \tag{3-2}$$

The initial conditions represent a stationary particle, that is, a localized pulse with zero initial velocity. The initial position is a Gaussian envelope modulated by a symmetric phase, which induces velocity in $+k$ and $-k$. This initial condition can be manipulated consciously so that the network needs to learn high and low frequencies at the same time, by introducing a set of wavenumbers $\mathcal{K} = \{k_1, k_2, \ldots, k_n\}$ with $k_1 < k_2 < \ldots < k_n$. Because of the dispersion relation Equation (1-59), each wavenumber travels with a different group velocity: the pulse disperses inside the light cone $|x| \leq t$. The boundary condition is periodic, as required by the pseudo-spectral method; however, with the chosen parameters the wave does not reach the boundary during the simulation. In Experiment 1 we use $\mathcal{K} = \{1, 5, 10\}$, $\sigma = 1$, $L = 80$ and $T = 36$. [VERIFICAR: value of the amplitude A; the presentation writes $\Omega = [-L, L]$ but the figures show $x \in [-40, 40]$]

Choice of parameters. The solution of Equation (3-2) models the field of a free particle released into an (almost) infinite space. To achieve this, the space must be large enough to model an infinite space, $L \gg \sigma$, and the simulation time must be short enough so that the wave does not hit the boundary.

1. **Space:** $L \geq 40\sigma$, enough times σ so that the boundary is not reached.
2. **Time:** since the group velocity is bounded by $c = 1$ (Equation (1-61)), the fastest front reaches the boundary at $t = L/2$. The thickness of the pulse is about 4σ, so we take $T \approx L/2 - 4\sigma$; for $L = 80$ and $\sigma = 1$, $T = 36$.
3. **Grid:** in a classical explicit scheme we would need the CFL (Courant--Friedrichs--Lewy) condition $c\,\Delta t/\Delta x \leq 1$. Here the propagator is exact in time, so the scheme is unconditionally stable, and the time step only defines the temporal resolution of the stored solution. What matters is that the spatial grid resolves the largest wavenumber of the initial condition: with N grid points, $\Delta x = L/N$ and we need $k_{\max} < \pi/\Delta x$. With $N = 1000$ and $L = 80$, $\pi/\Delta x \approx 39 > k_{\max} = 10$.[VERIFICAR: whether $N = 1000$ is the number of spatial points or of time steps]

Propagator. The solution is exact in Fourier space, which means that, by means of a propagator [100], the solution is computed without numerical errors beyond the truncation of the Fourier series; all we have to do is to compute the propagator and iterate. With the change of variables $v = \partial_t\phi$ and the operator $\mathcal{L} = m^2 - \partial_{xx}$, the Klein--Gordon equation becomes the first-order system

$$\frac{d}{dt}\begin{pmatrix}\phi\\ v\end{pmatrix} = \underbrace{\begin{pmatrix}0 & 1\\ -\mathcal{L} & 0\end{pmatrix}}_{A}\begin{pmatrix}\phi\\ v\end{pmatrix}, \qquad \mathbf{y}(t) = e^{At}\mathbf{y}(0), \tag{3-3}$$

with $\mathbf{y} = (\phi, \partial_t\phi)^T$. The propagator e^{At} is responsible for the temporal evolution of the field. Expanding it in Taylor series and separating the even and odd powers of A (Appendix A.5), we obtain

$$e^{At} = \begin{pmatrix}\cos\big(t\sqrt{\mathcal{L}}\big) & \dfrac{\sin(t\sqrt{\mathcal{L}})}{\sqrt{\mathcal{L}}}\\ -\sqrt{\mathcal{L}}\sin\big(t\sqrt{\mathcal{L}}\big) & \cos\big(t\sqrt{\mathcal{L}}\big)\end{pmatrix}. \tag{3-4}$$

To evaluate the operator $\mathcal{L}$ numerically we take advantage of the periodic boundary condition: the Fourier transform diagonalizes it, $\widehat{\mathcal{L}} = m^2 - (ik)^2 = m^2 + k^2 = \omega_k^2$, and the propagator becomes a 2×2 rotation for each wavenumber:

$$\begin{pmatrix} \hat{\phi}_k(t) \\ \dot{\hat{\phi}}_k(t) \end{pmatrix} = \begin{pmatrix} \cos(\omega_k t) & \dfrac{\sin(\omega_k t)}{\omega_k} \\ -\omega_k \sin(\omega_k t) & \cos(\omega_k t) \end{pmatrix} \begin{pmatrix} \hat{\phi}_k(0) \\ \dot{\hat{\phi}}_k(0) \end{pmatrix}. \tag{3-5}$$

This is exactly the discrete version of the exact solution (1-62). The algorithm is Algorithm 2.

Algorithm 2 Pseudo-spectral method for the Klein--Gordon equation.

Require: Physical parameters m, L, T, N; initial conditions $\phi(x,0)$, $\dot{\phi}(x,0)$.

$\Delta x \leftarrow L/N$, $\Delta t \leftarrow T/N_t$ ▷ precomputation of the frequency space

$k \leftarrow 2\pi \cdot \texttt{fftfreq}(N, \Delta x)$; $\omega_k \leftarrow \sqrt{m^2 + k^2}$

$\hat{\phi}_k \leftarrow \mathrm{FFT}(\phi(x,0))$; $\dot{\hat{\phi}}_k \leftarrow \mathrm{FFT}(\dot{\phi}(x,0))$ ▷ initial transform

$C_k \leftarrow \cos(\omega_k \Delta t)$; $S_k \leftarrow \sin(\omega_k \Delta t)$

$t \leftarrow 0$

while $t < T$ **do** ▷ time evolution, Equation (3-5)

  $\hat{\phi}_k^{\text{new}} \leftarrow C_k \hat{\phi}_k + \frac{1}{\omega_k} S_k \dot{\hat{\phi}}_k$

  $\dot{\hat{\phi}}_k^{\text{new}} \leftarrow -\omega_k S_k \hat{\phi}_k + C_k \dot{\hat{\phi}}_k$

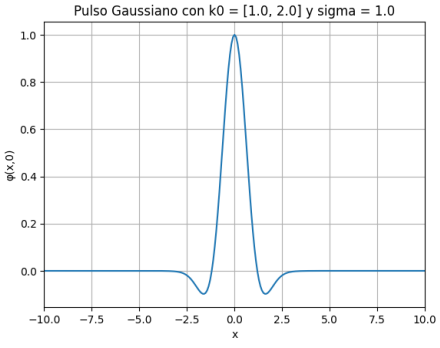
  $\hat{\phi}_k \leftarrow \hat{\phi}_k^{\text{new}}$; $\dot{\hat{\phi}}_k \leftarrow \dot{\hat{\phi}}_k^{\text{new}}$; $t \leftarrow t + \Delta t$

  store $\phi(x,t) \leftarrow \mathrm{Re}\left(\mathrm{IFFT}(\hat{\phi}_k)\right)$ ▷ recovery to real space

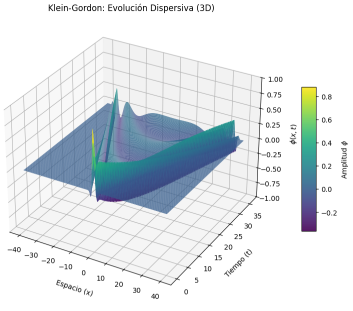
end while

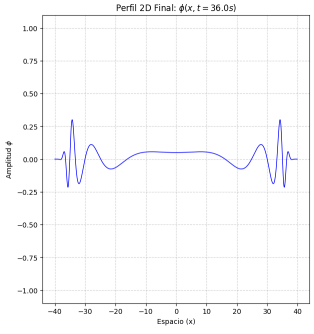
return history $\phi(x,t)$

Figure **3-1** shows the initial condition and the evolution of the field used in the preliminary experiment. The field disperses inside a light cone, at the same velocity to the left and to the right.



(a) Initial condition (Gaussian pulse).





(b) Evolution of the field and final profile.

Figura 3-1: Pseudo-spectral reference solution of the preliminary experiment. (a) Initial condition, Gaussian with $\sigma = 1$. (b) Evolution of the field during 36 time units: the field disperses inside a light cone at the same velocity to the left and to the right, and the final profile shows the dispersion of the pulse once released in the (almost) infinite domain. A video of the evolution is available in the repository.

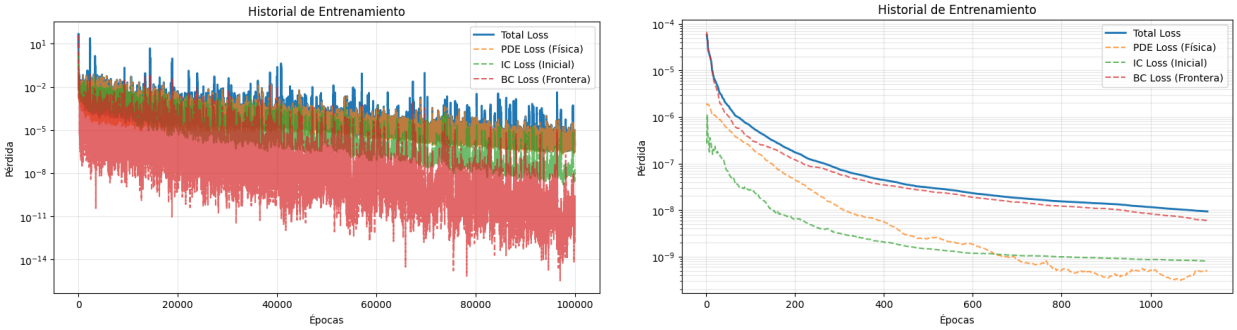
[VERIFICAR: the text of the first report says $k_0 = 0.5$, but the title of panel (a) says $k_0 = [1, 2]$]

3.3 Preliminary Experiment: ADAM + LRA and L-BFGS

The first experiment asks a simple question: how well can a PINN with a periodic activation, trained with the strategies of Section 2.5, reproduce the dispersion of a static spin-0 particle? We used a SIREN network ($\omega_0 = 30$) with 128 neurons per hidden layer and 6 hidden layers; this architecture was obtained by experimentation until a good performance was reached. We used ADAM + LRA as a first stage to characterize the minima within the set of achievable values: exponential decay of the learning rate with $\eta_0 = 10^{-4}$ and rate 0.95 every 1 500 iterations; the weights of the initial and boundary losses were recomputed every 10 iterations and updated with a moving average with $\alpha = 0.95$, starting from $(1, 1, 1)$. A random sampling algorithm generated new points $(\mathbf{x}_i, t_i)$, $i \in \{f, ic, bc\}$, at each iteration [27, 101], starting from 128 collocation points for each loss term; in this way, after 50 000 iterations the network was trained with a total of 6.4 million collocation points without saturating the memory.[VERIFICAR: the report mentions both 50 000 and 100 000 ADAM iterations] After ADAM + LRA found a region where the optimal value is, a variant of L-BFGS was used as final minimizer, with a stopping criterion of 5 000 iterations or a tolerance of machine epsilon; the weights and parameters of the first stage were loaded as initialization, so that the second stage has some ''memory'' of the first one. The physical conditions were the same as in the numerical simulation.

Results. The optimizer ADAM + LRA reduced the total loss to orders of 10^{-5} after the training, with a final learning rate of $\approx 3.27 \times 10^{-6}$ and final weights (1.0, 4.90, 0.82), reaching stability at a Pareto value (Figure **3-2**a). Although there seems to be convergence, ADAM is slow when converging to minima, but fast approaching them when the gradients are balanced; since they are not, LRA was used, and the curve shows a lot of noise produced by the change of weights every 10 epochs. The SIREN method favors the boundary conditions, so the boundary loss can reach values close to 10^{-12}; the total loss remained in the hands of the physics (PDE) loss and the initial condition.

The second-order optimizer L-BFGS refined the solution, reaching a total loss of order 10^{-8} (Figure **3-2**b), with smooth curves, no numerical explosions, and convergence in fewer than 500 iterations, which indicates that the inherited weights were good enough for the optimizer to reach an almost even convergence of all the loss terms. The PDE loss converged rapidly, indicating strong learning of the Klein--Gordon equation, that is, the network learned not to violate the physical law.

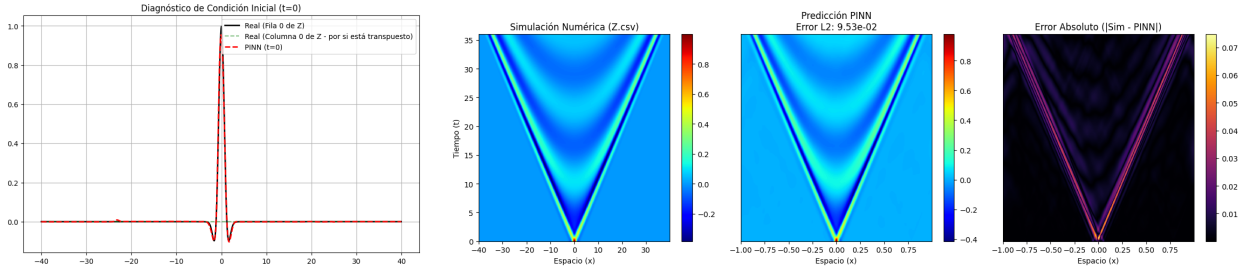


(a) ADAM + LRA. (b) L-BFGS.

Figura 3-2: Loss functions through the training epochs of the preliminary experiment. (a) ADAM + LRA: the noise comes from the update of the weights every 10 epochs. (b) L-BFGS: smooth curves, without numerical explosions.

Although the network learned the initial condition (Gaussian pulse) very well (Figure **3-3**a), the validation against the numerical simulation shows conflicts: the relative L^2 error is 9.53 % (Figure **3-3**b). The PINN generates the cone of the group velocity of the fastest waves (inside the light cone), but it is still inconsistent in amplitude. We interpret this result as an apparent Pareto value [88]: a training loss of order 10^{-8}

together with an error of almost 10 %. In the language of Theorem 2.3.1, the training error is small, but the generalization gap and the approximation error are not; this observation motivated the comparison of architectures of the next section.



(a) Initial condition ($t = 0$). (b) Simulation, PINN prediction and absolute error.

Figura 3-3: Validation of the preliminary experiment. (a) The network fits the static behavior at $t = 0$ very well. (b) The PINN reproduces the light cone, but the amplitude inside the cone is still inconsistent (L2RE $= 9.53\,\%$).

3.4 Experiment 1: Comparison of Architectures

Methodology. We replicated each of the architectures of Section 2.8 and implemented them on the Klein--Gordon problem (3-2) with $\mathcal{K} = \{1, 5, 10\}$, to analyze their behavior. All the networks were trained with the same strategies: input scaling (Equation (2-39)), learning rate annealing (Section 2.5.2), causal training (Equation (2-38)), focused stochastic sampling, exponential decay of the learning rate and mixed precision. The shared configuration is given in Table **3-1**, and the specific settings of each architecture in Table **3-2**.

Tabla 3-1: Shared training configuration of Experiment 1.

Component	Configuration
Optimizer	ADAM [93]
Loss weighting	LRA [27], $\alpha = 0.9$
Temporal strategy	Causal training [94]
Training epochs	100 000
Learning rate	10^{-3} initial, exponential decay 0.9 every 2 000 epochs
Physics collocation points (N_f)	12 000 (80 % inside the light cone)
Initial condition points (N_{ic})	$N_f/2 = 6\,000$
Boundary points (N_{bc})	$N_f/4 = 3\,000$
Normalization	min--max, $x \in [-1, 1]$, $t \in [-1, 1]$
Initial LRA weights	(1.0, 1.0, 0.1)
ω_0 (architectures with sin/cos)	15.0 (SIREN, AbuPINN, MABUR-PINN)

[VERIFICAR: the presentation reports $t \in [0, 1]$ and initial LRA weights $(1, 1, 1, 1)$; unify with the code]

Results. Table **3-3** summarizes the results, and Figure **3-4** shows the relative error against the complexity (number of parameters). MABUR-PINN V2 reaches the lowest error (1.70 % relative L^2) and the lowest final cost (3.81×10^{-7}), followed by AbuPINN (2.14 %), which does it with a moderate number of parameters (42 073), and by M4-PINN (4.41 %). In contrast, the vanilla PINN shows an almost total inability to solve the problem (71 %), while MSF-PINN, despite its high number of parameters (2.11×10^6), does not reduce the error significantly (43.2 %). MABUR-PINN V1, without Fourier embedding, reaches 11.2 %.

Tabla 3-2: Architectures of Experiment 1 (ϵ: causality parameter; chunks: number of temporal chunks of the causal loss).

Architecture	Depth	Width	ϵ	Chunks	Other
Vanilla [21]	6	64	10	16	tanh
SIREN [73]	6	64	10	16	SIREN activation
MSF [46]	5	200	10	16	Multiple embedding σ
M4 [27]	6	64	10	16	Gated architecture + embedding
Abu [99]	6	64	100	16	Blended activation
MABUR (ours)	6 (2 bl.)	64	100	16	M4 + Abu + adaptive residual

Tabla 3-3: Comparative results of the architectures on the Klein--Gordon equation (Experiment 1).

Architecture	Rel. L^2 error (%)	Abs. error	Parameters	Final cost
MABUR-PINN V2	**1.70**	**0.0006**	149 020	$\mathbf{3.81 \times 10^{-7}}$
AbuPINN	2.14	0.0008	42 073	4.60×10^{-7}
M4-PINN	4.41	0.0013	124 033	2.21×10^{-6}
MABUR-PINN V1	11.20	0.0037	51 177	7.25×10^{-6}
PINN-SIREN	21.20	0.0068	21 057	5.85×10^{-6}
MSF-PINN	43.20	0.0142	2 110 049	9.99×10^{-5}
Vanilla PINN	71.00	0.0232	21 057	6.72×10^{-5}

[VERIFICAR: single run or mean ± standard deviation over seeds]

Figure **3-5** compares the error maps of three representative networks; the error maps and loss curves of all the architectures are in Appendix E. The vanilla PINN has large errors distributed over the whole cone. AbuPINN has a homogeneous error without pronounced peaks, except near the origin. MABUR-PINN V2 concentrates its (small) error along the two fronts of the light cone.

[VERIFICAR: the panel titles of (a) say "ABU-PINN" and those of (c) "ABUR-PINN"; regenerate the figures with correct titles, in English]

Discussion. Several observations can be extracted.

1. **A low value of the cost function does not guarantee a low relative error.** PINN-SIREN reaches a lower final cost than MABUR-PINN V1 (5.85×10^{-6} vs. 7.25×10^{-6}), and its boundary loss reaches 10^{-12}, but its error is almost twice as large (21.2 % vs. 11.2 %). This is Theorem 2.3.1 in practice: the training error is only one of the three terms.

2. **Having many parameters does not solve, by itself, the optimization problems of hyperbolic PDEs if the architecture suffers from representability problems.** MSF-PINN has fifty times more parameters than AbuPINN and an error twenty times larger, with high variance in its training curves.

3. **Adaptive activations are the decisive ingredient.** The three best networks (MABUR V2, AbuPINN and M4) all have either adaptive activations or multiplicative gates. AbuPINN is the most balanced architecture: low error with a moderate computational cost. Its ability to learn the optimal combination of activations (including sinusoids) is key for problems with periodic and rapidly varying components.

4. **MABUR-PINN V2 improves the error, at a cost.** The improvement of V2 over AbuPINN is 0.44 percentage points, with 3.5 times more parameters. With a single run per configuration, this difference cannot yet be considered significant.

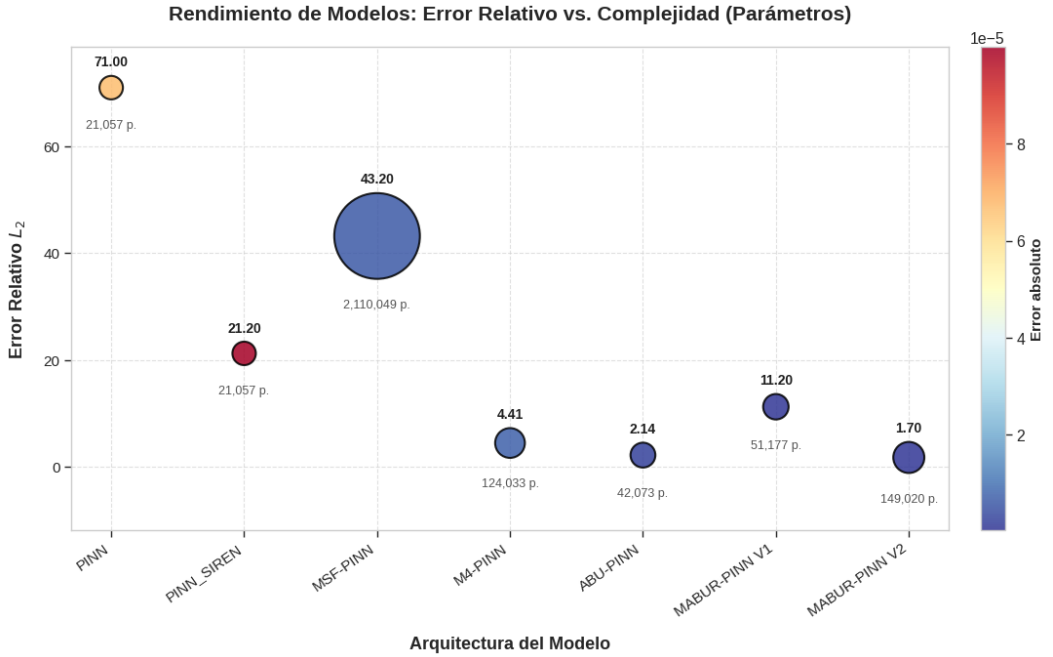


Figura 3-4: Relative L^2 error (%) against complexity: the size of each marker is the number of parameters and the color is the absolute error.

5. **The remaining error lives on the fronts.** For MABUR-PINN (V1 and V2) the error is concentrated along the light cone, where the fastest components (the largest wavenumbers) travel. This agrees with the spectral bias (Equation (2-11)) and with the conditioning of Theorem 2.3.4: the highest frequencies are the last ones to be learned. V1 also shows difficulties representing the frequencies inside the cone.

6. **The Fourier embedding still matters.** V1, without embedding, is clearly worse than V2, with an embedding of scale $\sigma = 1$. Therefore, in its current form, MABUR-PINN reduces but does not eliminate the dependence on σ; Experiment 2 quantifies this dependence.

3.5 Experiment 2: Regularization and Fourier Scale

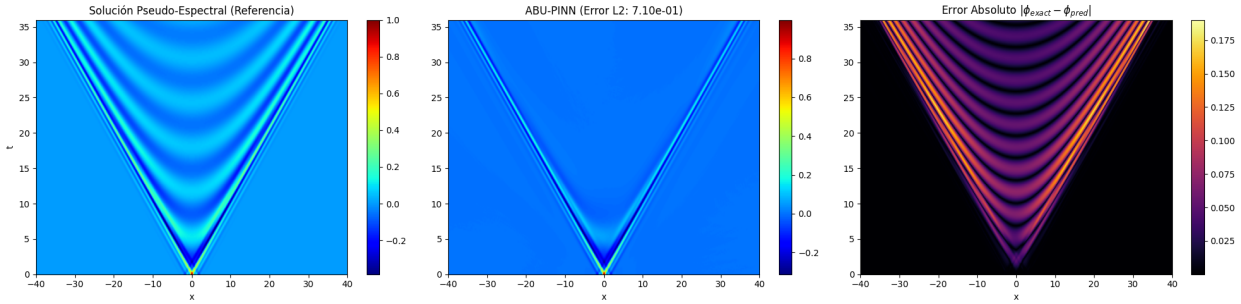
In this experiment, the model MABUR-PINN V2 was trained with and without regularization of the loss by the derivatives of the residuals (Section 2.3.6), and with two values of the scale of the Fourier embedding, to see the impact of each one (Table **3-4**).

Tabla 3-4: MABUR-PINN V2 with and without derivative regularization, and with two Fourier scales (depth 6 with 2 blocks, width 64, 100 000 epochs).

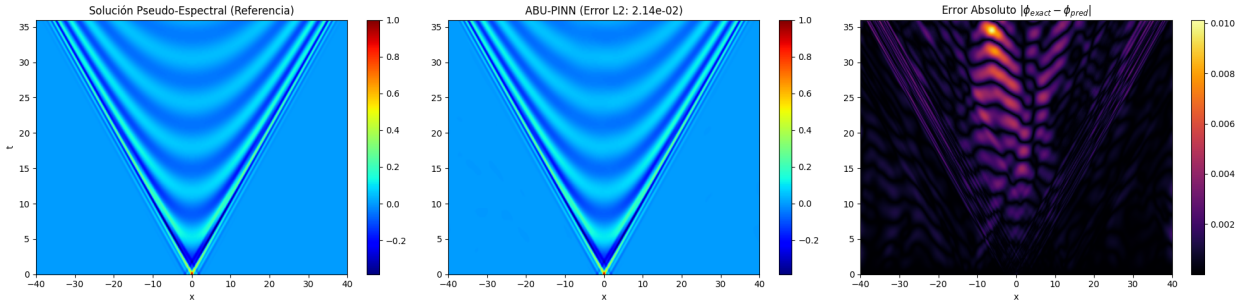
Model	Regularization	σ	Rel. L^2 error (%)
MABUR-PINN V2	No	1.0	1.70
MABUR-PINN V2	Yes	1.0	1.75
MABUR-PINN V2	No	5.0	3.32

[VERIFICAR: which gradient terms were included in the regularized loss]

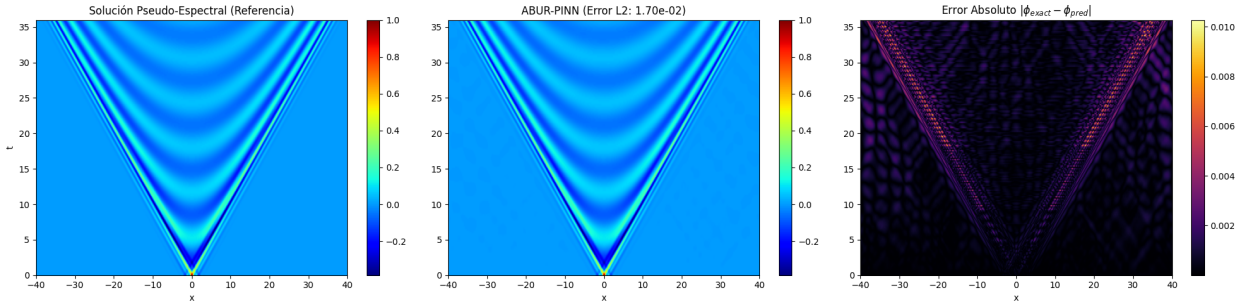
Two results stand out. First, the derivative regularization does not improve the error (1.75 % vs. 1.70 %).



(a) Vanilla PINN (L2RE 71.0 %).



(b) AbuPINN (L2RE 2.14 %).



(c) MABUR-PINN V2 (L2RE 1.70 %).

Figura 3-5: Reference solution (left), prediction (center) and absolute error (right) for three architectures on the Klein--Gordon equation.

This is consistent with Theorem 2.3.6: for periodic boundary conditions the boundary terms cancel, and the only extra term required by the energy estimate is the gradient of the initial position residual, which is already small because the initial condition is learned with high accuracy. Second, increasing the Fourier scale from $\sigma = 1$ to $\sigma = 5$ doubles the error. Large scales inject high frequencies from the beginning, which, according to Theorem 2.3.4 and the bound (2-47), worsens the conditioning. The dependence on σ is therefore reduced but still present in MABUR-PINN V2.

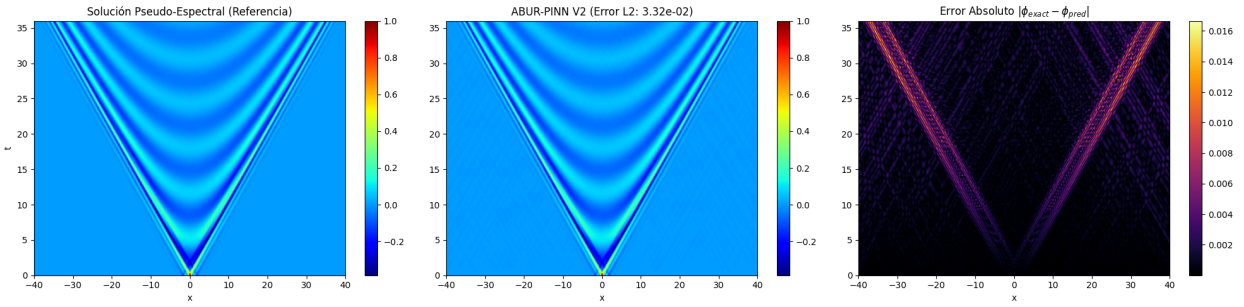


Figura 3-6: MABUR-PINN V2 with $\sigma = 5$ (L2RE 3.32 %): reference (left), prediction (center) and absolute error (right). The error is concentrated on the two fronts of the light cone.

[VERIFICAR: in the first report this figure is under the heading "V2 regularized"; check whether it corresponds to $\sigma = 5$ (its L2RE, 3.32×10^{-2}, matches $\sigma = 5$)]

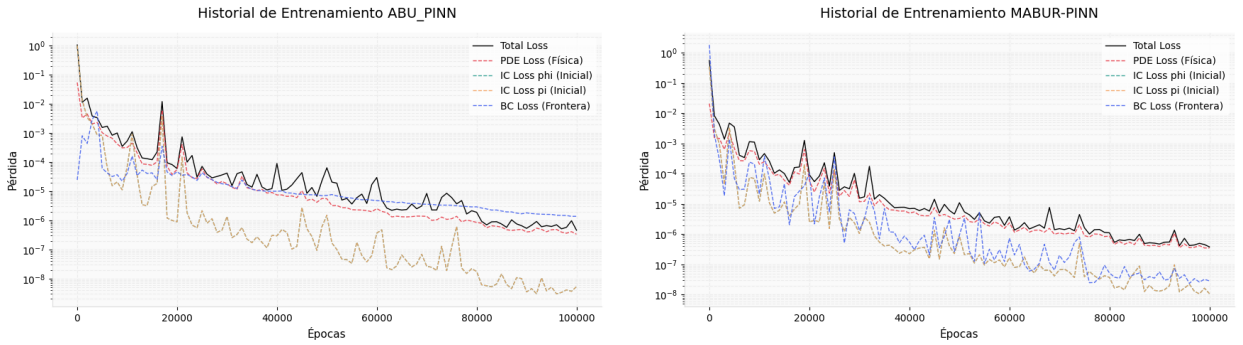
3.6 Behavior of the Optimizer ADAM + LRA

All the architectures were trained with ADAM and LRA to balance the terms of the composite loss. The final losses (Table **3-3**) show that MABUR-PINN V2 and AbuPINN reach the lowest values, followed by M4-PINN and PINN-SIREN, while the vanilla PINN and MSF-PINN present higher losses (of order 10^{-5} and 10^{-4}), indicating difficulties to minimize simultaneously the PDE residual and the initial and boundary conditions. The loss curves (Figure **3-7** and Appendix E) show that:

- ADAM + LRA reduces the total loss to orders between 10^{-7} and 10^{-5} after 10^5 epochs, but with noise due to the periodic readjustment of the weights λ_i;
- in AbuPINN, M4-PINN and MABUR-PINN the curves are more stable and the loss decreases rapidly, indicating a better balance of gradients; the total loss follows the physics loss, while the initial position loss decreases below 10^{-7};
- MSF-PINN shows high variance, probably because of its large number of parameters and the difficulty of training multi-scale embeddings.

3.7 Experiment 3: MABUR-PINN on the PINNacle Benchmark

Methodology. To test whether the conclusions obtained for the Klein--Gordon equation extend to other classes of PDEs, we adopted the experimental protocol of PINNacle [55]. MABUR-PINN was evaluated on a subset of 12 tasks of the benchmark, covering elliptic (Poisson), parabolic (heat), hyperbolic (wave) and nonlinear (Burgers, Navier--Stokes) equations in 1D and 2D (definitions in Appendix D). The results are



(a) AbuPINN. (b) MABUR-PINN V2.

Figura 3-7: Training history (total, PDE, initial position, initial velocity and boundary losses).

compared against three representative methods of the benchmark: LAAF, GAAF [68, 69] and FBPINN [102]. Base configuration: 5 layers of 100 neurons, 20 000 epochs, learning rate 10^{-3} (ADAM), 8 192 collocation points and 2 048 boundary/initial points. LRA, causal training and Fourier embedding were *not* used, in order to adopt the experimental parameters of the benchmark. The branches of the softmax blend were chosen for each family of PDEs (Table **3-5**).

Tabla 3-5: Branches of the softmax blend used for each PDE. $z = W\mathbf{x}+\mathbf{b}$ (dense branch, Glorot initialization); $z_s = \omega_0 W_s \mathbf{x}$ (SIREN branch, $\omega_0 = 20$); output $= \sum_k \mathrm{softmax}(\boldsymbol{\alpha})_k \sigma_k$.

PDE	Variant	Branches	No.
Burgers	1d-C, 2d-C	$\sin(\beta_1 z_s)$, $\tanh(\beta_2 z)$, $1/(1+(\beta_3 z)^2)$, $\ln(1+e^{\beta_4 z})$, $e^{-(\beta_5 z)^2}$	5
Poisson	2d-C, 2d-CG, 2d-MS	$\ln(1+e^{\beta_1 z})/\beta_1$, $z\,\sigma(\beta_2 z)$ (Swish), $\tanh(\beta_3 z)$, $1/(1+(\beta_4 z)^2)$, $e^{-(\beta_5 z)^2}$	5
Navier--Stokes	2d-C	$\sin(\beta_1 z_s)$, $\tanh(\beta_2 z)$, $z\,\sigma(\beta_3 z)$ (Swish), $e^{-(\beta_4 z)^2}$	4
Wave	1d-C	$\sin(z_s)$, $\cos(z_s)$, $\tanh(z)$, $1/(1+z^2)$	4

[VERIFICAR: branches used for Heat, NS2d-CG, Wave2d-CG and Wave2d-MS]

Results. Table **3-6** shows the relative L^2 errors. MABUR-PINN obtains the best result in 8 of the 12 tasks: Burgers1d-C, Poisson2d-CG, Heat2d-MS, Heat2d-VC, NS2d-C, NS2d-CG, Wave1d-C and Wave2d-MS. LAAF is the best in three tasks (Burgers2d-C, Poisson2d-MS, Heat2d-CG), FBPINN in Poisson2d-C and GAAF in Wave2d-CG.

[VERIFICAR: whether the PINNacle values are means over several runs and how many runs were made for MABUR-PINN]

Discussion by class.

- **Parabolic (heat).** MABUR-PINN is the best in the multi-scale and variable-coefficient problems, where the adaptive scales β_i seem to help, and is second in the complex geometry (behind LAAF). Parabolic problems are, as expected from Table **1-1**, the ones where the errors are smaller overall.
- **Elliptic (Poisson).** The results are mixed: the best result by one order of magnitude in the complex geometry (CG), second in the classic problem (C), and a failure (L2RE > 1) in the multi-scale problem (MS), as for all the methods except LAAF.
- **Nonlinear (Burgers, Navier--Stokes).** MABUR-PINN is the best in three of the four tasks. Burgers2d-C, whose solution develops steep fronts, remains difficult.

Tabla 3-6: Relative L^2 error (L2RE) on the PINNacle benchmark. In bold, the best result of each row. Data of LAAF, GAAF and FBPINN taken from Hao et al. [55, Table 3].

PDE	Variant	LAAF	GAAF	FBPINN	MABUR-PINN
Burgers	1d-C	1.43×10^{-2}	5.20×10^{-2}	2.32×10^{-1}	$\mathbf{6.35 \times 10^{-3}}$
	2d-C	$\mathbf{2.77 \times 10^{-1}}$	2.95×10^{-1}	--	4.67×10^{-1}
Poisson	2d-C	7.68×10^{-1}	6.04×10^{-1}	$\mathbf{4.49 \times 10^{-2}}$	6.03×10^{-2}
	2d-CG	4.80×10^{-1}	8.71×10^{-1}	2.90×10^{-2}	$\mathbf{2.98 \times 10^{-3}}$
	2d-MS	$\mathbf{5.93 \times 10^{-1}}$	9.31×10^{-1}	1.04	1.32
Heat	2d-CG	$\mathbf{2.39 \times 10^{-2}}$	4.61×10^{-1}	9.16×10^{-2}	3.71×10^{-2}
	2d-MS	7.40×10^{-2}	9.85×10^{-1}	8.20×10^{-2}	$\mathbf{1.87 \times 10^{-2}}$
	2d-VC	6.42×10^{-1}	8.49×10^{-1}	9.52×10^{-1}	$\mathbf{2.25 \times 10^{-1}}$
Navier--Stokes	2d-C	3.60×10^{-2}	3.79×10^{-2}	8.45×10^{-2}	$\mathbf{1.20 \times 10^{-2}}$
	2d-CG	8.24×10^{-2}	1.74×10^{-1}	8.27	$\mathbf{6.32 \times 10^{-2}}$
Wave	1d-C	4.54×10^{-1}	6.77×10^{-1}	5.91×10^{-1}	$\mathbf{3.53 \times 10^{-1}}$
	2d-CG	8.19×10^{-1}	$\mathbf{7.94 \times 10^{-1}}$	1.06	1.42
	2d-MS	1.06	1.06	1.03	$\mathbf{9.86 \times 10^{-1}}$

- **Hyperbolic (wave).** This is the hardest class, as predicted in Section 1.4. MABUR-PINN is the best in Wave1d-C (0.353), but all the methods fail in the 2D tasks (L2RE between 0.79 and 1.42), and MABUR-PINN is the worst in Wave2d-CG (heterogeneous medium $c(\mathbf{x})$); the reason for this failure has not been analyzed yet. In Wave2d-MS the comparison is not informative, because, as shown in Section 1.5 and Appendix D, its reference solution satisfies a Klein--Gordon equation with mass $2\pi^2$ instead of the stated wave equation; a network that minimizes the residual of the wave equation cannot approach that reference.

Threats to validity. The baselines are the published values of PINNacle and were not re-run by us; the branches of the activation were chosen per family of PDEs, which is an additional degree of tuning that the baselines did not have; and MABUR-PINN was run once per task. These three facts limit the strength of the comparison, and they are addressed in the checklist of future work.

3.8 Synthesis

The experiments confirm, at least qualitatively, the three intuitions of Section 1.4. First, smoothing helps the learner: the parabolic problems are the ones with the smallest errors, while the hyperbolic ones (wave and Klein--Gordon) concentrate the failures, and inside the Klein--Gordon problem the error of the best networks lives on the light-cone fronts, where the highest frequencies travel. Second, the energy estimate of Theorem 2.3.6 explains why the derivative regularization did not help in a periodic problem. Third, a small loss is not a small error: the preliminary experiment (10^{-8} loss, 9.5 % error) and PINN-SIREN show it clearly, as predicted by Theorem 2.3.1.

Regarding the proposed architecture, MABUR-PINN V2 obtained the lowest error on the Klein--Gordon equation and the best result in 8 of 12 tasks of the benchmark. At the same time, the experiments show that the Fourier embedding is still needed (V1 vs. V2) and that the scale σ still matters (Table **3-4**); the adaptive activations reduce, but do not remove, this dependence. The bound Equation (2-47) gives a theoretical reason for using bounded branches and a bounded range of scales, which suggests replacing Swish, Softplus and exponential branches, and constraining $\beta_{\max}$, in future versions.

4 Redes Neuronales Informadas por la física en un frame cuántico

4.1 Computación Cuántica Variacional

4.2 Circuitos Cuánticos Parametrizados (PQC) como aproximadores de funciones

4.3 Integración Híbrida: PINNs y Algoritmos Cuánticos

5 Conclusiones y recomendaciones

5.1 Conclusiones

5.2 Recomendaciones

A Anexo: Derivations of the Classical Representation Formulas

This appendix contains the complete derivations of the formulas of Section 1.1, and of the propagator used for the reference solution of Chapter 3. They follow Evans [43]; the equation numbers of the main text are cited where the result is used.

A.1 Transport Equation

The geometry of a function $u : \mathbb{R}^n \times [0,\infty) \to \mathbb{R}$ that satisfies $Lu = 0$, for the transport operator $L = \partial_t + \mathbf{b}\cdot\nabla$, is represented by the hypersurface $S \subset \mathbb{R}^{n+2}$ given by

$$S = \{(\mathbf{x}, t, z) \in \mathbb{R}^n \times [0,\infty) \times \mathbb{R} \mid \Phi(\mathbf{x}, t, z) = u(\mathbf{x}, t) - z = 0\}, \tag{A-1}$$

where S is simply the graph of the solution u. Transport inherently involves velocity and spatial displacement; with the constant space--time vector field $\mathbf{V} = (\mathbf{b}, 1)^T \in \mathbb{R}^{n+1}$ we arrive at the homogeneous transport equation $\partial_t u(\mathbf{x}, t) + \mathbf{b} \cdot \nabla u(\mathbf{x}, t) = 0$.

Homogeneous case. For the problem (1-1), the matrix form of the system is

$$\begin{pmatrix} \nabla u(\mathbf{x}, t) \\ \partial_t u(\mathbf{x}, t) \end{pmatrix} \cdot \begin{pmatrix} \mathbf{b} \\ 1 \end{pmatrix} = 0 \iff D_{\mathbf{V}} u = \nabla_{\mathbf{x},t} u \cdot \mathbf{V} = 0, \tag{A-2}$$

which establishes that the directional derivative of u in the direction of the vector $\mathbf{V}$ is identically zero throughout the space--time domain. This means that the velocity vector is perpendicular to the gradient and tangent to the level sets of u. In other words, if one moves in the direction of $\mathbf{V}$, the value of u remains constant along the characteristic trajectory. The solution to $D_{\mathbf{V}} u = 0$ is $u(\mathbf{x}, t) = g(\mathbf{x} - t\mathbf{b})$.

Non-homogeneous case. For the problem (1-2) we apply the same approach as in the homogeneous case. Consider the smooth function $u : \mathbb{R}^n \times [0,\infty) \to \mathbb{R}$; the graph of the solution is again a hypersurface, now in the $(n+2)$-dimensional space of triples $(\mathbf{x}, t, z)$. At any point $(\mathbf{x}, t, z) \in S$, the normal vector to the tangent plane of the surface S is given by the gradient of Φ,

$$D\Phi(\mathbf{x}, t, z) = \big(\nabla u(\mathbf{x}, t),\ \partial_t u(\mathbf{x}, t),\ -1\big)^T. \tag{A-3}$$

Thus, the PDE can be interpreted as a tangency condition. Inspired by the same idea as in the homogeneous case, we define $z(s) = u(\mathbf{x} + s\mathbf{b}, t + s)$ for $s \in [-t, 0]$. Calculating the derivative yields

$$\frac{dz}{ds} = \partial_t u(\mathbf{x} + s\mathbf{b}, t + s) + \mathbf{b} \cdot \nabla u(\mathbf{x} + s\mathbf{b}, t + s) = f(\mathbf{x} + s\mathbf{b}, t + s), \tag{A-4}$$

and integrating from $-t$ to 0 gives

$$z(0) - z(-t) = \int_{-t}^{0} \frac{dz}{ds}\, ds = \int_{-t}^{0} f(\mathbf{x} + s\mathbf{b}, t+s)\, ds. \tag{A-5}$$

Given that $z(0) = u(\mathbf{x}, t)$ and $z(-t) = u(\mathbf{x} - t\mathbf{b}, 0) = g(\mathbf{x} - t\mathbf{b})$, and using the change of variables $\tau = t + s$, we obtain the expression

$$u(\mathbf{x}, t) = g(\mathbf{x} - t\mathbf{b}) + \int_{0}^{t} f\big(\mathbf{x} + (\tau - t)\mathbf{b}, \tau\big)\, d\tau. \tag{A-6}$$

Through Duhamel's principle, this result means that the solution is the sum of the transported initial value $g(\mathbf{x} - t\mathbf{b})$ and the values accumulated from the source f, generated in the past times τ along the same characteristic curve.

A.2 Laplace's Equation

Green's representation formula. To integrate the boundary information of the problem (1-8), we employ a vector field that contains it; following Evans [43], we combine the harmonic function u and the fundamental solution Φ in the flux $\mathbf{F} = u\nabla\Phi - \Phi\nabla u$. By the divergence theorem (Green's second identity),

$$\int_{\Omega} \operatorname{div}(\mathbf{F})\, d\mathbf{y} = \int_{\Omega} \big(u(\mathbf{y})\Delta\Phi(\mathbf{x} - \mathbf{y}) - \Phi(\mathbf{x} - \mathbf{y})\Delta u(\mathbf{y})\big)\, d\mathbf{y} \tag{A-7}$$

$$= \int_{\partial\Omega} \Big(u(\mathbf{y})\frac{\partial\Phi}{\partial\mathbf{n}}(\mathbf{x} - \mathbf{y}) - \Phi(\mathbf{x} - \mathbf{y})\frac{\partial u}{\partial\mathbf{n}}(\mathbf{y})\Big)\, dS(\mathbf{y}), \tag{A-8}$$

where $\partial f/\partial\mathbf{n} = \nabla f \cdot \mathbf{n}$ for $f \in \{u, \Phi\}$ and $\mathbf{n}$ is the outward normal vector at $\partial\Omega$. On the right side of the first equation, $-\Delta\Phi(\mathbf{x} - \mathbf{y}) = \delta(\mathbf{x} - \mathbf{y})$ at the singular point and zero elsewhere, and $\Delta u(\mathbf{y}) = 0$ by definition; thus, only the singular point survives and the integral equals $-u(\mathbf{x})$. On the boundary, $u(\mathbf{y}) = g(\mathbf{y})$. Thus,

$$u(\mathbf{x}) = \int_{\partial\Omega} \Big(\Phi(\mathbf{x} - \mathbf{y})\frac{\partial u}{\partial\mathbf{n}}(\mathbf{y}) - g(\mathbf{y})\frac{\partial\Phi}{\partial\mathbf{n}}(\mathbf{x} - \mathbf{y})\Big)\, dS(\mathbf{y}). \tag{A-9}$$

However, $\partial u/\partial\mathbf{n}$ is unknown on the boundary. To complement the previous idea and eliminate the unknown term by the Green method, it is necessary to build an auxiliary (corrector) function $\phi^{\mathbf{x}} \in C^2(\Omega) \cap C(\bar{\Omega})$ that is harmonic in Ω and matches the fundamental solution on the boundary, $\phi^{\mathbf{x}}(\mathbf{y}) = \Phi(\mathbf{y} - \mathbf{x})$ for $\mathbf{y} \in \partial\Omega$. Applying the same identity to the flux $\mathbf{F} = u\nabla\phi^{\mathbf{x}} - \phi^{\mathbf{x}}\nabla u$, where now both functions are harmonic, we obtain

$$0 = \int_{\partial\Omega} \Big(g(\mathbf{y})\frac{\partial\phi^{\mathbf{x}}}{\partial\mathbf{n}}(\mathbf{y}) - \Phi(\mathbf{x} - \mathbf{y})\frac{\partial u}{\partial\mathbf{n}}(\mathbf{y})\Big)\, dS(\mathbf{y}). \tag{A-10}$$

By the superposition of Equations (A-9) and (A-10), the unknown normal derivative cancels and we finally find the solution u through Green's function $G(\mathbf{x}, \mathbf{y}) = \Phi(\mathbf{y} - \mathbf{x}) - \phi^{\mathbf{x}}(\mathbf{y})$:

$$u(\mathbf{x}) = -\int_{\partial\Omega} g(\mathbf{y})\frac{\partial G}{\partial\mathbf{n}}(\mathbf{x}, \mathbf{y})\, dS(\mathbf{y}), \tag{A-11}$$

where $\partial G/\partial\mathbf{n} = \nabla_{\mathbf{y}} G \cdot \mathbf{n}$. Finally, we have an analytical solution through an auxiliary function that imposes the cancellation on the boundary:

$$\begin{cases} -\Delta_{\mathbf{y}} G(\mathbf{x}, \mathbf{y}) = \delta(\mathbf{x} - \mathbf{y}) & \text{in } \mathbf{y} \in \Omega, \\ G(\mathbf{x}, \mathbf{y}) = 0 & \text{on } \mathbf{y} \in \partial\Omega. \end{cases} \tag{A-12}$$

Mean-value property (heuristic derivation). Consider a ball $B(\mathbf{x},r)\subset\Omega$ centered at a point $\mathbf{x}$ inside the domain. Applying Equation (A-11) to the ball itself (homogeneous case),

$$u(\mathbf{x}) = -\int_{\partial B(\mathbf{x},r)} u(\mathbf{y})\frac{\partial G}{\partial \mathbf{n}}(\mathbf{x},\mathbf{y})\,dS(\mathbf{y}). \tag{A-13}$$

Evaluating the $\partial G/\partial\mathbf{n}$ term on the sphere $\partial B(\mathbf{x},r)$ is easy for this geometry: since $\Phi(\mathbf{y}-\mathbf{x})$ only depends on $r=|\mathbf{x}-\mathbf{y}|$, it is constant on $\partial B(\mathbf{x},r)$, and the corrector is simply the constant $\phi^{\mathbf{x}}\equiv\Phi(r)$, whose gradient vanishes. The derivative with respect to the normal vector $\mathbf{n}$ is then

$$\frac{\partial G}{\partial \mathbf{n}}(\mathbf{x},\mathbf{y}) = \big(\nabla_r\Phi(r) - \nabla_r\phi^{\mathbf{x}}\big)\cdot\mathbf{n} = \nabla_r\Phi(r)\cdot\mathbf{n} \tag{A-14}$$

$$= \begin{cases} -\dfrac{1}{2\pi r} & (n=2),\\[2mm] -\dfrac{1}{n\alpha(n)}\dfrac{1}{r^{n-1}} & (n\geq 3); \end{cases} \tag{A-15}$$

in both cases the denominator is the surface area $|\partial B(\mathbf{x},r)| = n\alpha(n)r^{n-1}$, so

$$u(\mathbf{x}) = \frac{1}{|\partial B(\mathbf{x},r)|}\int_{\partial B(\mathbf{x},r)} u(\mathbf{y})\,dS(\mathbf{y}) = \fint_{\partial B(\mathbf{x},r)} u\,dS \qquad \text{(mean-value property)}. \tag{A-16}$$

This is the mean-value property used in Section 1.1.2.

A.3 Heat Equation

Fundamental solution. The temporal derivative is of first order while the spatial derivative is of second order; to treat this, we look for solutions that are invariant under the scaling $(\mathbf{x},t)\mapsto(\lambda^\beta\mathbf{x},\lambda t)$ for $\lambda>0$. Choosing $\lambda = 1/t$, we can write the ansatz

$$u(\mathbf{x},t) = \frac{1}{t^\alpha}\,v\Big(\frac{\mathbf{x}}{t^\beta}\Big); \tag{A-17}$$

it reveals the structure of the solution, but it is not sufficient. It should be noted that for heat phenomena the thermal mass is conserved at any time $0<t\leq T$. Usually, the conservation law is written as

$$\int_{\mathbb{R}^n}\frac{1}{t^\alpha}\,v\Big(\frac{\mathbf{y}}{t^\beta}\Big)\,d\mathbf{y} = \text{constant} \qquad \forall t>0. \tag{A-18}$$

Evans [43] shows through this treatment that $\beta=1/2$ and $\alpha=n/2$. Thus, we obtain *the fundamental solution of the heat equation*, Equation (1-15).

Green's function on a bounded domain. For the problem (1-18) we must appeal to a Green's function. Given that Φ in this case depends on the space--time coordinates, it is intuitive to apply the second Green's identity to the heat operator $L=\partial_s-\Delta$ and to its adjoint (backward) operator $L^*=-\partial_s-\Delta$. Fix $(\mathbf{x},t)$ and let $w(\mathbf{y},s)=\Phi(\mathbf{x}-\mathbf{y},t-s)$, which satisfies $L^*w=0$ for $s<t$. For a small $\varepsilon>0$,

$$0 = \int_0^{t-\varepsilon}\int_\Omega \big(w\,Lu - u\,L^*w\big)\,d\mathbf{y}\,ds = \int_0^{t-\varepsilon}\int_\Omega \big(w\partial_s u + u\partial_s w - w\Delta u + u\Delta w\big)\,d\mathbf{y}\,ds \tag{A-19}$$

$$= \int_0^{t-\varepsilon}\int_\Omega \partial_s(uw)\,d\mathbf{y}\,ds - \int_0^{t-\varepsilon}\int_\Omega \big(w\Delta u - u\Delta w\big)\,d\mathbf{y}\,ds \tag{A-20}$$

$$= \Big[\int_\Omega u(\mathbf{y},s)w(\mathbf{y},s)\,d\mathbf{y}\Big]_{s=0}^{s=t-\varepsilon} - \int_0^{t-\varepsilon}\int_{\partial\Omega}\Big(w\frac{\partial u}{\partial\mathbf{n}} - u\frac{\partial w}{\partial\mathbf{n}}\Big)\,dS(\mathbf{y})\,ds. \tag{A-21}$$

When $\varepsilon \to 0$, the first term at $s = t - \varepsilon$ becomes $u(\mathbf{x}, t)$ because $\Phi(\cdot, \varepsilon) \to \delta$; the term at $s = 0$ is $\int_\Omega h(\mathbf{y})\Phi(\mathbf{x} - \mathbf{y}, t)\, d\mathbf{y}$; and the boundary integral has an unknown factor $\partial u/\partial \mathbf{n}$. As in Laplace's equation, we must build an auxiliary function that eliminates that term: let $\phi^{\mathbf{x}}(\mathbf{y}, \tau)$ solve

$$\begin{cases} \partial_\tau \phi^{\mathbf{x}}(\mathbf{y}, \tau) - \Delta_{\mathbf{y}} \phi^{\mathbf{x}}(\mathbf{y}, \tau) = 0 & \text{in } \mathbf{y} \in \Omega,\ \tau \in (0, \infty), \\ \phi^{\mathbf{x}}(\mathbf{y}, 0) = 0 & \text{on } \mathbf{y} \in \Omega,\ \{\tau = 0\}, \\ \phi^{\mathbf{x}}(\mathbf{y}, \tau) = \Phi(\mathbf{x} - \mathbf{y}, \tau) & \text{on } \mathbf{y} \in \partial\Omega,\ \tau \in (0, \infty), \end{cases} \tag{A-22}$$

and define the Green's function $G(\mathbf{x}, \mathbf{y}, \tau) = \Phi(\mathbf{x} - \mathbf{y}, \tau) - \phi^{\mathbf{x}}(\mathbf{y}, \tau)$, which vanishes for $\mathbf{y} \in \partial\Omega$. Repeating Equations (A-19)--(A-21) with $w(\mathbf{y}, s) = G(\mathbf{x}, \mathbf{y}, t - s)$, the unknown term disappears and we obtain the closed form

$$u(\mathbf{x}, t) = \int_\Omega h(\mathbf{y}) G(\mathbf{x}, \mathbf{y}, t)\, d\mathbf{y} - \int_0^t \int_{\partial\Omega} g(\mathbf{y}, s) \frac{\partial G}{\partial \mathbf{n}}(\mathbf{x}, \mathbf{y}, t - s)\, dS(\mathbf{y})\, ds, \tag{A-23}$$

where $\partial G/\partial \mathbf{n} = \nabla_{\mathbf{y}} G \cdot \mathbf{n}$. The structure is exactly the same as Equation (A-11): an integral of the data against a kernel that already knows the boundary.

Interior regularity through a cutoff function. Let K and V be two space--time subsets of $\mathbb{R}^n \times (0, T)$ such that K is compact, V is open and $K \subset\subset V$. We define a cutoff function $\zeta \in C_c^\infty(\mathbb{R}^n \times (0, T))$ as

$$\zeta(\mathbf{x}, t) = \begin{cases} 1 & \text{if } (\mathbf{x}, t) \in K, \\ 0 & \text{if } (\mathbf{x}, t) \notin V. \end{cases} \tag{A-24}$$

Given that ζ equals one inside K and zero outside V, its derivatives $D^\alpha \zeta$ with $|\alpha| \geq 1$ are zero, except in the transition zone $V \setminus K$. We construct the function $v = u\zeta$. With the aim of analyzing what this transformation does to the heat equation, $\partial_t v(\mathbf{x}, t) - \Delta v(\mathbf{x}, t) = ?$, it is necessary to decompose each term:

$$\partial_t v = \zeta \partial_t u + u \partial_t \zeta, \tag{A-25}$$

$$\Delta v = \zeta \Delta u + 2\nabla u \cdot \nabla \zeta + u \Delta \zeta, \tag{A-26}$$

to find

$$\partial_t v - \Delta v = (\partial_t u)\zeta + u\partial_t \zeta - (\Delta u)\zeta - 2\nabla u \cdot \nabla \zeta - u\Delta\zeta \tag{A-27}$$

$$= (\partial_t u - \Delta u)\zeta + (\partial_t \zeta - \Delta \zeta)u - 2\nabla u \cdot \nabla \zeta \tag{A-28}$$

$$= (\partial_t \zeta - \Delta \zeta)u - 2\nabla u \cdot \nabla \zeta =: \tilde{f}(\mathbf{x}, t). \tag{A-29}$$

Since v vanishes near $t = 0$, writing the Duhamel solution (1-21) for v gives

$$v(\mathbf{x}, t) = \int_0^t \int_{\mathbb{R}^n} \Phi(\mathbf{x} - \mathbf{y}, t - s) \tilde{f}(\mathbf{y}, s)\, d\mathbf{y}\, ds \tag{A-30}$$

$$= \int_0^t \int_{\mathbb{R}^n} \Phi\, u\, (\partial_s \zeta - \Delta \zeta)\, d\mathbf{y}\, ds - 2 \int_0^t \int_{\mathbb{R}^n} \Phi\, (\nabla \zeta \cdot \nabla u)\, d\mathbf{y}\, ds, \tag{A-31}$$

where $\Phi = \Phi(\mathbf{x} - \mathbf{y}, t - s)$ and all derivatives act on $\mathbf{y}$. Using the identity $\nabla \cdot (u\Phi\nabla\zeta) = \Phi(\nabla u \cdot \nabla\zeta) + u\nabla \cdot (\Phi\nabla\zeta)$ and the divergence theorem (the boundary term vanishes because ζ has compact support), we move the derivative from u to the known functions:

$$v(\mathbf{x}, t) = \int_0^t \int_{\mathbb{R}^n} \Big[\Phi u(\partial_s \zeta - \Delta\zeta) + 2u\nabla \cdot (\Phi \nabla \zeta)\Big]\, d\mathbf{y}\, ds \tag{A-32}$$

$$= \int_0^t \int_{\mathbb{R}^n} u\Big[\Phi(\partial_s \zeta - \Delta\zeta) + 2\nabla\Phi \cdot \nabla\zeta + 2\Phi\Delta\zeta\Big]\, d\mathbf{y}\, ds \tag{A-33}$$

$$= \int_0^t \int_{\mathbb{R}^n} u\Big[\Phi(\partial_s \zeta + \Delta\zeta) + 2\nabla\Phi \cdot \nabla\zeta\Big]\, d\mathbf{y}\, ds \tag{A-34}$$

$$= \int_0^t \int_{\mathbb{R}^n} u(\mathbf{y}, s)\, \mathcal{K}(\mathbf{x}, \mathbf{y}, t, s)\, d\mathbf{y}\, ds. \tag{A-35}$$

The interpretation of this kernel is given in Section 1.1.3.

A.4 Wave Equation

d'Alembert's formula ($n = 1$). D'Alembert's formula takes advantage of the factorization of the wave operator $\Box = \partial_t^2 - \partial_x^2$; applying it to the function $u(x,t)$ we obtain

$$\Box u(x,t) = (\partial_t^2 - \partial_x^2)u(x,t) = (\partial_t + \partial_x)(\partial_t - \partial_x)u(x,t) \tag{A-36}$$

$$= (\partial_t + \partial_x)v(x,t), \tag{A-37}$$

where $v = (\partial_t - \partial_x)u$. The equation $(\partial_t + \partial_x)v = 0$ is the homogeneous transport equation with $b = 1$, and its solution is $v(x,t) = a(x-t)$ with $a(x) = v(x,0)$. We proceed with

$$(\partial_t - \partial_x)u(x,t) = a(x-t), \tag{A-38}$$

which is the non-homogeneous transport equation with $b = -1$; by Equation (A-6) its solution is

$$u(x,t) = \int_0^t a\big(x + (t-s) - s\big)\, ds + g(x+t) \tag{A-39}$$

$$= \frac{1}{2}\int_{x-t}^{x+t} a(y)\, dy + g(x+t). \tag{A-40}$$

Then, for $a(x) = v(x,0) = \partial_t u(x,0) - \partial_x u(x,0) = h(x) - g'(x)$, the previous equation is transformed to

$$u(x,t) = \frac{1}{2}\int_{x-t}^{x+t} \big(h(y) - g'(y)\big)\, dy + g(x+t) \tag{A-41}$$

$$= \frac{1}{2}\big[g(x+t) + g(x-t)\big] + \frac{1}{2}\int_{x-t}^{x+t} h(y)\, dy. \tag{A-42}$$

This is d'Alembert's formula.

Spherical means ($n \geq 2$). For finding a solution in dimensions $n \geq 2$, it is necessary to return to the concept of average introduced in Laplace's equation, to analyze what happens with the surrounding points of a particular point. Evans [43] defines the mean value over a sphere $\partial B(\mathbf{x}, r)$ centered at $\mathbf{x}$ as

$$U(\mathbf{x}, t; r) = \fint_{\partial B(\mathbf{x},r)} u(\mathbf{y}, t)\, dS(\mathbf{y}); \tag{A-43}$$

applying the wave operator $\Box$ to U, for $r > 0$, it is possible to find

$$\begin{cases} \partial_t^2 U - \partial_r^2 U - \dfrac{n-1}{r}\partial_r U = 0 & \text{in } \mathbb{R}_+ \times (0,\infty), \\ U(\mathbf{x}, 0; r) = G(\mathbf{x}; r) & \text{on } \mathbb{R}_+ \times \{t = 0\}, \\ \partial_t U(\mathbf{x}, 0; r) = H(\mathbf{x}; r) & \text{on } \mathbb{R}_+ \times \{t = 0\}, \end{cases} \tag{A-44}$$

where G and H are the spherical means of g and h. This is the Euler--Poisson--Darboux equation. It only provides a scheme for the structure of the solution; it is not clear regarding the closed form. Evans [43] distinguishes the solution between odd and even dimensions based on spherical averages.

A.5 Propagator of the Klein–Gordon Equation

For the first-order system (3-3), with $A = \begin{pmatrix} 0 & 1 \\ -\mathcal{L} & 0 \end{pmatrix}$ and $\mathcal{L} = m^2 - \partial_{xx}$, the propagator e^{At} is responsible for the temporal evolution of the field. To compute it, we expand it in Taylor series and look at the powers

of A:

$$A^2 = \begin{pmatrix} -\mathcal{L} & 0 \\ 0 & -\mathcal{L} \end{pmatrix}, \qquad A^{2n} = (-1)^n \mathcal{L}^n I,$$

$$A^{2n+1} = (-1)^n \mathcal{L}^n A = \begin{pmatrix} 0 & (-1)^n \mathcal{L}^n \\ -(-1)^n \mathcal{L}^{n+1} & 0 \end{pmatrix}.$$

Then, separating even and odd powers,

$$e^{At} = \sum_{n=0}^{\infty} \frac{t^{2n}}{(2n)!} A^{2n} + \sum_{n=0}^{\infty} \frac{t^{2n+1}}{(2n+1)!} A^{2n+1} = \begin{pmatrix} \cos\left(t\sqrt{\mathcal{L}}\right) & \dfrac{\sin(t\sqrt{\mathcal{L}})}{\sqrt{\mathcal{L}}} \\ -\sqrt{\mathcal{L}}\sin\left(t\sqrt{\mathcal{L}}\right) & \cos\left(t\sqrt{\mathcal{L}}\right) \end{pmatrix}, \tag{A-45}$$

where we recognized the series $\cos(t\sqrt{\mathcal{L}}) = \sum_n (-1)^n t^{2n} \mathcal{L}^n/(2n)!$ and $\sin(t\sqrt{\mathcal{L}})/\sqrt{\mathcal{L}} = \sum_n (-1)^n t^{2n+1} \mathcal{L}^n/(2n+1)!$.

B Anexo: Energy Estimates and Convergence Bound for Hyperbolic PDEs

B.1 Energy Estimate for a Klein–Gordon PINN

We prove Theorem 2.3.6. Multiply the error equation (2-15), $\partial_t^2 e - \Delta e + m^2 e = r$, by $\partial_t e$ and integrate over Ω. We analyze each term separately.

Term with $\partial_t^2 e$. Since $\partial_t\big((\partial_t e)^2\big) = 2\,\partial_t e\,\partial_t^2 e$,

$$\int_\Omega \partial_t^2 e\,\partial_t e\,d\mathbf{x} = \frac{1}{2}\frac{d}{dt}\int_\Omega (\partial_t e)^2\,d\mathbf{x}.$$

Term with $-\Delta e$ (spatial integration by parts). We apply Green's first identity:

$$-\int_\Omega \Delta e\,\partial_t e\,d\mathbf{x} = \int_\Omega \nabla e\cdot\nabla(\partial_t e)\,d\mathbf{x} - \int_{\partial\Omega}\frac{\partial e}{\partial \mathbf{n}}\,\partial_t e\,dS,$$

where $\partial e/\partial\mathbf{n} = \nabla e\cdot\mathbf{n}$ is the outward normal derivative. In addition, $\nabla e\cdot\nabla(\partial_t e) = \frac{1}{2}\partial_t\left|\nabla e\right|^2$; therefore,

$$-\int_\Omega \Delta e\,\partial_t e\,d\mathbf{x} = \frac{1}{2}\frac{d}{dt}\int_\Omega \left|\nabla e\right|^2\,d\mathbf{x} - \int_{\partial\Omega}\frac{\partial e}{\partial \mathbf{n}}\,\partial_t e\,dS. \tag{B-1}$$

Term with $m^2 e$.

$$\int_\Omega m^2 e\,\partial_t e\,d\mathbf{x} = \frac{1}{2}\frac{d}{dt}\int_\Omega m^2 e^2\,d\mathbf{x}.$$

Assembly and energy identity. Adding the three terms and using the energy $E(t)$ of Equation (2-16), the integrated equation is

$$\frac{dE}{dt} = \int_\Omega r\,\partial_t e\,d\mathbf{x} + \int_{\partial\Omega}\frac{\partial e}{\partial\mathbf{n}}\,\partial_t e\,dS. \tag{B-2}$$

With $r = 0$ and periodic or homogeneous boundary conditions, this is the conservation of energy Equation (1-63).

Bounds for the source and boundary terms. For the interior term we use Young's inequality ($ab \le \frac{1}{2}a^2 + \frac{1}{2}b^2$):

$$\int_\Omega r\,\partial_t e\,d\mathbf{x} \le \frac{1}{2}\|r\|^2_{L^2(\Omega)} + \frac{1}{2}\|\partial_t e\|^2_{L^2(\Omega)} \le \frac{1}{2}\|r\|^2_{L^2(\Omega)} + E(t),$$

since $\frac{1}{2}\|\partial_t e\|^2_{L^2} \le E(t)$. For the boundary integral we use that $\partial_t e|_{\partial\Omega} = \partial_t B$ and apply Young's inequality again:

$$\int_{\partial\Omega} \frac{\partial e}{\partial \mathbf{n}}\,\partial_t B\,dS \le \frac{1}{2}\Big\|\frac{\partial e}{\partial \mathbf{n}}\Big\|^2_{L^2(\partial\Omega)} + \frac{1}{2}\|\partial_t B\|^2_{L^2(\partial\Omega)}.$$

Substituting in Equation (B-2) we obtain

$$\frac{dE}{dt} \le E(t) + \frac{1}{2}\Big(\|r\|^2_{L^2(\Omega)} + \|\partial_t B\|^2_{L^2(\partial\Omega)} + \Big\|\frac{\partial e}{\partial \mathbf{n}}\Big\|^2_{L^2(\partial\Omega)}\Big) =: E(t) + \frac{1}{2}\rho(t). \tag{B-3}$$

Gronwall's inequality. Multiplying by e^{-t}, $\frac{d}{dt}\big(e^{-t}E(t)\big) \le \frac{1}{2}e^{-t}\rho(t) \le \frac{1}{2}\rho(t)$, and integrating from 0 to t,

$$E(t) \le \Big(E(0) + \frac{1}{2}\int_0^t \rho(s)\,ds\Big)e^t,$$

which is Equation (2-17). (A cruder use of Young's inequality gives the factor e^{2t} with a constant C, which is also a valid, but weaker, bound.) The initial energy is

$$E(0) = \frac{1}{2}\int_\Omega \Big(|I_1|^2 + |\nabla I_0|^2 + m^2|I_0|^2\Big)\,d\mathbf{x} = \frac{1}{2}\Big(\|I_1\|^2_{L^2} + \|\nabla I_0\|^2_{L^2} + m^2\|I_0\|^2_{L^2}\Big).$$

With this, we obtain a bound of the energy of the error in terms of the L^2 norms of the residuals, of the gradient of the initial position residual, and of the boundary terms. In practice, for the convergence of the PINN in the energy norm, the loss must control $\|r\|^2_{L^2}$, $\|I_1\|^2_{L^2}$, $\|I_0\|^2_{L^2}$ and $\|\nabla I_0\|^2_{L^2}$ (plus the boundary terms for non-periodic boundaries), which guarantees $E(t) \to 0$ and, therefore, convergence in H^1; in one spatial dimension, by the Sobolev embedding $H^1 \subset C^{0,1/2}$, the convergence is also uniform.

B.2 From the Dirac Equation to the Klein–Gordon Equation

Since each component of the Dirac spinor individually satisfies the Klein--Gordon equation [50], the study of the dynamics of wave packets and relativistic dispersion can be done validly in the scalar formalism of Klein--Gordon. Let the Dirac equation (natural units, metric signature $(+,-,-,-)$) be

$$(i\gamma^\mu\partial_\mu - m)\psi = 0. \tag{B-4}$$

We apply the operator $(i\gamma^\nu\partial_\nu + m)$:

$$\begin{aligned} 0 &= (i\gamma^\nu\partial_\nu + m)(i\gamma^\mu\partial_\mu - m)\psi \\ &= \underbrace{-\gamma^\nu\gamma^\mu\partial_\nu\partial_\mu\psi}_{\text{kinetic term}} \underbrace{- im\gamma^\nu\partial_\nu\psi + im\gamma^\mu\partial_\mu\psi}_{\text{cancel}} \underbrace{- m^2\psi}_{\text{mass term}}. \end{aligned}$$

Since $\partial_\nu\partial_\mu$ is symmetric, only the symmetric part of $\gamma^\nu\gamma^\mu$ survives, $\frac{1}{2}\{\gamma^\nu,\gamma^\mu\} = g^{\nu\mu}$, and $g^{\nu\mu}\partial_\nu\partial_\mu = \partial^\mu\partial_\mu$. So we are left with

$$(\partial^\mu\partial_\mu + m^2)\psi = 0 \iff (\partial_t^2 - \nabla^2 + m^2)\psi = 0,$$

and we arrive at the Klein--Gordon equation for each component of ψ.

C Anexo: Derivations for the Optimization Problem of PINNs

This appendix contains the intermediate computations of Sections 2.4 and 2.5.

C.1 Gradients and KKT System in Matrix Form

For the KKT conditions (2-21)--(2-22), expanding the gradient of the initial condition term yields

$$\nabla_\theta \mathcal{L}_{ic}(\theta) = \nabla_\theta \Big(\frac{1}{|\mathcal{T}_{ic}|} \sum_{\mathbf{x}\in\mathcal{T}_{ic}} |u_\theta(\mathbf{x},0) - g_0(\mathbf{x})|^2 \Big) \tag{C-1}$$

$$= \frac{2}{|\mathcal{T}_{ic}|} \sum_{\mathbf{x}\in\mathcal{T}_{ic}} \big(u_\theta(\mathbf{x},0) - g_0(\mathbf{x})\big) \nabla_\theta u_\theta(\mathbf{x},0). \tag{C-2}$$

The terms $2u_\theta \nabla_\theta u_\theta$ and $2g_0 \nabla_\theta u_\theta$ appear; both are nonlinear in θ, because u_θ is nonlinear and so is its derivative. The same behavior is found in the gradient of the boundary condition. For the physics loss, the gradient is even more complex because of the differential operator; for the Klein--Gordon operator $\mathcal{D}_{KG} = \partial_{tt} - \nabla^2 + m^2$,

$$\nabla_\theta \mathcal{L}_f(\theta) = \frac{1}{|\mathcal{T}_f|} \sum_{(\mathbf{x},t)\in\mathcal{T}_f} \nabla_\theta \left| \mathcal{D}_{KG}[u_\theta](\mathbf{x},t) \right|^2 \tag{C-3}$$

$$= \frac{2}{|\mathcal{T}_f|} \sum_{(\mathbf{x},t)\in\mathcal{T}_f} \mathcal{D}_{KG}[u_\theta](\mathbf{x},t)\, \mathcal{D}_{KG}\big[\nabla_\theta u_\theta\big](\mathbf{x},t). \tag{C-4}$$

Notice that $\nabla_\theta \mathcal{L}_f$ requires differentiating with respect to the parameters through the temporal and spatial derivatives of the network (∂_{tt}, ∇^2), which increases the computational complexity compared with a standard neural network.

The KKT system in matrix form. Writing $N_{ic} = |\mathcal{T}_{ic}|$, $N_{bc} = |\mathcal{T}_{bc}|$, $N_f = |\mathcal{T}_f|$, the expanded KKT conditions show that the optimizer must solve a system of high-dimensional nonlinear equations (the dimensionality is induced by the neural network):

$$\frac{2}{N_{ic}} J_{ic}^T R_{ic}(\theta) + \frac{2}{N_{bc}} J_{bc}^T R_{bc}(\theta) + \frac{2v}{N_f} J_f^T R_f(\theta) = 0, \tag{C-5}$$

$$\mathcal{L}_f(\theta) = 0, \tag{C-6}$$

where $R_{ic}(\theta) \in \mathbb{R}^{N_{ic}}$ has entries $u_\theta(\mathbf{x}_i,0) - g_0(\mathbf{x}_i)$; $J_{ic}(\theta) \in \mathbb{R}^{N_{ic}\times P}$ has rows $\nabla_\theta u_\theta(\mathbf{x}_i,0)^T$ (analogously for R_{bc}, J_{bc}); $R_f(\theta) \in \mathbb{R}^{N_f}$ has entries $\mathcal{D}_{KG}[u_\theta](\mathbf{x}_k,t_k)$; and $J_f(\theta) \in \mathbb{R}^{N_f\times P}$ has rows $\mathcal{D}_{KG}[\nabla_\theta u_\theta](\mathbf{x}_k,t_k)^T$.

One multiplier per collocation point. Since the single constraint $\mathcal{L}_f(\theta) = 0$ is degenerate (Section 2.4), the natural hard-constrained problem imposes $R_f(\mathbf{x}_k, t_k; \theta) = 0$ at each of the N_f collocation points, with N_f multipliers v_k, and Equation (C-5) becomes $\frac{2}{N_{ic}} J_{ic}^T R_{ic} + \frac{2}{N_{bc}} J_{bc}^T R_{bc} + J_f^T \mathbf{v} = 0$. Whether this system has a solution depends on whether the network is expressive enough to reach that optimum.

C.2 Non-Convexity of the Loss

For the initial condition term,

$$\nabla_\theta^2 \mathcal{L}_{ic}(\theta) = \frac{2}{N_{ic}} \sum_{\mathbf{x} \in \mathcal{T}_{ic}} \Big[\nabla_\theta u_\theta(\mathbf{x}, 0)\, \nabla_\theta u_\theta(\mathbf{x}, 0)^T + \big(u_\theta(\mathbf{x}, 0) - g_0(\mathbf{x})\big) \nabla_\theta^2 u_\theta(\mathbf{x}, 0) \Big]. \tag{C-7}$$

The first term, an outer product, is always positive semidefinite (it is the Gauss--Newton part). The interesting analysis is in the second term, for which we do not know the sign of $u_\theta(\mathbf{x}, 0) - g_0(\mathbf{x})$, and $\nabla_\theta^2 u_\theta$ is, in general, indefinite. Near the optimum this factor is small, but globally nothing guarantees positivity. In fact, writing $\mathcal{L}_{ic} = F \circ G$ with $G(\theta) = u_\theta(\cdot, 0) - g_0$ and $F = \|\cdot\|_2^2$, the composition rules of Boyd and Vandenberghe [85] would require G to be affine (or convex with F non-decreasing); but even for a single layer $u_\theta(\mathbf{y}) = b + V\sigma(c + \mathbf{w} \cdot \mathbf{y})$, the dependence on $(\mathbf{w}, c)$ passes through an activation that is neither convex nor concave, so $\mathcal{L}_{ic}$ is not convex. The cost function is non-convex, and non-convex optimization methods must be used.

C.3 Infeasible-Start Newton Method

The Newton step $\nabla^2 \mathscr{L}\, \delta = -\nabla \mathscr{L}$ applied to both variables (θ, v) with the linearized constraint is the linear system [85]

$$\begin{pmatrix} \nabla_\theta^2 \mathcal{L}_{ic} + \nabla_\theta^2 \mathcal{L}_{bc} + v \nabla_\theta^2 \mathcal{L}_f & \nabla_\theta \mathcal{L}_f \\ \nabla_\theta \mathcal{L}_f^T & 0 \end{pmatrix} \begin{pmatrix} \delta\theta_{\mathrm{nt}} \\ \delta v_{\mathrm{nt}} \end{pmatrix} = - \begin{pmatrix} \nabla_\theta \mathcal{L}_{ic} + \nabla_\theta \mathcal{L}_{bc} + v \nabla_\theta \mathcal{L}_f \\ \mathcal{L}_f \end{pmatrix}. \tag{C-8}$$

Its matrix needs the Hessian of the approximating function u_θ for the losses of the initial and boundary data, and the Hessian of the physics loss, which in turn contains spatial and temporal derivatives of u_θ; its cost is estimated in Section 2.4.

C.4 Stiffness of Gradient Descent

For the steepest descent step (2-30), the Taylor expansion of the loss function around the current parameters, up to second order, is

$$\mathcal{L}(\theta_{n+1}) = \mathcal{L}(\theta_n) + (\theta_{n+1} - \theta_n)^T \nabla_\theta \mathcal{L}(\theta_n) + \frac{1}{2} (\theta_{n+1} - \theta_n)^T \nabla_\theta^2 \mathcal{L}(\xi) (\theta_{n+1} - \theta_n). \tag{C-9}$$

Using the gradient descent update rule $\theta_{n+1} - \theta_n = -\eta \nabla_\theta \mathcal{L}(\theta_n)$, the loss difference becomes

$$\mathcal{L}(\theta_{n+1}) - \mathcal{L}(\theta_n) = -\eta \left\| \nabla_\theta \mathcal{L}(\theta_n) \right\|_2^2 + \frac{1}{2} \eta^2 \nabla_\theta \mathcal{L}(\theta_n)^T \nabla_\theta^2 \mathcal{L}(\xi) \nabla_\theta \mathcal{L}(\theta_n). \tag{C-10}$$

It is worth analyzing each term. The first one is the squared gradient times the learning rate. For the second one, let $\mathbf{x} = \nabla_\theta \mathcal{L}(\theta_n) / \left\| \nabla_\theta \mathcal{L}(\theta_n) \right\|$ be the unit direction of the gradient. Since the Hessian $\nabla_\theta^2 \mathcal{L}(\xi)$ is a real

symmetric matrix, by the spectral theorem [89] it can be diagonalized by an orthogonal matrix Q that contains its eigenvectors, $\nabla^2_\theta \mathcal{L}(\xi) = Q^T \operatorname{diag}(\lambda_1, \dots, \lambda_P) Q$, with $\lambda_1 \leq \cdots \leq \lambda_P$. With $\mathbf{y} = Q\mathbf{x}$,

$$\nabla_\theta \mathcal{L}(\theta_n)^T \nabla^2_\theta \mathcal{L}(\xi) \nabla_\theta \mathcal{L}(\theta_n) = \|\nabla_\theta \mathcal{L}(\theta_n)\|^2 \, \mathbf{y}^T \operatorname{diag}(\lambda_1, \dots, \lambda_P) \mathbf{y} = \|\nabla_\theta \mathcal{L}(\theta_n)\|^2 \sum_{i=1}^{P} \lambda_i y_i^2. \tag{C-11}$$

Substituting in Equation (C-10), we obtain Equation (2-31).

C.5 Bounds on the Gradients

Wang et al. [27] develop the analysis of bounds starting from inequalities of Sobolev type for $\nabla_\theta \mathcal{L}_f$ and $\nabla_\theta \mathcal{L}_{\text{data}}$:

$$\|\nabla_\theta \mathcal{L}_f(\theta)\|_{L^\infty} \leq C \left\|\nabla^2_\theta \mathcal{L}_f(\theta)\right\|_{L^\infty}, \qquad \|\nabla_\theta \mathcal{L}_{\text{data}}(\theta)\|_{L^\infty} \leq C \left\|\nabla^2_\theta \mathcal{L}_{\text{data}}(\theta)\right\|_{L^\infty}. \tag{C-12}$$

[VERIFICAR: exact form and hypotheses of these inequalities in Wang et al. (2021)] This imposes that the maximum value of the gradient is bounded by the maximum value of the curvature of the loss function. To connect this with the eigenvalues of the Hessian, we use the equivalence of matrix norms [89]: for $H \in \mathbb{R}^{P \times P}$, $P^{-1/2} \|H\|_\infty \leq \|H\|_2 \leq P^{1/2} \|H\|_\infty$. Then the inequalities are directly related to the eigenvalues of the Hessians of the physics and data losses:

$$\|\nabla_\theta \mathcal{L}_f(\theta)\|_{L^\infty} \leq C\sqrt{P} \left\|\nabla^2_\theta \mathcal{L}_f(\theta)\right\|_2 = C\sqrt{P} \sqrt{\lambda_{\max}(H_f^T H_f)}, \tag{C-13}$$

$$\|\nabla_\theta \mathcal{L}_{\text{data}}(\theta)\|_{L^\infty} \leq C\sqrt{P} \left\|\nabla^2_\theta \mathcal{L}_{\text{data}}(\theta)\right\|_2 = C\sqrt{P} \sqrt{\lambda_{\max}(H_{\text{data}}^T H_{\text{data}})}. \tag{C-14}$$

So the size of the gradient is bounded by the largest eigenvalue of its curvature. A large eigenvalue of $H^T H$ allows large values of the gradient, while a small eigenvalue restricts the gradient to small values. Therefore, the imbalance between gradients can be reaffirmed.

C.6 Fisher Matrix and KL Divergence

Let the Kullback--Leibler (KL) divergence between $P(\cdot \mid \theta)$ and $P(\cdot \mid \theta + \delta)$ be [91]

$$\mathrm{KL}(\theta \,\|\, \theta + \delta) = \mathbb{E}_{\mathbf{x} \sim P(\cdot \mid \theta)} \big[\log P(\mathbf{x} \mid \theta) - \log P(\mathbf{x} \mid \theta + \delta) \big].$$

Expanding $\log P(\mathbf{x} \mid \theta + \delta)$ to second order, we obtain

$$\log P(\mathbf{x} \mid \theta + \delta) \approx \log P(\mathbf{x} \mid \theta) + \nabla_\theta \log P(\mathbf{x} \mid \theta)^T \delta + \frac{1}{2} \delta^T \nabla^2_\theta \log P(\mathbf{x} \mid \theta) \, \delta,$$

and substituting,

$$\mathrm{KL}(\theta \,\|\, \theta + \delta) \approx -\mathbb{E}\big[\nabla_\theta \log P(\mathbf{x} \mid \theta)\big]^T \delta - \frac{1}{2} \delta^T \, \mathbb{E}\big[\nabla^2_\theta \log P(\mathbf{x} \mid \theta)\big] \, \delta. \tag{C-15}$$

The first term vanishes for every θ (not only at the optimum), because of the following properties:

$$\int P(\mathbf{x} \mid \theta) \, d\mathbf{x} = 1 \;\Rightarrow\; \int \nabla_\theta P(\mathbf{x} \mid \theta) \, d\mathbf{x} = 0,$$

$$\nabla_\theta \log P = \frac{\nabla_\theta P}{P} \;\Rightarrow\; \mathbb{E}\big[\nabla_\theta \log P(\mathbf{x} \mid \theta)\big] = \int P \, \nabla_\theta \log P \, d\mathbf{x} = 0.$$

Differentiating the last identity once more with respect to θ,

$$0 = \int \Big(\nabla_\theta P \, \nabla_\theta \log P^T + P \, \nabla_\theta^2 \log P \Big) \, d\mathbf{x} = \int P \Big(\nabla_\theta \log P \, \nabla_\theta \log P^T + \nabla_\theta^2 \log P \Big) \, d\mathbf{x},$$

that is, the expectation of the second moment satisfies

$$\mathbb{E}\big[\nabla_\theta \log P \, \nabla_\theta \log P^T \big] = -\mathbb{E}\big[\nabla_\theta^2 \log P \big].$$

Therefore,

$$\mathrm{KL}(\theta \,\|\, \theta + \delta) \approx \frac{1}{2} \delta^T \, \mathbb{E}\big[\nabla_\theta \log P(\mathbf{x} \mid \theta) \, \nabla_\theta \log P(\mathbf{x} \mid \theta)^T \big] \, \delta = \frac{1}{2} \delta^T F \delta.$$

D Anexo: Tasks of the PINNacle Benchmark

This appendix lists the twelve tasks of Hao et al. [55] used in Section 3.7 (CI: initial condition; CF: boundary condition).

D.1 Burgers Equation

Burgers1d-C. $u_t + uu_x = \nu u_{xx}$, $(x,t) \in [-1,1]\times[0,1]$, $u(x,0) = -\sin(\pi x)$, $u(-1,t) = u(1,t) = 0$, $\nu = 0.01/\pi$.

Burgers2d-C (coupled). $\mathbf{u}_t + \mathbf{u}\cdot\nabla\mathbf{u} - \nu\Delta\mathbf{u} = 0$, $(x,y,t) \in [0,L]^2 \times [0,1]$, $\mathbf{u}(x,y,0) = 2\mathbf{w}(x,y) + \mathbf{c}$ with $\mathbf{w}$ a random Fourier series (Gaussian random field), periodic boundary conditions, $L = 4$, $\nu = 0.001$.

D.2 Poisson Equation (elliptic)

Poisson2d-C. $-\Delta u = 0$ in $\Omega = [-0.5, 0.5]^2 \setminus \bigcup_{i=1}^{4} R_i$, where R_i are four circles of radius 0.1 centered at $(\pm 0.3, \pm 0.3)$; $u = 0$ on ∂R_i and $u = 1$ on the outer rectangle.

Poisson2d-CG (Poisson–Boltzmann). $-\Delta u + k^2 u = f(x,y)$ in $[-1,1]^2 \setminus \bigcup_{i=1}^{4} R_i$, $f(x,y) = A\sum_i(\mu_i^2 + x_i^2)\sin(\mu_1\pi x)\sin(\mu_2\pi y)$; $u = 0.2$ on the rectangle and $u = 1$ on the circles; $\mu_1 = 1$, $\mu_2 = 4$, $k = 8$, $A = 10$.

Poisson2d-MS (multi-scale). $-\nabla\cdot(a(\mathbf{x})\nabla u) = f(x,y)$ in $[-10,10]^2$, $\partial u/\partial\mathbf{n} + u = 0$ on $\partial\Omega$, with $a(\mathbf{x})$ piecewise linear on square subdomains.

D.3 Heat Equation (parabolic)

Heat2d-CG (complex geometry). $u_t - \Delta u = 0$ in $\left([-8,8]\times[-12,12] \setminus \bigcup_i R_i\right) \times [0,3]$, Robin condition $-\mathbf{n}\cdot(-c\nabla u) = g - qu$ with $c = 1$; large circles centered at $(\pm 4, \pm 3)$, $(\pm 4, \pm 9)$, $(0,0)$, $(0, \pm 6)$ with $r = 1$,

$g = 5$, $q = 1$; small circles centered at $(\pm 3.2, \pm 6)$, $(\pm 3.2, 0)$ with $r = 0.4$, $g = 1$, $q = 1$.

Heat2d-MS (multi-scale). $u_t - \frac{1}{(500\pi)^2} u_{xx} - \frac{1}{\pi^2} u_{yy} = 0$ in $[0,1]^2 \times [0,5]$, $u(x,y,0) = \sin(20\pi x)\sin(\pi y)$, $u = 0$ on $\partial\Omega$.

Heat2d-VC (variable coefficients). $u_t - \nabla\cdot(a(\mathbf{x})\nabla u) = f(\mathbf{x},t)$ in $[0,1]^2 \times [0,5]$, $f = A\sin(m_1\pi x)\sin(m_2\pi y)\sin(m_3\pi t)$ with $A = 200$, $m_1 = 1$, $m_2 = 5$, $m_3 = 1$; $a(\mathbf{x})$ generated by an exponential Gaussian random field; $u(x,y,0) = 0$, $u = 0$ on $\partial\Omega$.

D.4 Navier–Stokes Equations

NS2d-C (lid-driven cavity). $\mathbf{u}\cdot\nabla\mathbf{u} + \nabla p - \frac{1}{Re}\Delta\mathbf{u} = 0$, $\nabla\cdot\mathbf{u} = 0$ in $[0,1]^2$; on the top lid $\mathbf{u} = (4x(1-x), 0)$; elsewhere $\mathbf{u} = (0,0)$, and $p = 0$ at $(0,0)$; $Re = 100$.

NS2d-CG (backward-facing step). Same equations in $\Omega = [0,4]\times[0,2] \setminus \big([0,2]\times[1,2]\big)$; inflow $u_{\text{in}} = 4y(1-y)$; $p = 0$ at the outlet and no-slip $\mathbf{u} = 0$ on the walls; $Re = 100$.

D.5 Wave Equation (hyperbolic)

Wave1d-C. $u_{tt} - 4u_{xx} = 0$, $(x,t) \in [0,1]\times[0,1]$, $u(x,0) = \sin(\pi x) + \frac{1}{2}\sin(4\pi x)$, $u_t(x,0) = 0$, $u(0,t) = u(1,t) = 0$; exact solution

$$u(x,t) = \sin(\pi x)\cos(2\pi t) + \tfrac{1}{2}\sin(4\pi x)\cos(8\pi t).$$

Wave2d-CG (heterogeneous medium). $\nabla^2 u - \frac{1}{c(\mathbf{x})}\frac{\partial^2 u}{\partial t^2} = 0$, $(x,y) \in [-1,1]^2$, $t \in [0,5]$, $u(\mathbf{x},0) = \exp(-\|\mathbf{x}-\boldsymbol{\mu}\|^2/(2\sigma^2))$, $u_t(\mathbf{x},0) = 0$, Neumann condition $\partial u/\partial\mathbf{n} = 0$; $\boldsymbol{\mu} = (-0.5, 0)$, $\sigma = 0.3$; $c(\mathbf{x})$ a Gaussian random field.

Wave2d-MS (multi-scale). Stated equation $u_{tt} - (u_{xx} + a^2 u_{yy}) = 0$, $(x,y,t) \in [0,1]^2 \times [0,100]$, with reference solution

$$u = c_1 \sin(m_1\pi x)\sinh(n_1\pi y)\cos(p_1\pi t) + c_2\sinh(m_2\pi x)\sin(n_2\pi y)\cos(p_2\pi t),$$

$a = \sqrt{2}$, $m_1 = 1$, $m_2 = 3$, $n_1 = 1$, $n_2 = 2$, $p_1 = p_2 = 1$, $c_1 = c_2 = 1$.

The reference solution of Wave2d-MS is a Klein–Gordon solution. For the first term, $u_1 = \sin(\pi x)\sinh(\pi y)\cos(\pi t)$, we have $\partial_{tt}u_1 = -\pi^2 u_1$, $\partial_{xx}u_1 = -\pi^2 u_1$ and $\partial_{yy}u_1 = \pi^2 u_1$, so

$$\partial_{tt}u_1 - (\partial_{xx}u_1 + 2\partial_{yy}u_1) = (-\pi^2 + \pi^2 - 2\pi^2)u_1 = -2\pi^2 u_1.$$

For the second term, $u_2 = \sinh(3\pi x)\sin(2\pi y)\cos(\pi t)$, we have $\partial_{tt}u_2 = -\pi^2 u_2$, $\partial_{xx}u_2 = 9\pi^2 u_2$ and $\partial_{yy}u_2 = -4\pi^2 u_2$, so

$$\partial_{tt}u_2 - (\partial_{xx}u_2 + 2\partial_{yy}u_2) = (-\pi^2 - 9\pi^2 + 8\pi^2)u_2 = -2\pi^2 u_2.$$

Therefore, by linearity, $u = u_1 + u_2$ satisfies

$$u_{tt} - (u_{xx} + 2u_{yy}) + 2\pi^2 u = 0, \tag{D-1}$$

which is a Klein--Gordon equation with mass $m^2 = 2\pi^2$, and not the stated wave equation. (The computation was also checked symbolically.) This would explain why all the methods of Table **3-6** report L2RE ≈ 1 in this task, and it is one more motivation for studying the Klein--Gordon equation (Section 1.5).

E Anexo: Error Maps and Training Curves of Experiment 1

Each figure shows the pseudo-spectral reference solution, the prediction and the absolute error (top), and the history of the loss terms during training (bottom).

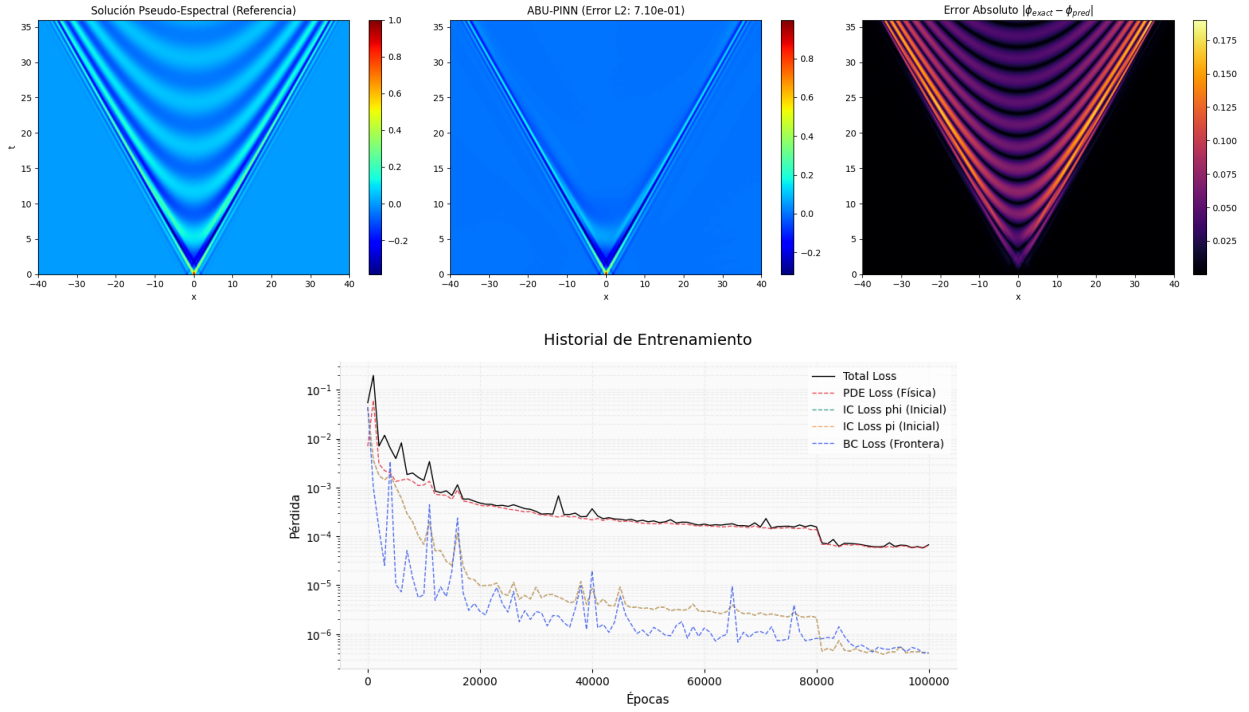


Figura E-1: Vanilla PINN: L2RE 71.0 %. Large errors distributed over the whole region; the total loss oscillates and converges slowly, to values of order 10^{-5}.

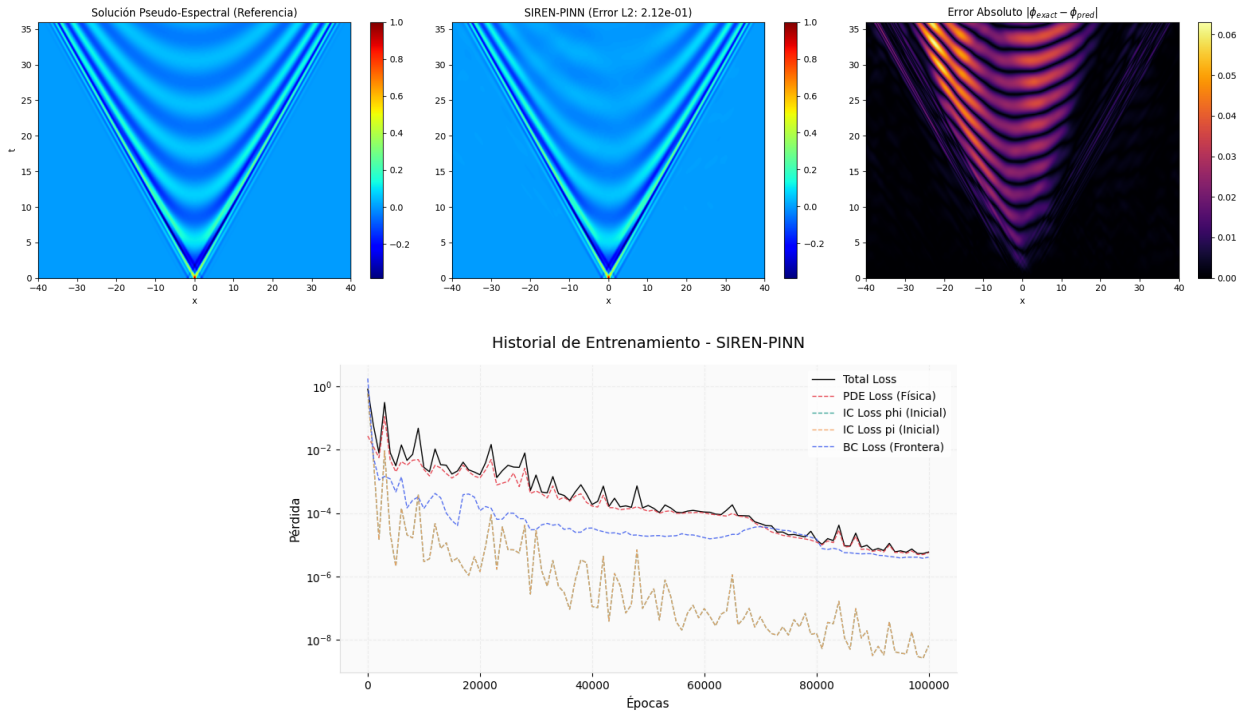


Figura E-2: PINN-SIREN: L2RE 21.2 %. The maximum errors are reduced with respect to the vanilla PINN, with a better capture of the oscillations; the initial position loss reaches values close to 10^{-8}.

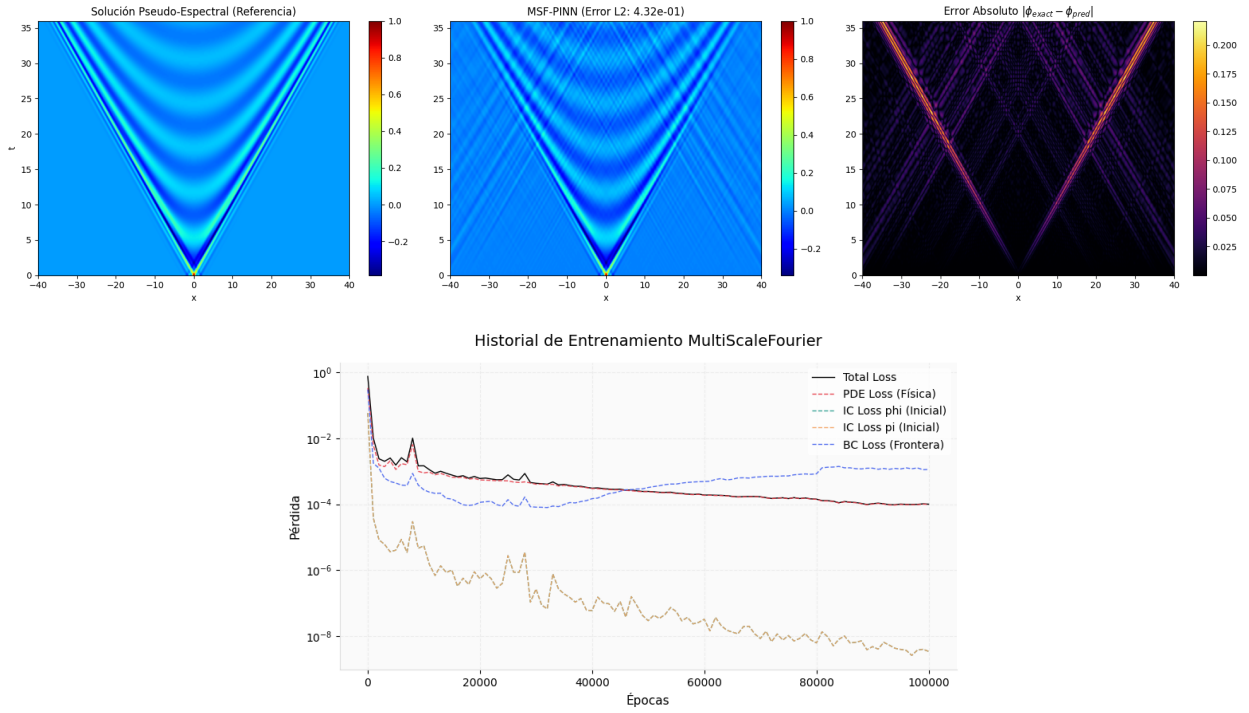


Figura E-3: MSF-PINN: L2RE 43.2 %. Despite the high number of parameters, the errors are significant, especially where the solution has high frequencies; the loss curves show high variance and poor convergence (order 10^{-4}).

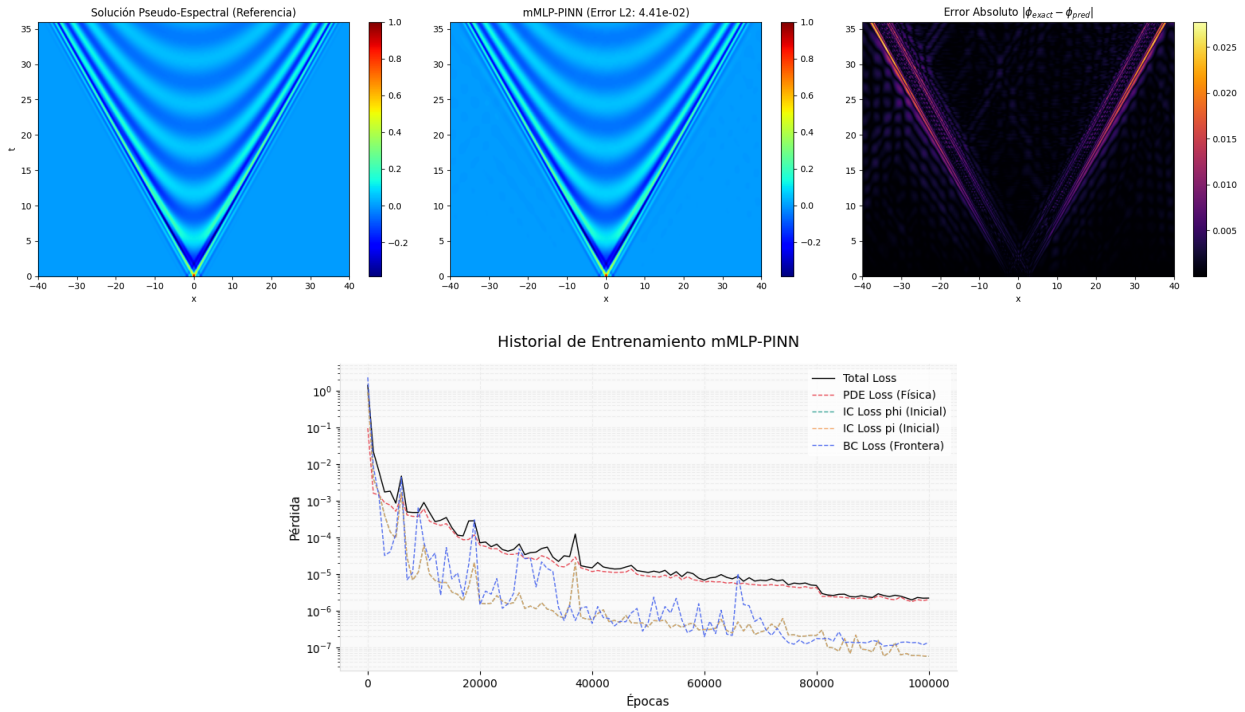


Figura E-4: M4-PINN: L2RE 4.41 %. The maximum errors are drastically reduced, with a more homogeneous distribution; the total loss decreases rapidly and stabilizes at about 2.21×10^{-6}.

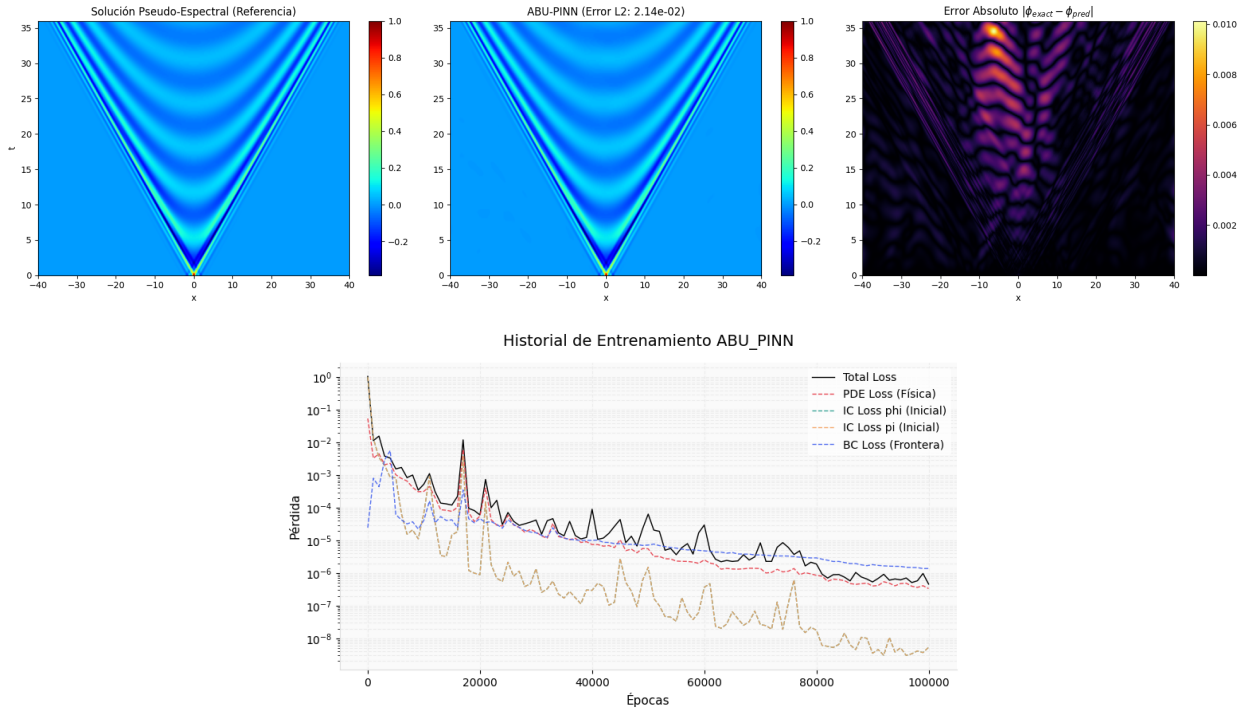


Figura E-5: AbuPINN: L2RE 2.14 %. Homogeneous error without pronounced peaks; the total loss reaches 4.60×10^{-7} with an almost stable and fast convergence.

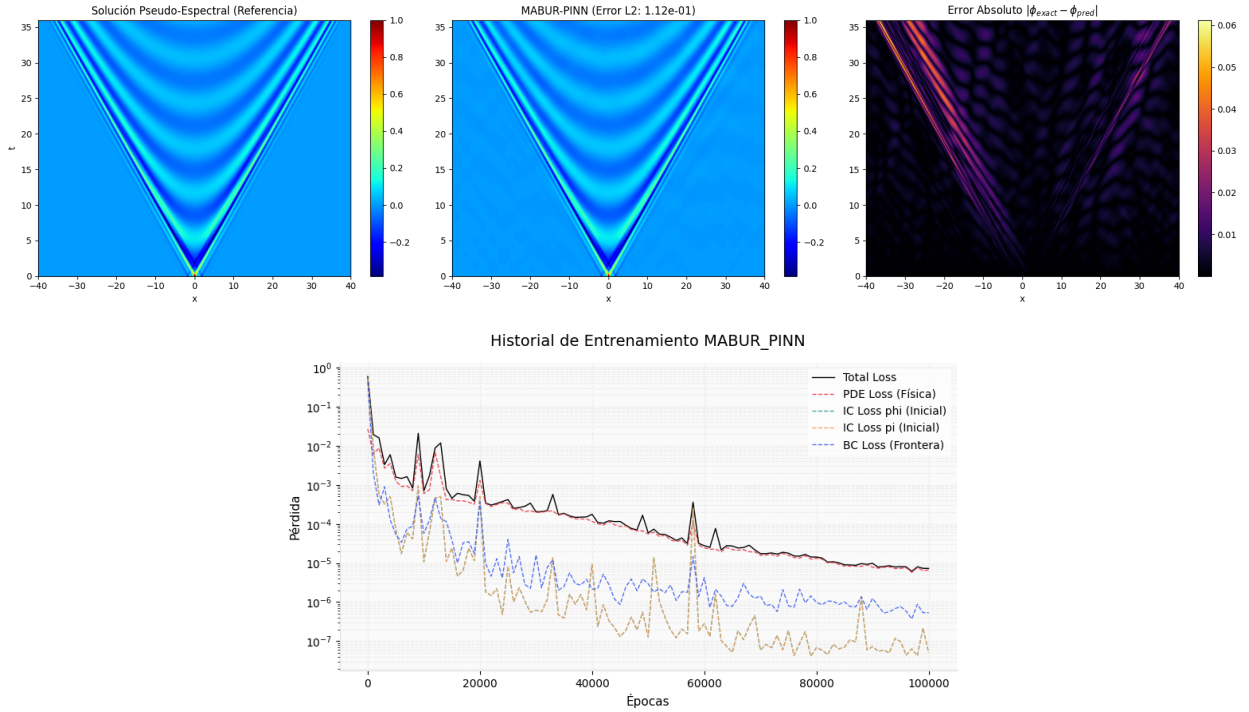


Figura E-6: MABUR-PINN V1: L2RE 11.2 %. Improvements in the high-frequency regions with respect to MSF-PINN, but difficulties representing the frequencies inside the light cone; the total loss converges to about 7.25×10^{-6}.

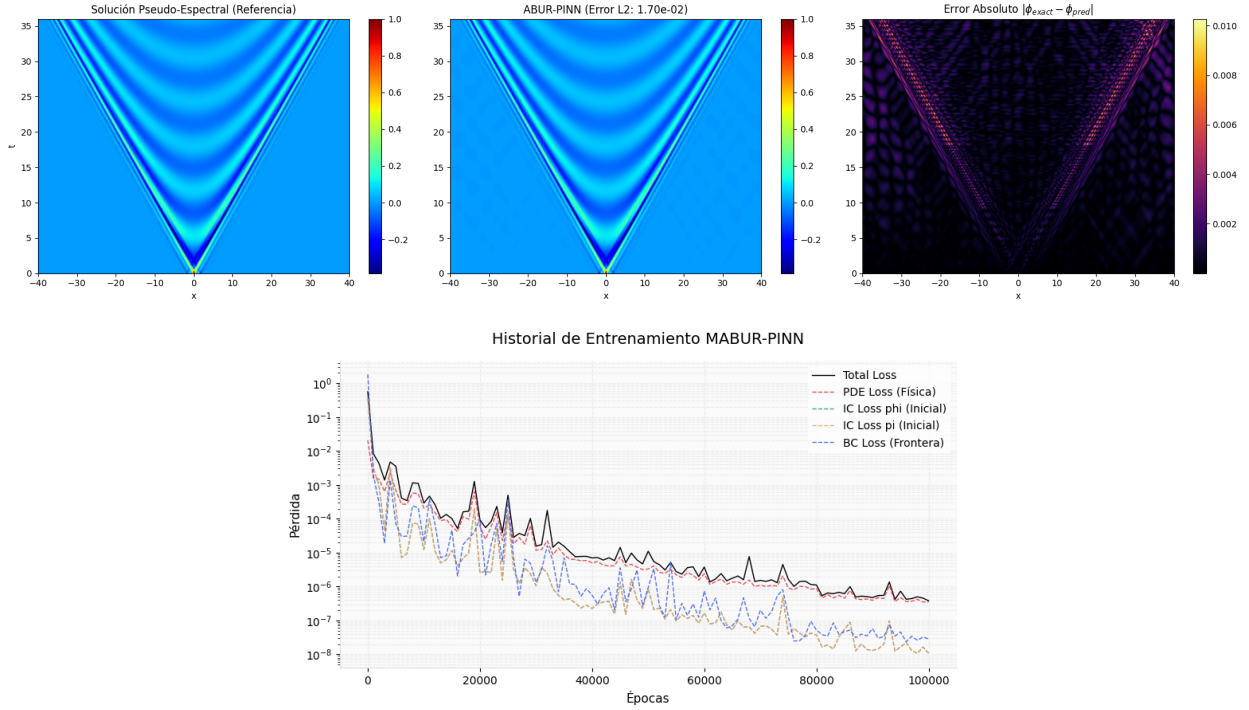


Figura E-7: MABUR-PINN V2: L2RE 1.70 %. The error is concentrated on the fronts of the light cone; the total loss converges to 3.81×10^{-7}.

E. Error Maps of Experiment 1

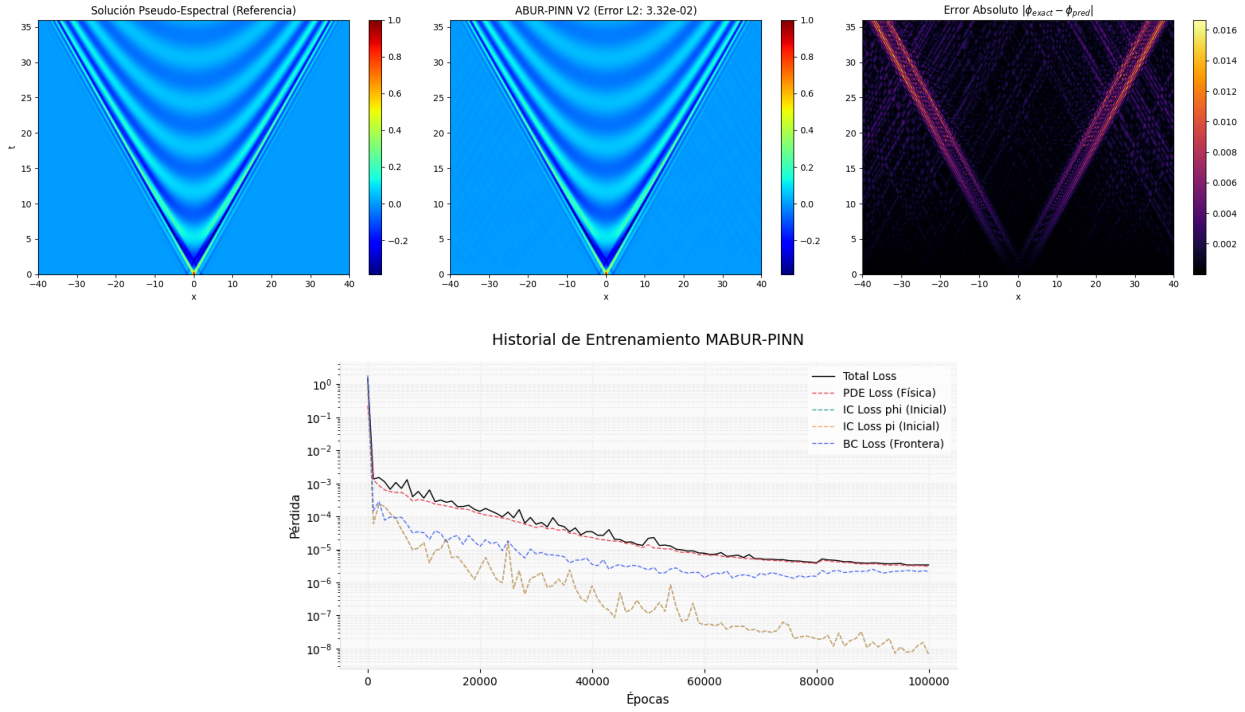


Figura E-8: MABUR-PINN V2 ($\sigma = 5$): L2RE 3.32 %. Same architecture with a larger Fourier scale.

F Anexo: Código y repositorio

All the code and results of this thesis are available in the following repositories:

- Architectures and Klein--Gordon experiments (Chapter 3): `https://github.com/Emmaquantum/HPINN`
- Optimization study and pseudo-spectral reference solution: `https://github.com/Emmaquantum/Optimization_PINN`

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